

PolyKybd

RoHS Compliance Appendix

Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

CE Technical File — Restriction of Hazardous Substances evidence for all populated components and the bare printed circuit board.

Prepared: 2026-07-12

Parts → Certificate Reference

Sec.	Designators	Part / Value	Qty	Manufacturer	Certificate document
1	C1_1-C1_36	2.2uF 0603 25V	36	Samsung	2304140030_Samsung-Electro-Mechanics-CL05 A105KA5NQNC_C52923.pdf
	C2,C3_1-C3_36,C5_1-C5_36,C6_1-C6_36,C24,C25,C30-C32	1uF 0402 16V	114	Samsung	
	C2_1-C2_36	2.2uF 0603 16V	36	Samsung	
	C3,C8	27pF 0402 16V	2	Samsung	
	C4,C10	4.7uF 0805 25V	2	Samsung	
	C5,C11	6.8pF 0603 50V	2	Samsung	
	C6,C12-C14,C26	10uF 0805 16V	5	Samsung	
	C9,C15-C23	100nF 0402 16V	10	Samsung	
2	C4_1-C4_36,C33,C34	4.7uF 3216 25V Tantalum	38	Vishay	vishay-rohs-20250901.pdf
	R5,R6,R16,R19,R21	1k 0402	5	Vishay	
3	D1	Red 0805 C84256	1	NationStar	2409272203_Foshan-NationStar-Optoelectronics -NCD0805R1_C84256.pdf
4	D1_1-D1_36,D12	1N4148WSX	37	Shenzhen JingYang	2405091035_Shenzhen-JingYang-1N4148WSX_C6423741.pdf
5	D2	1N5819WS	1	Hottech	ROHS-2025-HottechElectronics-CTI-A22502473 6310100101.pdf
6	D5	Green 0805 C2297	1	Hubei KENTO	C2297.pdf
7	D6	Yellow 0805 C2296	1	Hubei KENTO	1806151129_Hubei-KENTO-Elec-KT-0805Y_C2296.pdf
8	FB1,FB2	GZ2012D601TF	2	Sunlord	2310301640_Sunlord-GZ2012D601TF_C1017.pdf
9	J1-J36	FH34SRJ-14S-0.5SH	36	JUSHUO	FH34SRJ-14S-0.5SH(50)_CL0580-1252-8-50_SpecSheet_0000414509.pdf
10	J37	FH34SRJ-12S-0.5SH	1	Hirose	FH34SRJ-12S-0.5SH(99)_CL0580-1253-0-99_SpecSheet_0000414526.pdf
11	J39	FH12-30S-0.5SH	1	Hirose	FH12-30S-05SH.pdf
12	L1,L2	2.2uH	2	SXN	2108131530_SXN-Shun-Xiang-Nuo-Elec-SMMS 0420-2R2M_C133191.pdf
13	LED1-LED36	XL-3030RGBC	36	Xinglight	2402181502_XINGLIGHT-XL-3030RGBC-WS2812B_C5349958.pdf
14	R1,R2	22 0603	2	Yageo	2304140030_YAGEO-RC0603JR-0710ML_C141675.pdf
	R1_1-R1_36,R15,R20,R25	10k 0402	39	Yageo	
	R8	5.6K 0402	1	Yageo	
	R10,R11,R30	390k 0603 1%	3	Yageo	
	R12,R17	150k 0603	2	Yageo	
	R13	12k 0603	1	Yageo	
	R14	680k 0603	1	Yageo	
	R22,R23	27 0603	2	Yageo	
	R24	2M2 0603	1	Yageo	
15	R3,R4,R7,R9	5.1K 0402	4	Yageo	50630577_YAGEO-AC0603FR-0710ML_C227547.pdf
	R18	10M 0603	1	Yageo	
16	SW2,SW3	EVQPUC02K	2	Panasonic	panasonic_rohs_2ma_2000379.pdf
17	SW_K_1-SW_K_36	Kailh CPG151101S11-16	36	Kailh	2208291600_Kailh-CPG151101S11-16_C5156480.pdf
18	U1,U2	TLV62569DBVR	2	TI	TexasTLV62569DBVR.pdf
19	U3	SY6280AAC	1	Silergy	ROHS-2024-SilergyCorp.pdf
20	U4-U8	74HC595BQ	5	Nexperia	Nexperia - Statement on RoHS 20241008.pdf
	U12-U25	74AHC1G125GW,125	14	Nexperia	
21	U9	BY25Q64ESCIG(R)	1	BOYAMICRO	2006180000_BOYAMICRO-BY25Q64ESCIG-R_C50176394.pdf
22	U10	RP2040	1	Raspberry Pi	rp2040-rohs.png
23	U11,U26	TPD4E05U06DQAR	2	MSKSEMI	C2836386-TPD4E05U06DQAR-MS.pdf

Sec.	Designators	Part / Value	Qty	Manufacturer	Certificate document
24	USB1,USB2	TYPE-C-31-M-12	2	Hroparts	2410010003_Korean-Hroparts-Elec-TYPE-C-31-M-12_C165948.pdf
25	Y1	7M12000044	1	TXC	TXC-7M12000044.pdf
26	PCB / bare board	Printed circuit board (ENIG & lead-free HASL finish); incl. exposed copper of mounting holes & test points	-	JLCPCB	rohs-certificate-of-conformity.pdf

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Section 2 (p.11)	— Vishay — C4_1-C4_36,C33,C34, R5,R6,R16,R19,R21
Section 3 (p.13)	— NationStar — D1
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Section 1

Samsung

Designators	Part / Value	Qty
C1_1-C1_36	2.2uF 0603 25V	36
C2,C3_1-C3_36,C5_1-C5_36,C6_1-C6_36,C24,C25,C30-C32	1uF 0402 16V	114
C2_1-C2_36	2.2uF 0603 16V	36
C3,C8	27pF 0402 16V	2
C4,C10	4.7uF 0805 25V	2
C5,C11	6.8pF 0603 50V	2
C6,C12-C14,C26	10uF 0805 16V	5
C9,C15-C23	100nF 0402 16V	10

RoHS evidence document: **2304140030_Samsung-Electro-Mechanics-CL05A105KA5NQNC_C52923.pdf**

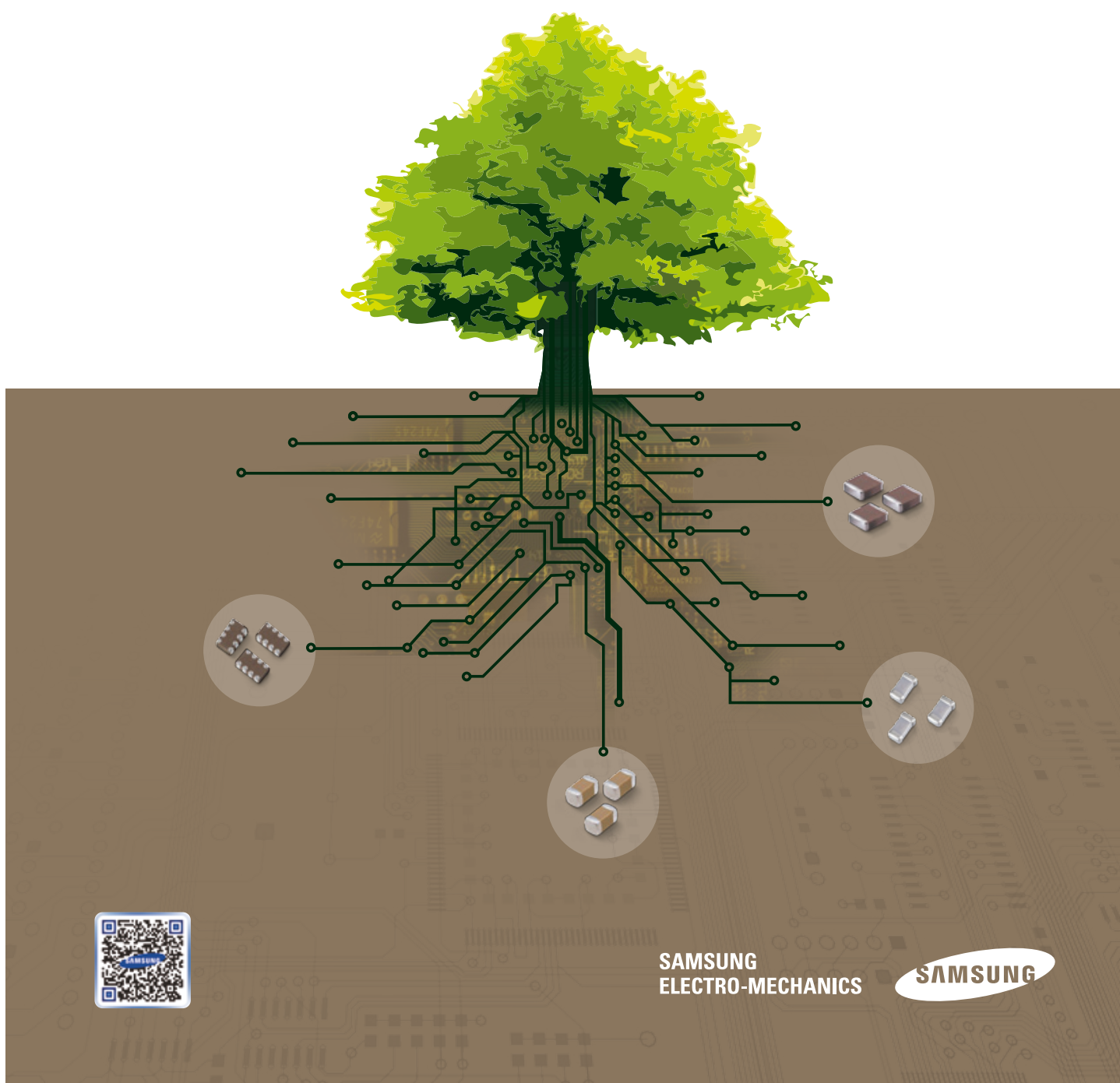
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

Excerpt: page(s) 1, 2, 13, 19, 25 of 84 — RoHS-relevant pages only; the complete document is retained in the project repository.

November 2015



MULTILAYER CERAMIC CAPACITORS



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We declare that all our MLCCs are produced in accordance with EU ROHS and REACH Directive.

1. RoHS Compliance and restriction of Br

The following restricted materials are not used in packaging materials as well as products in compliance with the law and restriction.
- Cd, Pb, Hg, Cr6+, As, Br and the compounds, PCB, asbestos

2. No use of materials breaking Ozone layer

The following ODS materials are not used in our fabrication process.
- ODS material : Freon, Haron, 1-1-1 TCE, CCl4, HCFC

If you want more detailed Information, Please Visit Samsung Electro-mechanics Website
[<http://www.semclcr.com>]

Please, see the last page of this catalog for our environmental certification list.

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Product Lineup (Standard & High Capacitors-C0G)

	Part Number	Size L x W (mm)	Capacitance	Rated Voltage (Vdc)	Capacitance Tolerance	Thickness Max.(mm)
1	CL05C0R5CB5NNN □	1.00×0.50	0.5pF	50	±0.25pF	0.55
2	CL05CR75CB5NNN □		0.75pF	50	±0.25pF	0.55
3	CL05C010CB5NNN □		1.0pF	50	±0.25pF	0.55
4	CL05C1R2CB5NNN □		1.2pF	50	±0.25pF	0.55
5	CL05C1R5CB5NNN □		1.5pF	50	±0.25pF	0.55
6	CL05C1R8CB5NNN □		1.8pF	50	±0.25pF	0.55
7	CL05C020CB5NNN □		2.0pF	50	±0.25pF	0.55
8	CL05C2R2CB5NNN □		2.2pF	50	±0.25pF	0.55
9	CL05C2R4CB5NNN □		2.4pF	50	±0.25pF	0.55
10	CL05C2R5CB5NNN □		2.5pF	50	±0.25pF	0.55
11	CL05C2R7CB5NNN □		2.7pF	50	±0.25pF	0.55
12	CL05C030CB5NNN □		3.0pF	50	±0.25pF	0.55
13	CL05C3R3CB5NNN □		3.3pF	50	±0.25pF	0.55
14	CL05C3R5CB5NNN □		3.5pF	50	±0.25pF	0.55
15	CL05C3R6CB5NNN □		3.6pF	50	±0.25pF	0.55
16	CL05C3R9CB5NNN □		3.9pF	50	±0.25pF	0.55
17	CL05C040CB5NNN □		4.0pF	50	±0.25pF	0.55
18	CL05C4R3CB5NNN □		4.3pF	50	±0.25pF	0.55
19	CL05C4R7CB5NNN □		4.7pF	50	±0.25pF	0.55
20	CL05C050DB5NNN □		5.0pF	50	±0.5pF	0.55
21	CL05C5R6DB5NNN □		5.6pF	50	±0.5pF	0.55
22	CL05C060DB5NNN □		6.0pF	50	±0.5pF	0.55
23	CL05C6R2DB5NNN □		6.2pF	50	±0.5pF	0.55
24	CL05C6R8DB5NNN □		6.8pF	50	±0.5pF	0.55
25	CL05C070DB5NNN □		7.0pF	50	±0.5pF	0.55
26	CL05C080DB5NNN □		8.0pF	50	±0.5pF	0.55
27	CL05C8R2DB5NNN □		8.2pF	50	±0.5pF	0.55
28	CL05C090DB5NNN □		9.0pF	50	±0.5pF	0.55
29	CL05C9R1DB5NNN □		9.1pF	50	±0.5pF	0.55
30	CL05C100JB5NNN □		10pF	50	±5%	0.55
31	CL05C110JB5NNN □		11pF	50	±5%	0.55
32	CL05C120JB5NNN □		12pF	50	±5%	0.55
33	CL05C130JB5NNN □		13pF	50	±5%	0.55
34	CL05C150JB5NNN □		15pF	50	±5%	0.55
35	CL05C160JB5NNN □		16pF	50	±5%	0.55
36	CL05C180JB5NNN □		18pF	50	±5%	0.55
37	CL05C200JB5NNN □		20pF	50	±5%	0.55
38	CL05C220JA5NNN □		22pF	25	±5%	0.55
39	CL05C220JB5NNN □		22pF	50	±5%	0.55
40	CL05C240JB5NNN □		24pF	50	±5%	0.55
41	CL05C270JB5NNN □		27pF	50	±5%	0.55
42	CL05C270JA5NNN □		27pF	25	±5%	0.55
43	CL05C300JB5NNN □		30pF	50	±5%	0.55
44	CL05C330JB5NNN □		33pF	50	±5%	0.55
45	CL05C360JB5NNN □		36pF	50	±5%	0.55
46	CL05C390JB5NNN □		39pF	50	±5%	0.55
47	CL05C430JB5NNN □		43pF	50	±5%	0.55
48	CL05C470JB5NNN □		47pF	50	±5%	0.55
49	CL05C510JB5NNN □		51pF	50	±5%	0.55
50	CL05C560JB5NNN □		56pF	50	±5%	0.55

Part Numbering
SystemStandard &
High CapacitorsSuper Small Size
CapacitorsHigh-Q
CapacitorsMedium-High
Voltage CapacitorsArray Type
CapacitorsLow ESL
CapacitorsReliability Test
ConditionPremium Capacitors
for Automotive
ApplicationsPackaging
SpecificationApplication Manual
for Surface Mounting

※ □ mark means packaging code. If you want to learn the code or quantity in detail, please see p74.

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Product Lineup (Standard & High Capacitors-X5R)

	Part Number	Size L x W (mm)	Capacitance	Rated Voltage (Vdc)	Capacitance Tolerance	Thickness Max.(mm)
1	CL05A223KO5NNN□	1.00×0.50	22 nF	16	±10%	0.55
2	CL05A104KA5NNN□		0.1 μF	25	±10%	0.55
3	CL05A104KO5NNN□		0.1 μF	16	±10%	0.55
4	CL05A104KP5NNN□		0.1 μF	10	±10%	0.55
5	CL05A224KA5NNN□		0.22 μF	25	±10%	0.55
6	CL05A224KP5NNN□		0.22 μF	10	±10%	0.55
7	CL05A334KA5NNN□		0.33 μF	25	±10%	0.55
8	CL05A334KP5NNN□		0.33 μF	10	±10%	0.55
9	CL05A474KA5NNN□		0.47 μF	25	±10%	0.55
10	CL05A474KO5NNN□		0.47 μF	16	±10%	0.55
11	CL05A474KP5NNN□		0.47 μF	10	±10%	0.55
12	CL05A474KQ5NNN□		0.47 μF	6.3	±10%	0.55
13	CL05A474KR5NNN□		0.47 μF	4	±10%	0.55
14	CL05A105KA5NQ□		1 μF	25	±10%	0.60
15	CL05A105KO5NNN□		1 μF	16	±10%	0.55
16	CL05A105KO3LQ□		1 μF	16	±10%	0.33
17	CL05A105KP5NNN□		1 μF	10	±10%	0.55
18	CL05A105KP3LNN□		1 μF	10	±10%	0.33
19	CL05A105KQ5NNN□		1 μF	6.3	±10%	0.55
20	CL05A105KQ3LNN□		1 μF	6.3	±10%	0.33
21	CL05A105KR5NNN□		1 μF	4	±10%	0.55
22	CL05A105KR3LNN□		1 μF	4	±10%	0.33
23	CL05A225MA5NUN□		2.2 μF	25	±20%	0.70
24	CL05A225KO5NQ□		2.2 μF	16	±10%	0.60
25	CL05A225MP5NS□		2.2 μF	10	±20%	0.57
26	CL05A225KP3LRN□		2.2 μF	10	±10%	0.33
27	CL05A225MQ5NNN□		2.2 μF	6.3	±20%	0.55
28	CL05A225KQ3LRN□		2.2 μF	6.3	±10%	0.33
29	CL05A225MR5NNN□		2.2 μF	4	±20%	0.55
30	CL05A225KR3LRN□		2.2 μF	4	±10%	0.33
31	CL05A475MO5NUN□		4.7 μF	16	±20%	0.70
32	CL05A475MP5NRN□		4.7 μF	10	±20%	0.65
33	CL05A475MQ5NRN□		4.7 μF	6.3	±20%	0.65
34	CL05A475MQ3LUN□		4.7 μF	6.3	±20%	0.35
35	CL05A106MP5NUN□		10 μF	10	±20%	0.70
36	CL05A106MQ5NUN□		10 μF	6.3	±20%	0.70
37	CL05A106MR5NRN□		10 μF	4	±20%	0.65
38	CL05A156MR5NUN□		15 μF	4	±20%	0.70
39	CL05A226MR5NZN□		22 μF	4	±20%	0.90
1	CL10A474KB8NNN□	1.60×0.80	0.47 μF	50	±10%	0.90
2	CL10A474KA8NNN□		0.47 μF	25	±10%	0.90
3	CL10A474KP8NNN□		0.47 μF	10	±10%	0.90
4	CL10A474KQ8NNN□		0.47 μF	6.3	±10%	0.90
5	CL10A474KR8NNN□		0.47 μF	4	±10%	0.90
6	CL10A105KB8NNN□		1 μF	50	±10%	0.90
7	CL10A105KA5LNN□		1 μF	25	±10%	0.50
8	CL10A105KA8NNN□		1 μF	25	±10%	0.90
9	CL10A105KO8NNN□		1 μF	16	±10%	0.90
10	CL10A105KO5LNN□		1 μF	16	±10%	0.50
11	CL10A105KP8NNN□		1 μF	10	±10%	0.90
12	CL10A105KP5LNN□		1 μF	10	±10%	0.50
13	CL10A105KQ8NNN□		1 μF	6.3	±10%	0.90
14	CL10A105KQ5LNN□		1 μF	6.3	±10%	0.50

Part Numbering
SystemStandard &
High CapacitorsSuper Small Size
CapacitorsHigh-Q
CapacitorsMedium-High
Voltage CapacitorsArray Type
CapacitorsLow ESL
CapacitorsReliability Test
ConditionPremium Capacitors
for Automotive
ApplicationsPackaging
SpecificationApplication Manual
for Surface Mounting

※ □ mark means packaging code. If you want to learn the code or quantity in detail, please see p74.

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Product Lineup (Standard & High Capacitors-X7R, X7S)

	Part Number	Size L x W (mm)	Capacitance	Rated Voltage (Vdc)	Capacitance Tolerance	Thickness Max.(mm)
1	CL05B151KB5NNN □	1.00×0.50	150pF	50	±10%	0.55
2	CL05B181KB5NNN □		180pF	50	±10%	0.55
3	CL05B221KB5NNN □		220pF	50	±10%	0.55
4	CL05B271KB5NNN □		270pF	50	±10%	0.55
5	CL05B331KB5NNN □		330pF	50	±10%	0.55
6	CL05B391KB5NNN □		390pF	50	±10%	0.55
7	CL05B471KB5NNN □		470pF	50	±10%	0.55
8	CL05B561KB5NNN □		560pF	50	±10%	0.55
9	CL05B681KB5NNN □		680pF	50	±10%	0.55
10	CL05B102KB5NNN □		1nF	50	±10%	0.55
11	CL05B122KB5NNN □		1.2nF	50	±10%	0.55
12	CL05B152KB5NNN □		1.5nF	50	±10%	0.55
13	CL05B182KB5NNN □		1.8nF	50	±10%	0.55
14	CL05B222KB5NNN □		2.2nF	50	±10%	0.55
15	CL05B272KB5NNN □		2.7nF	50	±10%	0.55
16	CL05B332KB5NNN □		3.3nF	50	±10%	0.55
17	CL05B472KB5NNN □		4.7nF	50	±10%	0.55
18	CL05B562KB5NNN □		5.6nF	50	±10%	0.55
19	CL05B682KB5NNN □		6.8nF	50	±10%	0.55
20	CL05B822KB5NNN □		8.2nF	50	±10%	0.55
21	CL05B103KB5NNN □		10nF	50	±10%	0.55
22	CL05B123KA5NNN □		12nF	25	±10%	0.55
23	CL05B153KA5NNN □		15nF	25	±10%	0.55
24	CL05B223KA5NNN □		22nF	25	±10%	0.55
25	CL05B273KO5NNN □		27nF	16	±10%	0.55
26	CL05B333KO5NNN □		33nF	16	±10%	0.55
27	CL05B393KO5NNN □		39nF	16	±10%	0.55
28	CL05B473KO5NNN □		47nF	16	±10%	0.55
29	CL05B563KO5NNN □		56nF	16	±10%	0.55
30	CL05B683KO5NNN □		68nF	16	±10%	0.55
31	CL05B823KO5NNN □		82nF	16	±10%	0.55
32	CL05B104KO5NNN □		100nF	16	±10%	0.55
33	CL05B224KO5NNN □		220nF	16	±10%	0.55
34	CL05B474KP5NNN □		470nF	10	±10%	0.55
35	CL05B105KQ5NQ □		1μF	6.3	±10%	0.60
1	CL05Y474KP5NNN □	1.00×0.50	470nF	10	±10%	0.55
37	CL10B101KB8NNN □	1.60×0.80	100pF	50	±10%	0.90
38	CL10B121KB8NNN □		120pF	50	±10%	0.90
39	CL10B151KB8NNN □		150pF	50	±10%	0.90
40	CL10B181KB8NNN □		180pF	50	±10%	0.90
41	CL10B201KB8NNN □		200pF	50	±10%	0.90
42	CL10B221KB8NNN □		220pF	50	±10%	0.90
43	CL10B271KB8NNN □		270pF	50	±10%	0.90
44	CL10B331KB8NNN □		330pF	50	±10%	0.90
45	CL10B391KB8NNN □		390pF	50	±10%	0.90
46	CL10B471KB8NNN □		470pF	50	±10%	0.90
47	CL10B561KB8NNN □		560pF	50	±10%	0.90
48	CL10B681KB8NNN □		680pF	50	±10%	0.90
49	CL10B751KB8NNN □		750pF	50	±10%	0.90
50	CL10B821KB8NNN □		820pF	50	±10%	0.90
51	CL10B102KB8NNN □		1nF	50	±10%	0.90
52	CL10B122KB8NNN □		1.2nF	50	±10%	0.90

※ □ mark means packaging code. If you want to learn the code or quantity in detail, please see p74.

Part Numbering
SystemStandard &
High CapacitorsSuper Small Size
CapacitorsHigh-Q
CapacitorsMedium-High
Voltage CapacitorsArray Type
CapacitorsLow ESL
CapacitorsReliability Test
ConditionPremium Capacitors
for Automotive
ApplicationsPackaging
SpecificationApplication Manual
for Surface Mounting

Section 2
Vishay

Designators	Part / Value	Qty
C4_1-C4_36,C33,C34	4.7uF 3216 25V Tantalum	38
R5,R6,R16,R19,R21	1k 0402	5

RoHS evidence document: **vishay-rohs-20250901.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863



ROHS Compliance Statement

Directives 2002/95/EG, 2011/65/EU, 2015/863/EU
(RoHS I – III)

September 1st, 2025

Dear Valued Customer,

Vishay Intertechnology, Inc. is committed to be in compliance with all applicable global laws, directives and regulations.

Unless otherwise specified, Vishay Intertechnology, Inc. hereby certifies that all its products that are identified as RoHS compliant satisfy the requirements of the above-listed directives of the European Parliament respecting the restriction of the use of certain hazardous substances in electrical and electronic equipment.

Vishay Intertechnology, Inc. is closely following the various RoHS exemption renewal requests, which as of today, are still pending and awaiting a decision by the European Commission.

According to RoHS Article 5. 5, existing exemptions shall remain valid until a decision on the renewal application is taken by the European Commission and a new expiry date is set. Where necessary, further renewal requests will be issued at least 18 months before the new expiry date.

Aside from satisfying its regulatory obligations, Vishay is actively searching for possible alternatives and scrutinizing viable options based on performance, quality and cost. Currently concerned exemptions as per Annex II, 2011/65/EU are 6 a, b, c, 7 a, 7 c(I) and 7 c(II).

Please mail your request to VPECM.Surveys@Vishay.com for specific products RoHS compliance reports.


Roy Shoshani
Executive Vice President
COO – Semiconductors and CTO
Vishay Intertechnology, Inc.

Section 3
NationStar

Designators	Part / Value	Qty
D1	Red 0805 C84256	1

RoHS evidence document: **2409272203_Foshan-NationStar-Optoelectronics-NCD0805R1_C84256.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 3 of 8 — RoHS-relevant pages only; the complete document is retained in the project repository.

表单编号: FGH/BD-0053.01-15A



佛山市国星光电股份有限公司

FOSHAN NATIONSTAR OPTOELECTRONICS CO., LTD

产品规格书

SPECIFICATION

顾客名称 Customer		产品名称 Product	Chip LED
顾客型号 Customer Type		产品型号 Type	NCD0805R1
顾客部品号 Customer No.		版本号 Version NO	A 版



地址: 广东省佛山市禅城区华宝南路 18 号

Add: NO.18 South Huabao Rd, Foshan, Guangdong, China

电话 (Tel): 0757-82100219

传真 (Fax): 0757-82100220

邮编 (Zip): 528000

邮箱 (Email): chipLED@nationstar.com

<http://www.nationstar.com>

研发中心 Research & Development Center			客户 (加盖公章) Customer (Stamp)
制定 DRAW	审核 CHECK	批准 APPROVE	确认 CONFIRM
发放日期 (Release Date): 2019-10-30			

保存期限: 长期

表单编号: FGH/BD-0053.01-15A



片式发光二极管

NCD0805R1

Chip Light Emitting Diode

技术数据表 Technical Data Sheet

本产品主要作为信号指示及照明的电子元件广泛应用于各类使用表面贴装结构的电子产品中,如家用电器的开关指示灯、手机键盘灯、汽车仪表盘指示灯等。

This product is generally used as indicator and luminance for surface mounted electronic equipment, such as household appliance, communication equipment, and dashboard.

特性:

Features:

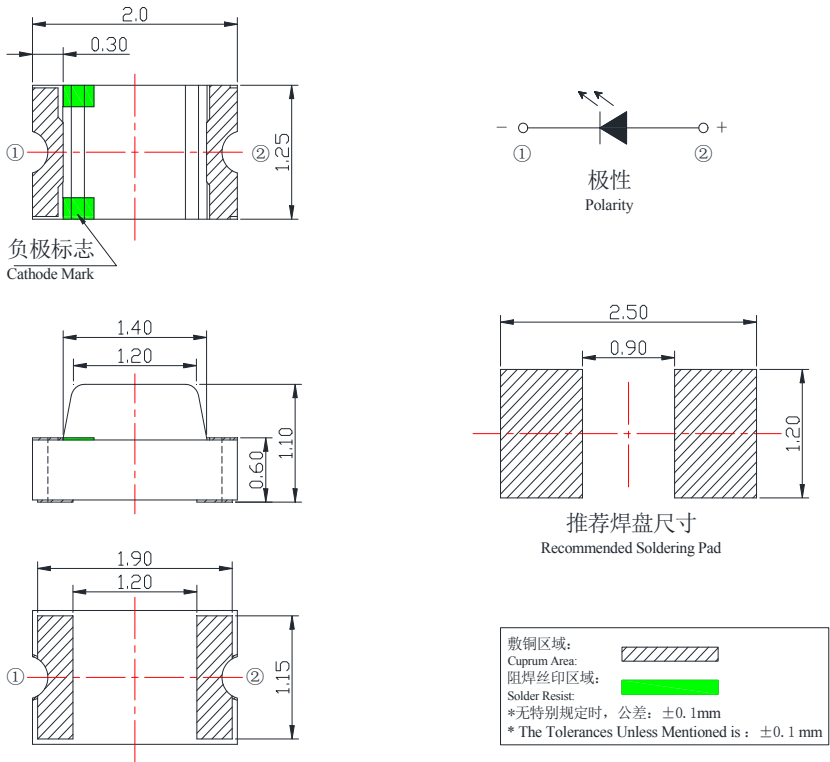
- 管芯材料: AlGaInP
Material:
- 封装材料: 环氧树脂
Encapsulation: Epoxy Resin
- 焊接方法: 无铅回流焊
Soldering methods: Pb-Free reflow soldering
- 光强度高, 功耗低, 可靠性好, 寿命长
High Luminous Intensity ,Low Power Dissipation, Good Reliability and Long Lifespan
- 符合欧盟公布的 ROHS 指令要求
Complied With ROHS Directive

发光颜色: 红色
Emitting Color: Red

* 产品规格如因工艺改进而有所改变, 恕不另行通知。
* The specifications of the product may be modified for improvement without notice.

外形尺寸

Outline Dimension



表单编号: FGH/BD-0053.01-15A



片式发光二极管

光电参数 (1)

Electro-Optical Characteristics(1)

✧ 极限参数 (温度=25℃)
Absolute Maximum Ratings (Temperature=25℃)

参数名称 Parameter	符号 Symbol	数值 Rating	单位 Unit
正向电流 Forward Current	I _F	25	mA
正向脉冲电流* Pulse Forward Current*	I _{FP}	50	mA
反向电压 Reverse Voltage	V _R	5	V
工作温度 Operating Temperature	T _{OPR}	-30 ~ +85	℃
贮存温度 Storage Temperature	T _{stg}	-40 ~ +100	℃
功耗 Power Dissipation	P _D	65	mW

* 注: 脉冲宽度≤0.1ms, 占空比≤1/10 * Note: Pulse Width≤0.1ms, Duty≤1/10

✧ 光电参数 (温度=25℃)
Electro-Optical Characteristics (Temperature=25℃)

参数名称 Parameter	符号 Symbol	条件 Condition	最小值 Min.	典型值 Typ.	最大值 Max.	单位 Unit
反向电流 Reverse Current	I _R	V _R =5V	-	-	10	μA
视角 View Angle	2θ _{1/2}	-	-	130	-	deg.
正向电压 Forward Voltage	V _F	I _F =20mA	1.6	2.0	2.6	V
峰值波长 Peak Wavelength	λ _P		-	630	-	nm
主波长 Dominant Wavelength	λ _d		615	620	630	nm
半波宽度 Spectrum Radiation Bandwidth	Δλ		-	15	-	nm
光强 Luminous Intensity	I _V		67	115	195	mcd

* 注 1: 光强偏差±15%; 压降偏差±0.1V; (X,Y)坐标偏差±0.01; 单色光波长偏差±1nm。
* Note1: Tolerance on each Luminous Intensity bin is ±15%; Tolerance on each Forward Voltage bin is ±0.1V;Tolerance on each Hue(X,Y) bin is ±0.01; Tolerance of Dominant Wavelength ±1nm.

* 注 2: 以上参数仅供参考, 请以实物标签为准。我司给出的参数均由国星测试系统测得。
* Note2: The parameters above are only for your reference. In case of any discrepancy, please adhere to the label of our actual products. All parameters are tested by the standard testing system of NationStar.

Section 4
Shenzhen JingYang

Designators	Part / Value	Qty
D1_1-D1_36,D12	1N4148WSX	37

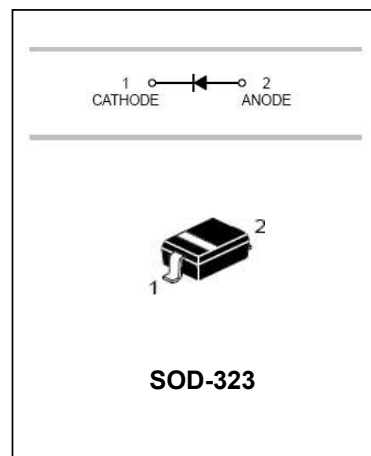
RoHS evidence document: **2405091035_Shenzhen-JingYang-1N4148WSX_C6423741.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

**1N4148WSX****Silicon Epitaxial Planar Diode****FEATURES**

- Fast Switching Speed: $t_{rr}=4ns$
- Surface Mount Package Ideally Suited For Automatic Insertion
- For General Purpose Switching Applications
- High Conductance
- Available in Lead Free Version
- ROHS compliant
- MSL: 3



Lead-free

**SOD-323****APPLICATIONS**

- Surface mount fast switching diode

ORDERING INFORMATION

Type No.	Marking	Package Code
1N4148WSX	T4	SOD-323

MAXIMUM RATING @ $T_a=25^{\circ}C$ unless otherwise specified

Characteristic	Symbol	Value	Unit
Non-Repetitive Peak Reverse Voltage	V_{RM}	100	V
Peak Repetitive Reverse Voltage	V_{RRM}	75	V
Working Peak Reverse Voltage	V_{RWM}		
DC Reverse Voltage	V_R		
RMS Reverse Voltage	$V_{R(RMS)}$	53	V
Average Rectified Output Current	I_o	150	mA
Non-Repetitive Peak Forward Surge Current @ $t=1.0 \mu s$ @ $t=1.0 s$	I_{FSM}	2.0 1.0	A
Power Dissipation	P_d	200	mW
Thermal Resistance Junction to Ambient Air	$R_{\theta JA}$	625	$^{\circ}C/W$
Operating and Storage Temperature Range	T_j, T_{STG}	-65 to +150	$^{\circ}C$

service@jy-electronics.com.cn

Revision A2

1

www.jy-electronics.com.cn

 JYELECTRONICS[®]

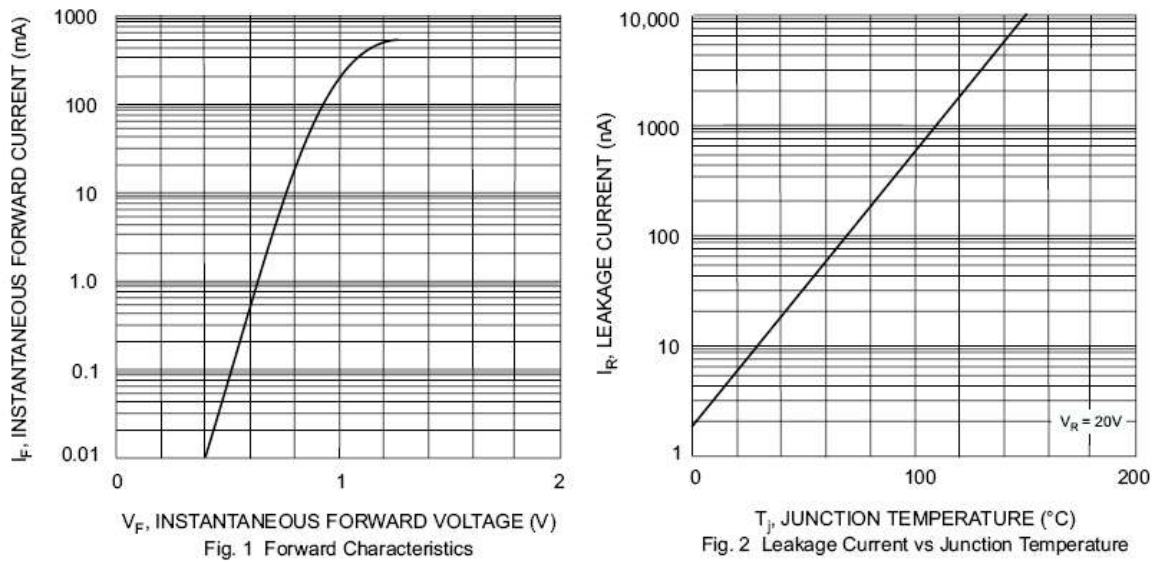
1N4148WSX

Silicon Epitaxial Planar Diode

ELECTRICAL CHARACTERISTICS @ Ta=25℃ unless otherwise specified

Characteristic	Symbol	Min	Max	Unit	Test Condition
Reverse Breakdown Voltage	$V_{(BR)R}$	75	-	V	$I_R=1.0\mu A$
Forward Voltage	V_F	-	0.715 0.855 1.0 1.25	V	$I_F=1.0mA$ $I_F=10mA$ $I_F=50mA$ $I_F=150mA$
Reverse Current	I_R	-	1.0 25	μA nA	$V_R=75V$ $V_R=20V$
Junction Capacitance	C_J	-	2.0	pF	$V_R=0, f=1.0MHz$
Reverse Recovery Time	t_{rr}	-	6.0	ns	$I_F=I_R=10mA,$ $I_{rr}=0.1\times I_R, R_L=100\Omega$

TYPICAL CHARACTERISTICS @ Ta=25℃ unless otherwise specified



JYELECTRONICS®

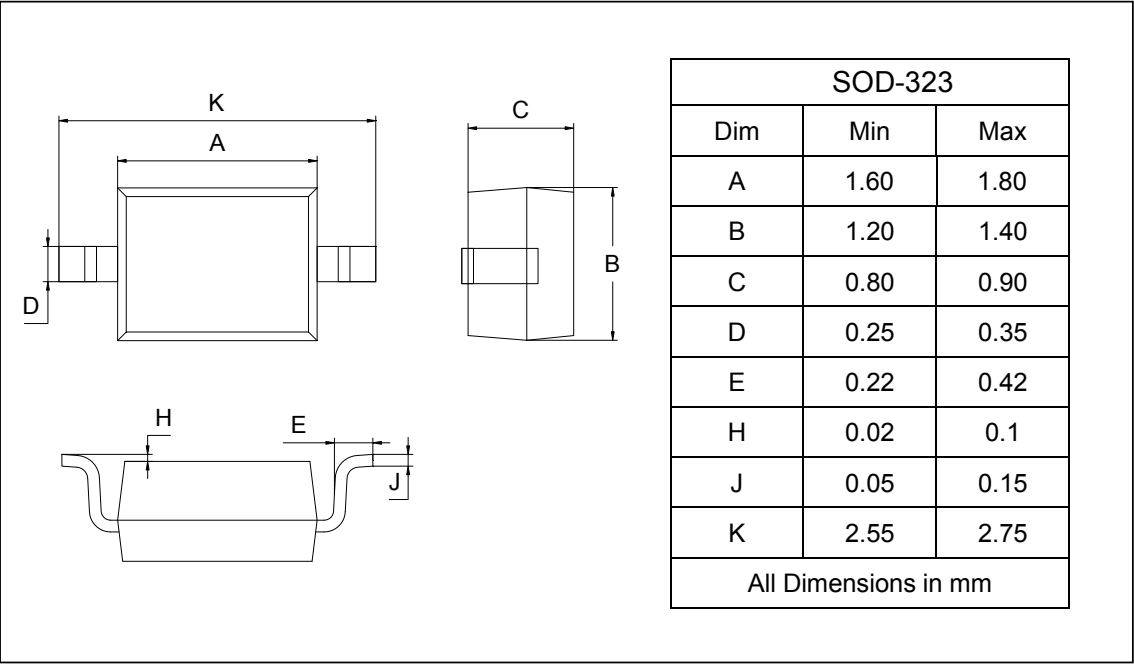
1N4148WSX

Silicon Epitaxial Planar Diode

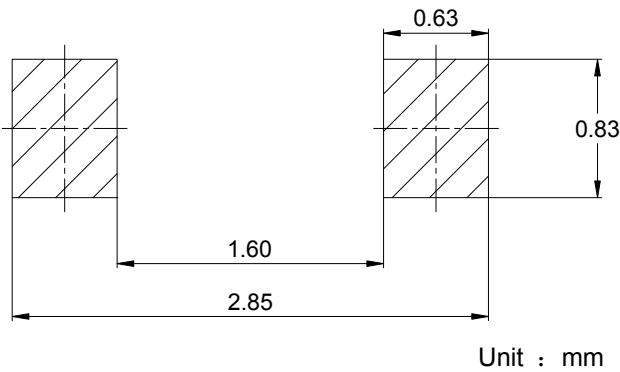
PACKAGE OUTLINE

Plastic surface mounted package

SOD-323



SOLDERING FOOTPRINT



PACKAGE INFORMATION

Device	Package	Shipping
1N4148WSX	SOD-323	3000/Tape&Reel

Section 5
Hottech

Designators	Part / Value	Qty
D2	1N5819WS	1

RoHS evidence document: **ROHS-2025-HottechElectronics-CTI-A225024736310100101.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863



中国认可
国际互认
检测
TESTING
CNAS L1910



Test Report

Report No. A225024736310100101

Page 1 of 7

Company Name
shown on Report

SHENZHEN HOTTECH ELECTRONICS CO., LTD.

Address

ROOM NO.N4C618,4TH FLOOR, NEW ASIA GUOLI BUILDING, FUTIAN DISTRICT,
SHENZHEN, GUANGDONG.

The following sample(s) and sample information was/were submitted and identified by/on the behalf of the applicant

Sample Name

二极管/三极管/MOS管/线性稳压管

Model No.

SOT-23/23-3L/323/523/723/923/363/23-5/23-6/89
SOD-123/323/523/723/TO-92/92L/92S/126/251/252/220/220F
SOP-8/SOP-7/TSSOP-8/SOP-14/DFN3*3/PDFN5*6/DIP-8/DIP-14

Sample Received Date

Apr. 16, 2025

Testing Period

Apr. 16, 2025 to Apr. 17, 2025

Test Requested

As specified by client, to test Lead (Pb), Cadmium (Cd), Mercury (Hg), Hexavalent Chromium (Cr(VI)), Polybrominated Biphenyls (PBBs), Polybrominated Diphenyl Ethers (PBDEs), Phthalates (DBP, BBP, DEHP, DIBP) in the submitted sample(s).

Test Method

Please refer to the following page(s).

Test Result(s)

Please refer to the following page(s).

Conclusion

Tested Sample	According to standard/directive	Result
Submitted Sample	RoHS Directive 2011/65/EU with amendment (EU) 2015/863	PASS

PASS means that the results shown on the report comply with the limits set by RoHS Directive 2011/65/EU with amendment (EU) 2015/863.



Approved by

Helen Liu

Helen Liu

Lab Authorized Signatory

Date

Apr. 17, 2025

No. R393386937

CTI Building, Xing Dong Community, Xin'an Sub-district, Bao'an District, Shenzhen City, Guangdong Province, P.R. China



Test Report

Report No. A225024736310100101

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Test Method

Test Item(s)	Test Method	Measured Equipment(s)
Lead (Pb)	IEC 62321-5:2013	ICP-OES
Cadmium (Cd)	IEC 62321-5:2013	ICP-OES
Mercury (Hg)	IEC 62321-4:2013+AMD1:2017 CSV	ICP-OES
Hexavalent Chromium (Cr(VI))	IEC 62321-7-2:2017 and/or determination of Total Chromium by IEC 62321-5:2013	UV-Vis/ICP-OES
Polybrominated Biphenyls (PBBs)	IEC 62321-12:2023	GC-MS
Polybrominated Diphenyl Ethers (PBDEs)	IEC 62321-12:2023	GC-MS
Phthalates (DBP, BBP, DEHP, DIBP)	IEC 62321-12:2023	GC-MS





Test Report

Report No. A225024736310100101

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Test Result(s)

Tested Item(s)	Result	MDL	Limit
	001		
Lead (Pb)	N.D.	2 mg/kg	1000 mg/kg
Cadmium (Cd)	N.D.	2 mg/kg	100 mg/kg
Mercury (Hg)	N.D.	2 mg/kg	1000 mg/kg
Hexavalent Chromium (Cr(VI))	N.D.	8 mg/kg	1000 mg/kg
Tested Item(s)	Result	MDL	Limit
	001		
Polybrominated Biphenyls (PBBs)			
Monobromobiphenyl	N.D.	25 mg/kg	1000 mg/kg
Dibromobiphenyl	N.D.	25 mg/kg	
Tribromobiphenyl	N.D.	25 mg/kg	
Tetrabromobiphenyl	N.D.	25 mg/kg	
Pentabromobiphenyl	N.D.	25 mg/kg	
Hexabromobiphenyl	N.D.	25 mg/kg	
Heptabromobiphenyl	N.D.	25 mg/kg	
Octabromobiphenyl	N.D.	25 mg/kg	
Nonabromobiphenyl	N.D.	25 mg/kg	
Decabromobiphenyl	N.D.	25 mg/kg	
Tested Item(s)	Result	MDL	Limit
	001		
Polybrominated Diphenyl Ethers (PBDEs)			
Monobromodiphenyl ether	N.D.	25 mg/kg	1000 mg/kg
Dibromodiphenyl ether	N.D.	25 mg/kg	
Tribromodiphenyl ether	N.D.	25 mg/kg	
Tetrabromodiphenyl ether	N.D.	25 mg/kg	
Pentabromodiphenyl ether	N.D.	25 mg/kg	
Hexabromodiphenyl ether	N.D.	25 mg/kg	
Heptabromodiphenyl ether	N.D.	25 mg/kg	
Octabromodiphenyl ether	N.D.	25 mg/kg	
Nonabromodiphenyl ether	N.D.	25 mg/kg	
Decabromodiphenyl ether	N.D.	25 mg/kg	

10%
专
10%



Test Report

Report No. A225024736310100101

Page 4 of 7

Test Result(s)

Tested Item(s)	Result	MDL	Limit
	001		
Phthalates (DBP, BBP, DEHP, DIBP)			
Dibutyl phthalate (DBP) CAS#:84-74-2	N.D.	50 mg/kg	1000 mg/kg
Butyl benzyl phthalate (BBP) CAS#:85-68-7	N.D.	50 mg/kg	1000 mg/kg
Di-(2-ethylhexyl) phthalate (DEHP) CAS#:117-81-7	N.D.	50 mg/kg	1000 mg/kg
Diisobutyl phthalate (DIBP) CAS#:84-69-5	N.D.	50 mg/kg	1000 mg/kg

Sample/Part Description

No.	CTI Sample ID	Description
1	001	Electronic components(Tested as a whole)

Remark:

-The sample(s) had been dissolved totally tested for Lead, Cadmium, Mercury.

-The sample(s) was tested as a whole, because it's impossible to disassemble or separate it by current equipment and technology. The result(s) shown on this report may be different from the content of any homogeneous material.

-According to the client's statement, the reasons for the multiple information in the "sample information" of this report may include (but are not limited to): supplying to different buyers, being sold to different countries or regions, former names, or mixtures of several substances.

-According to the client's statement, the Company Name shown on Report in this report and the Company Name shown on Report in the report A2250247363101001 are the Group-subsidiary relations, the test result(s) of this report is/are presented in reference to the result(s) that reported in A2250247363101001.

-MDL = Method Detection Limit

-N.D. = Not Detected (<MDL)

-mg/kg = ppm = parts per million

-1000 mg/kg = 0.1%

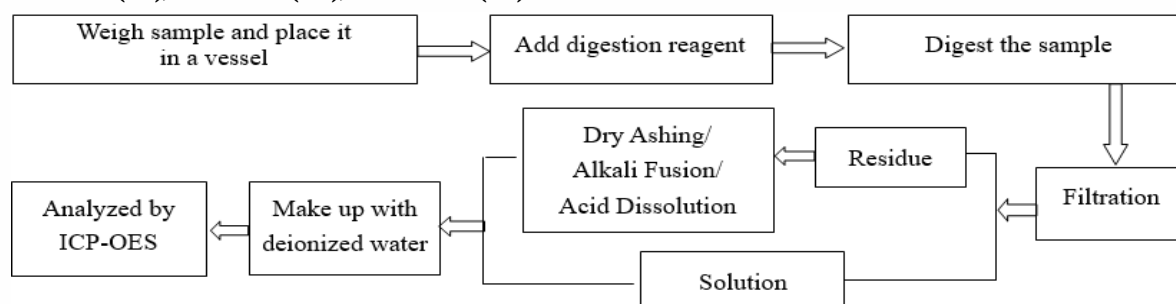
Test Report

Report No. A225024736310100101

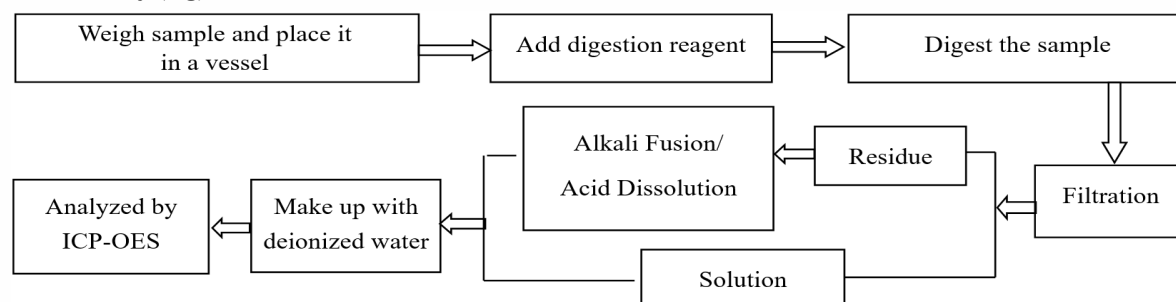
Page 5 of 7

Test Process

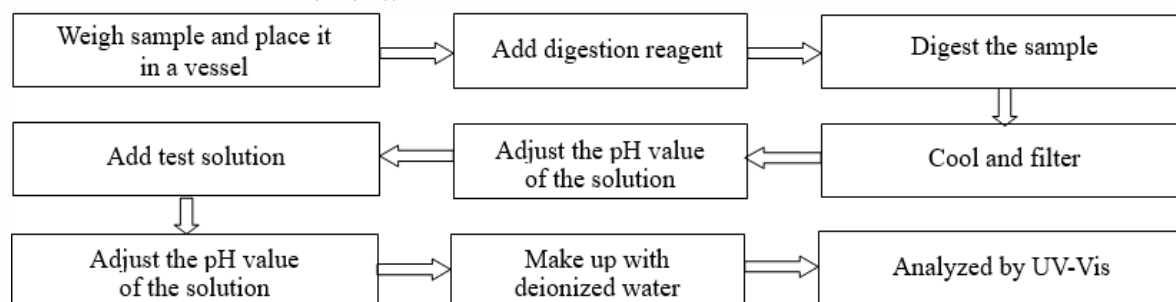
1. Lead (Pb), Cadmium (Cd), Chromium (Cr)



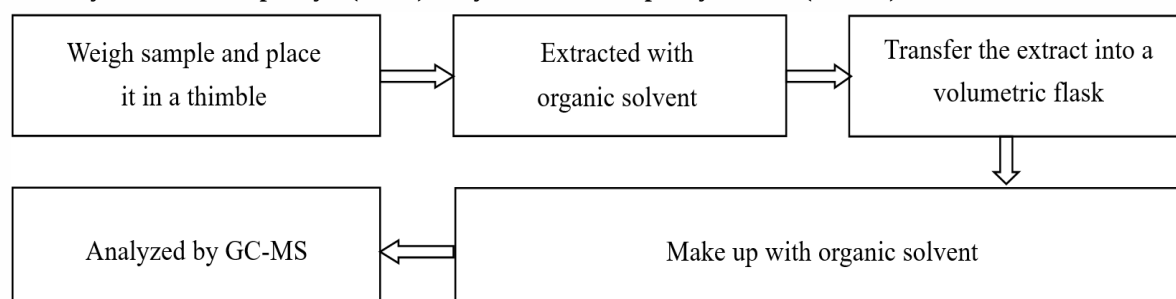
2. Mercury (Hg)



3. Hexavalent Chromium (Cr(VI))



4. Polybrominated Biphenyls (PBBs), Polybrominated Diphenyl Ethers (PBDEs)



1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20. 21. 22. 23. 24. 25. 26. 27. 28. 29. 30. 31. 32. 33. 34. 35. 36. 37. 38. 39. 40. 41. 42. 43. 44. 45. 46. 47. 48. 49. 50. 51. 52. 53. 54. 55. 56. 57. 58. 59. 60. 61. 62. 63. 64. 65. 66. 67. 68. 69. 70. 71. 72. 73. 74. 75. 76. 77. 78. 79. 80. 81. 82. 83. 84. 85. 86. 87. 88. 89. 90. 91. 92. 93. 94. 95. 96. 97. 98. 99. 100.

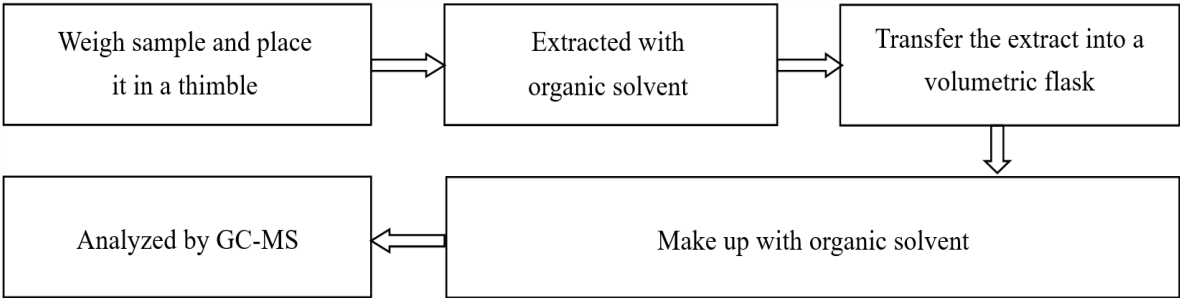


Test Report

Report No. A225024736310100101

Page 6 of 7

5. Phthalates (DBP, BBP, DEHP, DIBP)



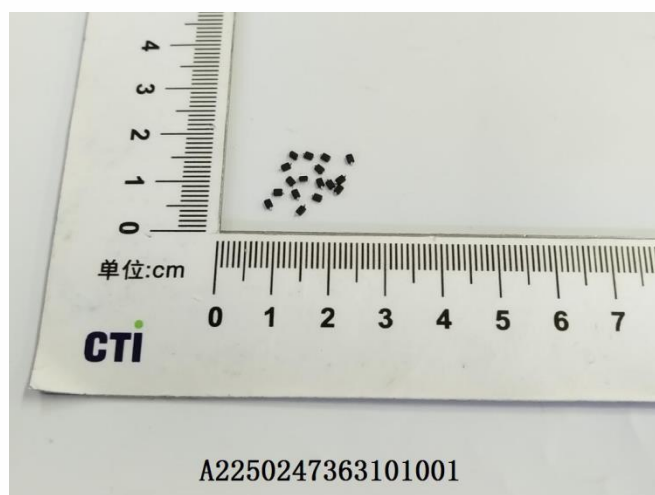
70241544

Test Report

Report No. A225024736310100101

Page 7 of 7

Photo(s) of the sample(s)



Statement:

1. This report is considered invalid without approved signature, special seal and the seal on the perforation;
2. The Company Name shown on Report and Address, the sample(s) and sample information was/were provided by the applicant who should be responsible for the authenticity which CTI hasn't verified;
3. The result(s) shown in this report refer(s) only to the sample(s) tested;
4. Unless otherwise stated, the decision rule for conformity reporting is based on Binary Statement for Simple Acceptance Rule (w=0) stated in ILAC-G8:09/2019 / CNAS-GL015:2022;
5. Without written approval of CTI, this report can't be reproduced except in full;
6. In case of any discrepancy between the English version and Chinese version of the testing reports (if generated), the Chinese version shall prevail.

*** End of report ***

/ 华测公司



Appendix

Client Reference Photo (Non-tested sample)



Statement:

1. The Appendix Information was/were provided by the applicant who should be responsible for the authenticity which CTI hasn't verified.
2. The Appendix Information is/are the supplement(s) for the Report A2250247363101001.

CTI

Section 6
Hubei KENTO

| Designators | Part / Value | Qty |
|-------------|------------------|-----|
| D5 | Green 0805 C2297 | 1 |

RoHS evidence document: **C2297.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 3, 8, 9, 10 of 11 — RoHS-relevant pages only; the complete document is retained in the project repository.



湖北匡通电子股份有限公司

0805 翠绿

承 认 书

Specification for approval

客户名称(Customer name): _____

经办者(Director): _____

职称(title): _____

客户料号(Customer part NO): _____

版本(Revision): _____ A.0

发件日期(Issue date): _____ 2018-12-06

回文日期(Return date): _____

一、谨致执事者：兹提供�公司产品之有关详细规格及图面数据，敬请给予办理测试认定手续，
同时敬请送返一份附有贵公司签认之测试认定后之样品认定书。

(We are please in sending you herewith our specification and drawings for your approval.)

(Please return to us one copy "For Approval" with your approved signatures.)

二、附件(Accessory):

☐样品 ☐出货检验记录表 ☐封装尺寸图 ☐电气特性曲线

☐内部线路图 ☐焊性建议 ☐PAD 建议 ☐包装方式

三、客户意见栏(Customer's Proposal)

☐ 同意(Agree): (请于认可栏中签名)

☐ 不同意(Disagree):

原因(Reason): _____

客户认可签章(Customer Signature): _____



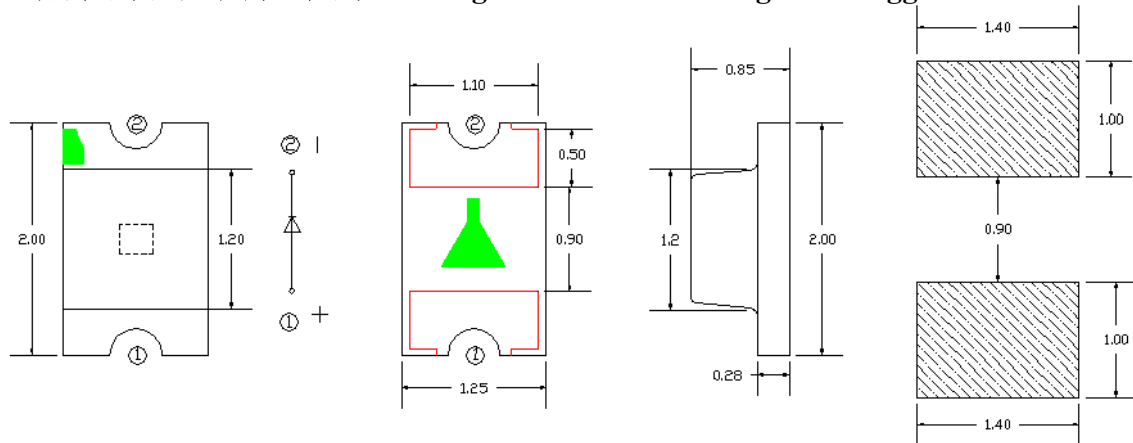
湖北匡通电子股份有限公司

0805 翠绿

1. 产品描述/ Features

- 外观尺寸/ Package (L/W/H) : 2.0*1.25*0.85 mm
- 颜色/ Color: 翠绿 / Green light
- 胶体/ Lens: 透明平面胶体/ Transparent planar colloid
- EIA规范标准包装/ EIA STD Package
- 环保产品, 符合ROHS要求/ Meet ROHS, Green Product
- 适用于自动贴片机/ Compatible With SMT Automatic Equipment
- 适用于红外线回流焊制程/ Compatible With Infrared Reflow Solder Process

2. 外形尺寸及建议焊盘尺寸/ Package Profile & Soldering PAD Suggested

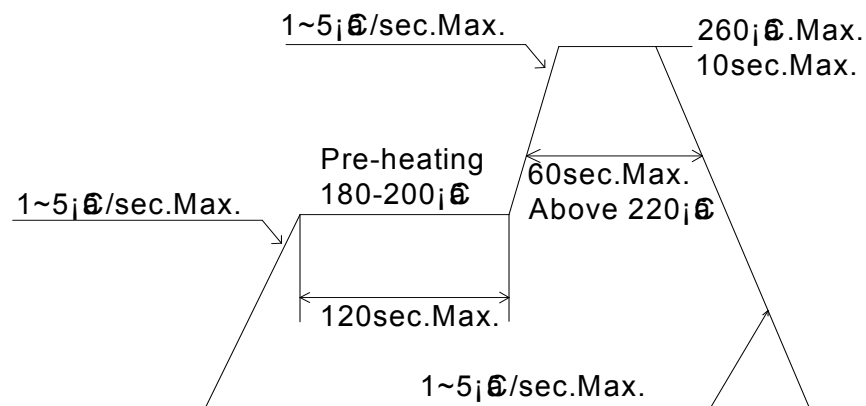


注/ Notes: 1. 单位 : 毫米 (mm) / All dimensions are in millimeters

2. 公差 : 如无特别标注则为 ± 0.1 mm Tolerance is ± 0.10 mm unless otherwise noted

3. 建议焊接温度曲线 / Soldering Profile Suggested

lead-free solder





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4. 最大绝对额定值/ Absolute Maximum Ratings (Ta=25°C)

| 参 数/ Parameter | 符号Symbol | 最大额定值/ Rating | 单 位/ Unit |
|--|----------|---|-----------|
| 消耗功率/ Power Dissipation | Pd | 100 | mW |
| 最大脉冲电流/ Peak Forward Current
(1/10占空比, 0.1ms脉宽) | IFP | 60 | mA |
| 正向直流工作电流/ DC Forward Current | IF | 30 | mA |
| 反向电压/Backward Voltage | VR | 5 | V |
| 工作温度范围
Operating Temperature Range | Topr | -40°C ~ +85°C | |
| 存储温度范围
Storage Temperature Range | Tstg | -40°C ~ +85°C | |
| 焊接条件
Soldering Condition | Tsol | 回流焊/ Reflow soldering : 260°C , 10s
手动焊/ Hand soldering : 300°C , 3s | |
| 抗静电能力
Electrostatic Discharge | ESD | | V |

5. 光电参数/ Electrical Optical Characteristics (Ta=25°C)

| 参数
Parameter | 符号
Symbol | 最小值
Min. | 代表值
Typ. | 最大值
Max. | 单位
Unit | 测试条件
Test Condition |
|---------------------------------|--------------|-------------|-------------|-------------|------------|------------------------|
| 光强
Light Intensity | IV | 175 | -- | 430 | mcd | IF = 5mA |
| 半光强视角
Viewing Angle | 2θ1/2 | --- | 120 | --- | deg | IF = 5mA |
| 主波长
Dominant Wavelength | λd | 513 | | 528 | nm | IF = 5mA |
| 峰值波长
Peak Wavelength | λp | 516 | | 525 | nm | IF = 5mA |
| 正向电压
Forward Voltage | VF | 2.6 | | 3.1 | V | IF = 5mA |
| 反向电压
Backward Voltage | IR | --- | --- | 5 | μA | VR = 5V |
| 半波宽
Spectral Line Half-Width | Δλ | | 15 | | nm | IF = 5mA |



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11. 注意事项/ Cautions:

11.1. 焊接/welding

11.1.1 SMD LED 灌封胶较软，外力易损坏发光面及塑料壳，焊接时要轻拿轻放。

SMD LED is soft and easy to damage the luminous surface and plastic shell by external force. It should be handled lightly when welding

11.1.2 建议使用易洗型的助焊剂，依照回流曲线条件回流焊接，回流次数最多两次，确保 LED 发光面干净，异物会影响发光颜色。

It is recommended to use soldering flux with tin wash type, reflow soldering according to the condition of reflux curve, reflow twice at most, ensure the LED luminous surface is clean, foreign matter will affect the luminous color.

11.1.3 只建议在修理和重工的情况下使用手工焊接，最高焊接温度不应超过 300 度，且须在 3 秒内完成（手工焊接只可焊接一次）烙铁最大功率应不超过 25W。

Manual welding is only recommended for repair and heavy industry; The maximum welding temperature should not exceed 300 degrees, and must be completed within 3 seconds (manual welding can only be welded once) soldering iron maximum power should not exceed 25W.

11.1.4 焊接过程中，严禁在高温情况下碰触胶体；焊接后，禁止对胶体施加外力，禁止弯折 PCB，避免元件受到撞击。

During the soldering process, do not touch the lens at high temperature, After soldering, any mechanical force on the lens or any excessive vibration shall not be accepted to apply, also the circuit board shall not be bent as well.

11.1.5 请不要将不同 BIN 级的 LED 使用于同一个产品上，否则可能会导致产品的严重色差。

Please do not use different BIN LED on the same product, otherwise it may cause serious color difference.

11.2. 清洗/cleaning

11.2.1 不能用超声波清洗，建议使用异丙醇（isopropyl alcohol）、纯酒精擦拭或浸渍（浸渍不超过 1 分钟）在室温下放置 15 分钟再使用；清洗后，确保 LED 发光面干净，异物会影响发光颜色。

/No ultrasonic cleaning. It is recommended to use isopropyl alcohol, pure alcohol to wipe or soak, not more than 1 minute, and leave at room temperature for 15 minutes before use. After cleaning, make sure the LED luminous surface is clean and the foreign matter will affect the luminous color.

11.2.2 应避免接触或污染天那水，三氯乙烯、丙酮、硫化物、氮化物、酸、碱、盐类，这些物质会损伤 LED。Avoid touching or contaminating the water, trichloroethylene, acetone, sulfide, nitride, acid, alkali, and salts that can damage leds.



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11.3. 灌封/enbedment

- 11.3.1 挥发性物质会渗透到 LED 内部, 在通电产生光子及热的条件下, 会导致 LED 变色, 进而造成严重光衰, 严禁使用任何对 LED 器件的性能或者可靠性有害的物质或材料, 针对特定的用途和使用环境, 建议对所有的物质和材料进行相容性的测试。在贴装 LED 时候, 不要使用能产生有机挥发性气体的粘结剂。

Volatile substances to leach into the LED inside, photons in electricity and heat conditions, will lead to the LED color, thus causing serious droop, it is forbidden to use any of the LED device performance or reliability of harmful substances or materials, for a specific purpose and use of the environment, advice on all the material and the material compatibility test. When attaching LED, do not use adhesive that can produce volatile organic gas.

- 11.3.2 使用正常灌密封胶时, 建议先以少量试验, 常温点亮 168 小时, 确定没有问题再作业。

It is recommended to light up for 168 hours at room temperature for a small amount of test before using normal filling and sealing glue.

11.4. 保存/save

- 11.4.1 打开包装前, LED 应存储在温度 30℃或以下, 相对湿度在 RH60%以下, 一年内使用。

Before opening the package, LED should be stored in a temperature 30 °C or below, under RH60 % relative humidity, used in a year.

- 11.4.2 LED 是湿度敏感元件, 为避免元件吸湿, 打开包装后, LED 应在温度 30℃或以下, 相对湿度在 60%以内, 使用时间 7 天。LED 吸潮后, 回流焊时可能裂胶, 影响发光颜色。对于未使用的散件, 请去潮处理 (卷装品: 烘烤 60℃±5℃/24H; 散装品: 烘烤 105℃±5℃/1H), 然后再用铝箔袋密封后保存或者储存在氮气防潮柜内。

LED is humidity sensitive element, element to avoid moisture absorption, after open the packing, the LED should be in temperature 30 °C or below, within 60% relative humidity, using time 7 days. After moisture absorption, LED may crack when reflow soldering, influence the luminous color. For bulk is not used, please deal with the tide (for package product: bake 60 °C + / - 5 °C / 24 h. For bulk goods: baking 105 °C + 5 °C, 1 hours), and then save after sealed with aluminum foil bag or stored in nitrogen moistureproof enclosure

- 11.4.3 保存环境中避免有酸、碱以及腐蚀气体存在, 同时避免强烈震动及强磁场作用。

Avoid the presence of acid, alkali and corrosive gas in the preservation environment, and avoid strong vibration and strong magnetic field.



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11.5.静电/electrostatic

- 11.5.1 静电或峰值浪涌电压会损坏 LED,避免在开灯、关灯时产生瞬时电压。

Static electricity or peak surge voltage will damage the LED, avoiding instantaneous voltage when the lamp is turned on or off.

- 11.5.2 建议使用 LED 时佩戴防静电手腕带,防静电手套,穿防静电鞋,使用的设备、仪器正确接地。LED 损坏后,表现出漏电流明显增加,低电流正向电压变低,低电流点不亮等现象。

It is recommended to wear anti-static wrist bands, anti-static gloves and anti-static shoes when using LED. The equipment and instruments used are properly grounded. After the LED was damaged, the leakage current increased obviously, the forward voltage of low current became lower, and the low current point did not light, etc.

11.6 测试/test

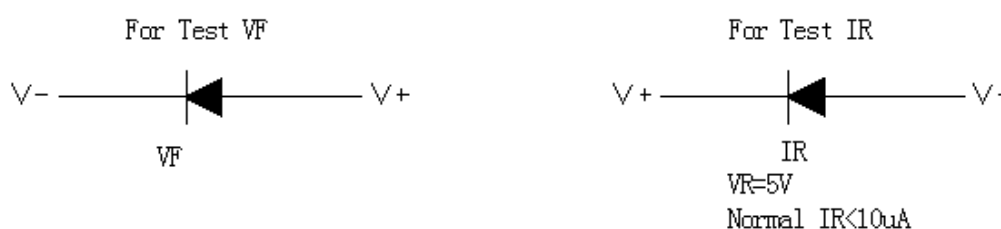
- 11.6.1 LED 要在额定电流下驱动,同时电路中需要加限流电阻保护;否则,轻微的电压变化就会引起较大的电流变化,从而破坏 LED。

LED shall be driven at rated current, and shall be protected by current-limiting resistance in the circuit. Otherwise, slight voltage changes will cause large current changes, which will damage the LED.

- 11.6.2 在电路导通或关闭情况下,要避免瞬间浪涌电压的产生,否则 LED 将被烧坏。

When the circuit is on or off, avoid sudden surge voltage. Otherwise, the LED will be burnt out

请参照下图示检测 LED:/Please check the LED as shown



- 11.6.3 顺向电压 V_F 过高或反向电压 V_R 过高, 均会损坏 LED.

If the forward voltage V_F is too high or the reverse voltage V_R is too high, the LED will be damaged.

- 11.6.4 点亮或测试 LED 时, 加在 LED 两端的反向电压不得高于 5V, 否则容易击伤 LED.

When lighting or testing the LED, the reverse voltage added on both ends of the LED shall not be higher than 5V, otherwise it is easy to damage the LED.

Section 7
Hubei KENTO

| Designators | Part / Value | Qty |
|-------------|-------------------|-----|
| D6 | Yellow 0805 C2296 | 1 |

RoHS evidence document: **1806151129_Hubei-KENTO-Elec-KT-0805Y_C2296.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 3 of 8 — RoHS-relevant pages only; the complete document is retained in the project repository.

承 认 书

Specification for approval

客户名称:

CUSTOMER NAME: _____

经办者:

职称:

DIRECTOR: _____ TITLE: _____

客户料号:

CUSTOMER PART NO.: _____

品名:

版本:

PART NUMBER: KT -0805Y REVISION: KT-0A

发件日期:

回文日期:

ISSUE DATE: _____ RETURN DATE: ____/____/____

- 一、 谨致执事者：兹提供敝公司产品之有关详细规格及图面数据，
敬请给予办理测试认定手续。
同时敬请送返一份附有贵公司签认之测试认定后之样品认定书。
We are please in sending you herewith our specification and drawings for your approval.
Please return to us one copy "For Approval" with your approved signatures.

二、 附件:

ACCESSORY: ☐样品 ☐出货检验记录表 ☐封装尺寸图 ☐电气特性曲线
☐内部线路图 ☐焊性建议 ☐PAD 建议 ☐包装方式

三、 客户意见栏 CUSTOMER'S PROPOSAL

- ☐ AGREE 同意 (请于认可栏中签名)
☐ DISAGREE 不同意

REASON 原因: _____

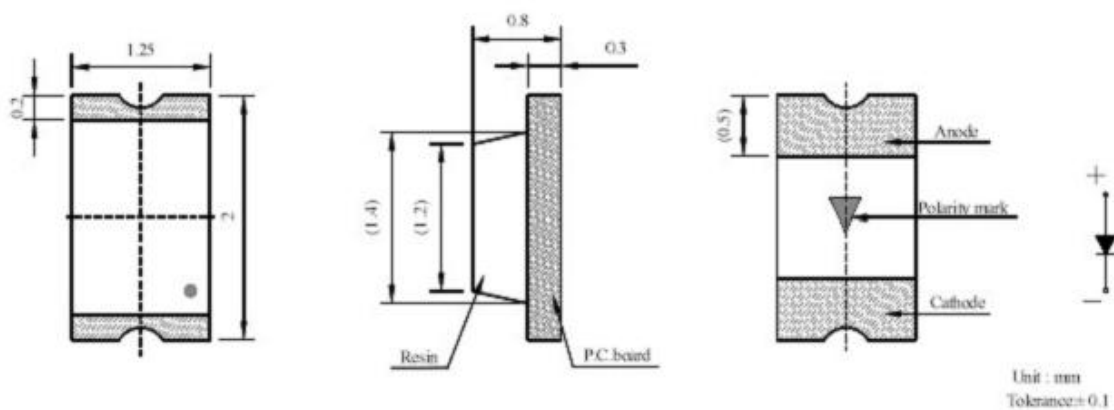
客户认可签章:

CUSTOMER SIGNATURE: _____

1、特性:

- 1.1 封装尺寸: 2.0*1.2*0.8mm
- 1.2 发光颜色: 黄色
- 1.3 发光类型: 单色型
- 1.4 焊接方式: 回流焊
- 1.5 符合 RoHS 标准

2、成品外观尺寸



备注:

- 1. 所有尺寸均以 mm 为单位
- 2. 在没有明确标注的情况下, 公差均为 $\pm 0.10\text{mm}$

3、最大绝对标称值（环境温度=25℃）

| 参数 | 缩写 | 标称值 | 单位 |
|-----------|-----------------|--------------------------------------|----|
| 顺向电流 | I _F | 20 | mA |
| 顺向峰值电流 *1 | I _{FP} | 100 | mA |
| 反向电压 | VR | 5 | V |
| 焊接温度 | Tsol | 回流焊：250 °C，8sec.
手工焊：300 °C，3sec. | |
| 使用温度 | Topr | -40°C~+85 | |
| 储存温度 | Tstg | -40°C~+85 | |

* I_{FP} 条件：脉宽≤0.1msec, 周期≤1/10

4、光电特性参数（环境温度=25℃）：

| 参数 | 缩写 | 最小值 | 典型值 | 最大值 | 单位 | 条件 |
|------|----------------|-----|-----|-----|-----|---------|
| 顺向电压 | V _f | 2.0 | | 2.2 | V | |
| 亮度 | I _v | 95 | - | 113 | mcd | IF=20mA |
| 发光角度 | 2Θ1/2 | - | 120 | - | deg | |
| 反向电流 | IR | - | - | 1 | μA | VR=5V |
| 波长 | WLD | 592 | | 594 | nm | IF=20mA |

备注：1. 亮度偏差：±5%

2. 电压偏差：±0.03V

3. 波长偏差：±1nm

Section 8
Sunlord

| Designators | Part / Value | Qty |
|-------------|--------------|-----|
| FB1,FB2 | GZ2012D601TF | 2 |

RoHS evidence document: **2310301640_Sunlord-GZ2012D601TF_C1017.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

Shenzhen Sunlord Electronics Co., Ltd.

Multilayer Chip Ferrite Bead – GZ Series

Operating temp. : -55°C ~+125°C



FEATURES

- ◆ Internal silver printed layers and magnetic shielded structures to minimize crosstalk.
- ◆ Can be used in a wide range of frequency (from dozens of MHz to hundreds of MHz) to suppress EMI.
- ◆ Three types material and wide range of impedance values for various applications.

APPLICATIONS

- ◆ Noise suppression for low speed signal of electric equipments such as computers and peripheral devices, smart wearable device, LCD TVs, communication equipments, OA equipments, etc.

PRODUCT IDENTIFICATION

| | | | | | | |
|----|------|---|-----|---|---|-------|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| GZ | 3216 | D | 121 | T | F | (A99) |

| 1 Type | |
|--------|-----------------------------------|
| GZ | Chip Ferrite Bead for General Use |

| 4 Nominal Impedance | |
|---------------------|---------------|
| Example | Nominal Value |
| 300 | 30Ω |
| 121 | 120Ω |
| 102 | 1000Ω |

| 2 External Dimensions (L×W) (mm) | |
|----------------------------------|----------|
| 0603 [0201] | 0.6×0.3 |
| 1005 [0402] | 1.0×0.5 |
| 1608 [0603] | 1.6×0.8 |
| 2012 [0805] | 2.0×1.25 |
| 3216 [1206] | 3.2×1.6 |

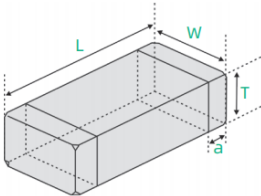
| 5 Packing | |
|-----------|-------------|
| T | Tape & Reel |

| 3 Material Code |
|-----------------|
| D, E, U |

| 6 Hazardous Substance Free Products |
|-------------------------------------|
| F |

| 7 Internal Code |
|-----------------|
| A99 |

SHAPE AND DIMENSIONS



| Unit: mm [inch] | | | | |
|-----------------|---|-------------------------|--------------------------|--------------------------|
| Type | L | W | T | a |
| GZ0603 [0201] | 0.6±0.05
[.024±.002] | 0.3±0.05
[.012±.002] | 0.3±0.05
[.012±.002] | 0.15±0.05
[.006±.002] |
| GZ1005 [0402] | 1.0±0.15
[.039±.006] | 0.5±0.15
[.020±.006] | 0.5±0.15
[.020±.006] | 0.25±0.1
[.010±.004] |
| GZ1608 [0603] | 1.6±0.15
[.063±.006] | 0.8±0.15
[.031±.006] | 0.8±0.15
[.031±.006] | 0.3±0.2
[.012±.008] |
| GZ2012 [0805] | 2.0 (+0.3, -0.1)
[.079 (+.012, -.004)] | 1.25±0.2
[.049±.008] | 0.85±0.2
[.033±.008] | 0.5±0.3
[.020±.012] |
| GZ3216 [1206] | 3.2±0.2
[.126±.008] | 1.6±0.2
[.063±.008] | 1.1±0.2
[0.043±0.008] | 0.5±0.3
[.020±.012] |

Shenzhen Sunlord Electronics Co., Ltd.

SPECIFICATIONS GZ0603 TYPE

| Part Number | Impedance | Z Test Frequency | Max. DC Resistance | Max. Rated Current | Thickness |
|--------------|-----------|------------------|--------------------|--------------------|-------------------------|
| Units | Ω | MHz | Ω | mA | mm [inch] |
| Symbol | Z | Freq. | DCR | I _r | T |
| GZ0603D600TF | 60±25% | 100 | 0.40 | 200 | 0.3±0.05
[.012±.002] |
| GZ0603D750TF | 75±25% | 100 | 0.40 | 200 | |
| GZ0603D800TF | 80±25% | 100 | 0.60 | 200 | |
| GZ0603D121TF | 120±25% | 100 | 0.80 | 200 | |
| GZ0603D241TF | 240±25% | 100 | 1.00 | 200 | |
| GZ0603D471TF | 470±25% | 100 | 1.40 | 200 | |
| GZ0603D601TF | 600±25% | 100 | 1.70 | 200 | |
| GZ0603D102TF | 1000±25% | 100 | 2.50 | 100 | |
| GZ0603U100TF | 5~15 | 100 | 0.10 | 500 | |
| GZ0603U700TF | 70±25% | 100 | 0.40 | 200 | |
| GZ0603U800TF | 80±25% | 100 | 0.40 | 200 | |
| GZ0603U121TF | 120±25% | 100 | 0.50 | 200 | |
| GZ0603U241TF | 240±25% | 100 | 0.80 | 200 | |
| GZ0603U601TF | 600±25% | 100 | 1.50 | 100 | |
| GZ0603U102TF | 1000±25% | 100 | 2.50 | 100 | |

GZ1005 TYPE

| Part Number | Impedance | Z Test Frequency | Max. DC Resistance | Max. Rated Current | Thickness |
|--------------|-----------|------------------|--------------------|--------------------|-------------------------|
| Units | Ω | MHz | Ω | mA | mm [inch] |
| Symbol | Z | Freq. | DCR | I _r | T |
| GZ1005D100TF | 0~15 | 100 | 0.05 | 500 | 0.5±0.15
[.020±.006] |
| GZ1005D310TF | 31±25% | 100 | 0.20 | 300 | |
| GZ1005D600TF | 60±25% | 100 | 0.30 | 200 | |
| GZ1005D800TF | 80±25% | 100 | 0.35 | 200 | |
| GZ1005D121TF | 120±25% | 100 | 0.40 | 200 | |
| GZ1005D221TF | 220±25% | 100 | 0.45 | 150 | |
| GZ1005D301TF | 300±25% | 100 | 0.50 | 100 | |
| GZ1005D421TF | 420±25% | 100 | 0.60 | 100 | |
| GZ1005D501TF | 500±25% | 100 | 0.80 | 100 | |
| GZ1005D601TF | 600±25% | 100 | 0.90 | 100 | |
| GZ1005D751TF | 750±25% | 100 | 1.00 | 100 | |
| GZ1005D102TF | 1000±25% | 100 | 1.20 | 100 | |
| GZ1005D152TF | 1500±25% | 100 | 1.60 | 100 | |
| GZ1005D182TF | 1800±25% | 100 | 2.00 | 50 | |
| GZ1005E800TF | 80±25% | 100 | 0.35 | 200 | |
| GZ1005E121TF | 120±25% | 100 | 0.40 | 200 | |
| GZ1005E241TF | 240±25% | 100 | 0.50 | 200 | |
| GZ1005E601TF | 600±25% | 100 | 0.90 | 100 | |
| GZ1005U100TF | 0~15 | 100 | 0.05 | 500 | |
| GZ1005U300TF | 30±25% | 100 | 0.20 | 300 | |
| GZ1005U700TF | 70±25% | 100 | 0.30 | 200 | |
| GZ1005U121TF | 120±25% | 100 | 0.40 | 200 | |
| GZ1005U221TF | 220±25% | 100 | 0.50 | 100 | |
| GZ1005U301TF | 300±25% | 100 | 0.60 | 100 | |
| GZ1005U421TF | 420±25% | 100 | 0.80 | 100 | |
| GZ1005U601TF | 600±25% | 100 | 0.90 | 100 | |
| GZ1005U102TF | 1000±25% | 100 | 1.20 | 100 | |

Shenzhen Sunlord Electronics Co., Ltd.

SPECIFICATIONS GZ1608 TYPE

| Part Number | Impedance | Z Test Frequency | Max. DC Resistance | Max. Rated Current | Thickness |
|--------------|-----------|------------------|--------------------|--------------------|-------------------------|
| Units | Ω | MHz | Ω | mA | mm [inch] |
| Symbol | Z | Freq. | DCR | I _r | T |
| GZ1608D110TF | 0~15 | 100 | 0.05 | 2000 | 0.8±0.15
[.031±.006] |
| GZ1608D300TF | 30±25% | 100 | 0.05 | 2000 | |
| GZ1608D600TF | 60±25% | 100 | 0.10 | 500 | |
| GZ1608D800TF | 80±25% | 100 | 0.15 | 400 | |
| GZ1608D101TF | 100±25% | 100 | 0.20 | 300 | |
| GZ1608D121TF | 120±25% | 100 | 0.20 | 300 | |
| GZ1608D221TF | 220±25% | 100 | 0.30 | 300 | |
| GZ1608D301TF | 300±25% | 100 | 0.35 | 200 | |
| GZ1608D471TF | 470±25% | 100 | 0.45 | 200 | |
| GZ1608D601TF | 600±25% | 100 | 0.45 | 200 | |
| GZ1608D751TF | 750±25% | 100 | 0.50 | 200 | |
| GZ1608D102TF | 1000±25% | 100 | 0.60 | 200 | |
| GZ1608D152TF | 1500±25% | 100 | 0.70 | 150 | |
| GZ1608D182TF | 1800±25% | 100 | 0.90 | 100 | |
| GZ1608D202TF | 2000±25% | 100 | 1.20 | 100 | |
| GZ1608D222TF | 2200±25% | 100 | 1.20 | 100 | |
| GZ1608D252TF | 2500±25% | 100 | 1.20 | 100 | |
| GZ1608E121TF | 120±25% | 100 | 0.20 | 300 | |
| GZ1608E181TF | 180±25% | 100 | 0.30 | 300 | |
| GZ1608E601TF | 600±25% | 100 | 0.45 | 200 | |
| GZ1608E102TF | 1000±25% | 100 | 0.60 | 200 | |
| GZ1608U100TF | 0~15 | 100 | 0.05 | 2000 | |
| GZ1608U300TF | 30±25% | 100 | 0.05 | 2000 | |
| GZ1608U600TF | 60±25% | 100 | 0.10 | 500 | |
| GZ1608U121TF | 120±25% | 100 | 0.20 | 300 | |
| GZ1608U221TF | 220±25% | 100 | 0.30 | 300 | |
| GZ1608U301TF | 300±25% | 100 | 0.35 | 200 | |
| GZ1608U471TF | 470±25% | 100 | 0.40 | 200 | |
| GZ1608U601TF | 600±25% | 100 | 0.50 | 200 | |
| GZ1608U102TF | 1000±25% | 100 | 0.60 | 200 | |

GZ2012 TYPE

| Part Number | Impedance | Z Test Frequency | Max. DC Resistance | Max.Rated Current | Thickness |
|--------------|-----------|------------------|--------------------|-------------------|-------------------------|
| Units | Ω | MHz | Ω | mA | mm [inch] |
| Symbol | Z | Freq. | DCR | I _r | T |
| GZ2012D070TF | 0~15 | 100 | 0.04 | 2000 | 0.85±0.2
[.033±.008] |
| GZ2012D190TF | 19±25% | 100 | 0.04 | 2000 | |
| GZ2012D300TF | 30±25% | 100 | 0.05 | 1500 | |
| GZ2012D800TF | 80±25% | 100 | 0.10 | 1000 | |
| GZ2012D121TF | 120±25% | 100 | 0.15 | 800 | |
| GZ2012D181TF | 180±25% | 100 | 0.18 | 700 | |
| GZ2012D221TF | 220±25% | 100 | 0.20 | 600 | |
| GZ2012D301TF | 300±25% | 100 | 0.20 | 500 | |
| GZ2012D421TF | 420±25% | 100 | 0.30 | 500 | |
| GZ2012D501TF | 500±25% | 100 | 0.30 | 500 | |
| GZ2012D601TF | 600±25% | 100 | 0.30 | 500 | |
| GZ2012D751TF | 750±25% | 100 | 0.35 | 500 | |
| GZ2012D102TF | 1000±25% | 100 | 0.35 | 500 | |
| GZ2012D152TF | 1500±25% | 100 | 0.40 | 500 | |



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SPECIFICATIONS GZ2012 TYPE

| Part Number | Impedance | Z Test Frequency | Max. DC Resistance | Max.Rated Current | Thickness |
|--------------|-----------|------------------|--------------------|-------------------|-------------------------|
| Units | Ω | MHz | Ω | mA | mm [inch] |
| Symbol | Z | Freq. | DCR | Ir | T |
| GZ2012D202TF | 2000±25% | 100 | 0.50 | 500 | 0.85±0.2
[.033±.008] |
| GZ2012D252TF | 2500±25% | 100 | 0.70 | 200 | |
| GZ2012E800TF | 80±25% | 100 | 0.10 | 1000 | |
| GZ2012E181TF | 180±25% | 100 | 0.20 | 600 | |
| GZ2012E301TF | 300±25% | 100 | 0.20 | 500 | |
| GZ2012E501TF | 500±25% | 100 | 0.30 | 500 | |
| GZ2012E601TF | 600±25% | 100 | 0.30 | 500 | |
| GZ2012E102TF | 1000±25% | 100 | 0.35 | 500 | |
| GZ2012U100TF | 0~15 | 100 | 0.04 | 2200 | |
| GZ2012U170TF | 17±25% | 100 | 0.04 | 2000 | |
| GZ2012U300TF | 30±25% | 100 | 0.05 | 1500 | |
| GZ2012U700TF | 70±25% | 100 | 0.10 | 1000 | |
| GZ2012U121TF | 120±25% | 100 | 0.15 | 800 | |
| GZ2012U221TF | 220±25% | 100 | 0.20 | 600 | |
| GZ2012U301TF | 300±25% | 100 | 0.20 | 500 | |
| GZ2012U421TF | 420±25% | 100 | 0.25 | 500 | |
| GZ2012U601TF | 600±25% | 100 | 0.30 | 500 | |
| GZ2012U102TF | 1000±25% | 100 | 0.40 | 500 | |

GZ3216 TYPE

| Part Number | Impedance | Z Test Frequency | Max. DC Resistance | Max.Rated Current | Thickness |
|-----------------|-----------|------------------|--------------------|-------------------|---------------------------|
| Units | Ω | MHz | Ω | mA | mm [inch] |
| Symbol | Z | Freq. | DCR | Ir | T |
| GZ3216D000TF | 0~15 | 100 | 0.03 | 2200 | 0.85±0.2
[.033±.008] |
| GZ3216D310TF | 31±25% | 100 | 0.05 | 2000 | |
| GZ3216D600TF | 60±25% | 100 | 0.10 | 1000 | |
| GZ3216D301TF | 300±25% | 100 | 0.20 | 600 | |
| GZ3216D100TFA99 | 0~15 | 100 | 0.05 | 2000 | 1.10±0.2
[0.043±0.008] |
| GZ3216D190TFA99 | 19±25% | 100 | 0.05 | 2000 | |
| GZ3216D800TFA99 | 80±25% | 100 | 0.10 | 1000 | |
| GZ3216D101TFA99 | 100±25% | 100 | 0.10 | 1000 | |
| GZ3216D121TFA99 | 120±25% | 100 | 0.10 | 1000 | |
| GZ3216D151TFA99 | 150±25% | 100 | 0.15 | 800 | |
| GZ3216D221TFA99 | 220±25% | 100 | 0.20 | 600 | |
| GZ3216D501TFA99 | 500±25% | 100 | 0.30 | 600 | |
| GZ3216D601TFA99 | 600±25% | 100 | 0.30 | 600 | |
| GZ3216D102TFA99 | 1000±25% | 100 | 0.60 | 500 | |
| GZ3216D122TFA99 | 1200±25% | 100 | 0.60 | 300 | |
| GZ3216U601TFA99 | 600±25% | 100 | 0.30 | 600 | |

※: Products with other electrical characteristics can be provided upon customer's request. Please contact your local sales.

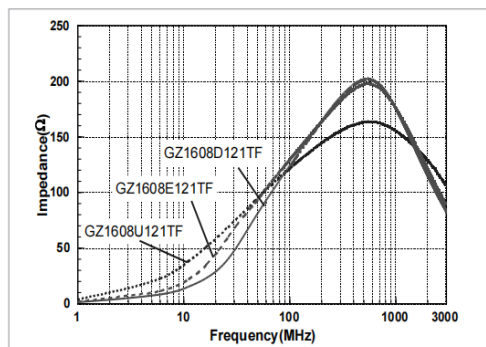


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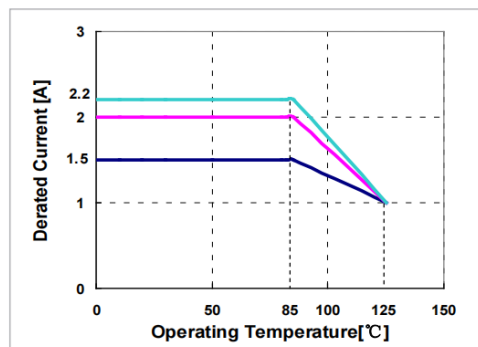
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TYPICAL ELECTRICAL CHARACTERISTICS

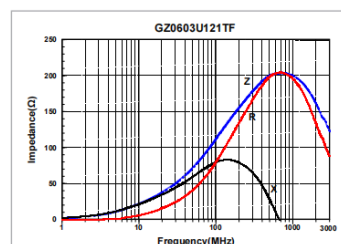
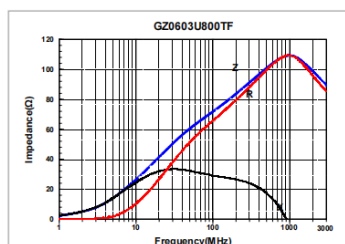
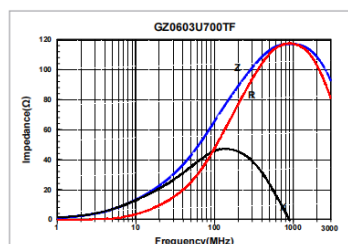
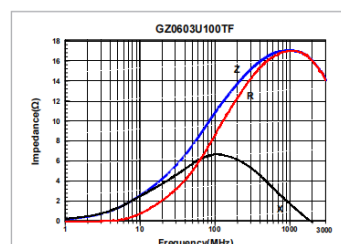
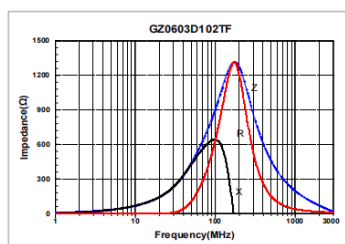
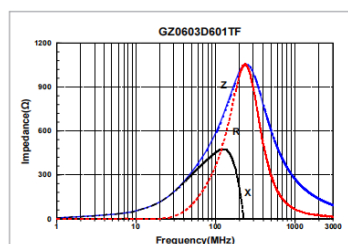
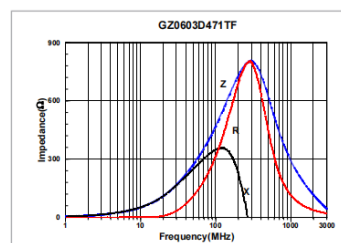
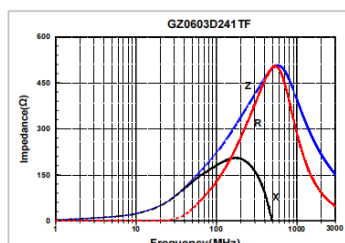
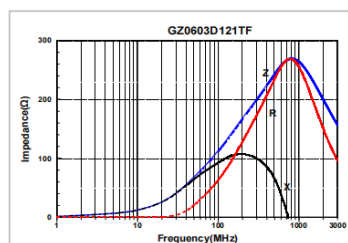
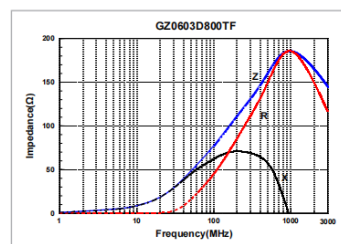
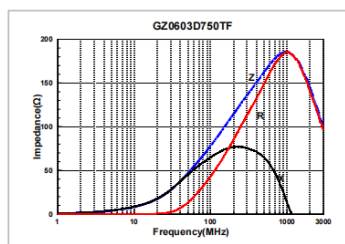
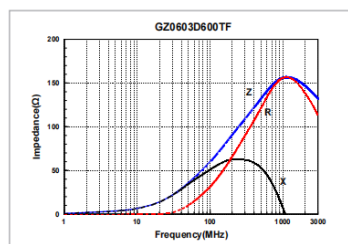
D, E, U Material Comparison

**Rated Current**

When operating temperatures exceeding +85°C, derating of current is necessary for chip ferrite beads for which rated current is 1000mA over. Please apply the derating curve shown in chart according to the operating temperature.

**DETAIL ELECTRICAL CHARACTERISTICS**

GZ0603 TYPE


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Multilayer Chip Ferrite Bead

Wire Wound Ferrite Bead

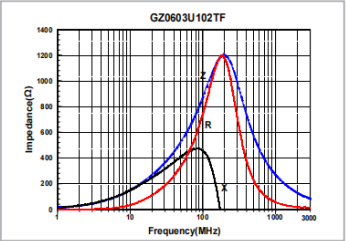
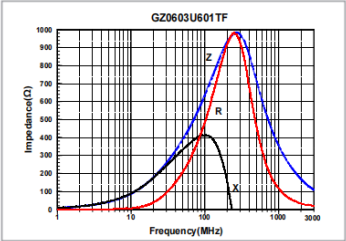
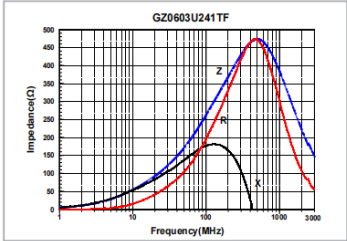
Multilayer Chip Common Mode Filter

Wire Wound Chip Common Mode Choke Coil for Signal Line

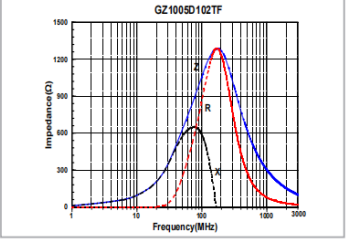
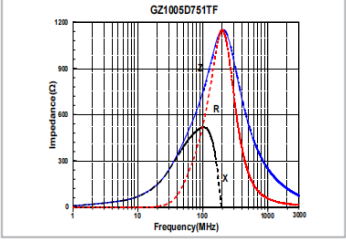
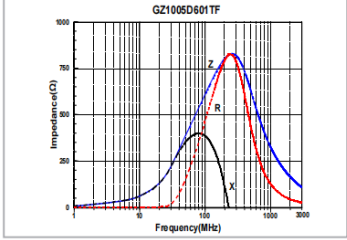
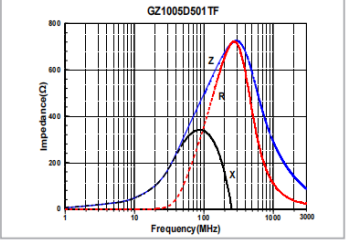
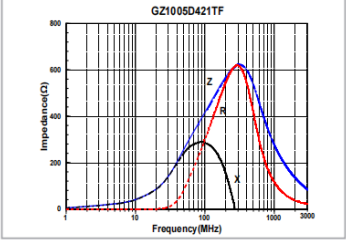
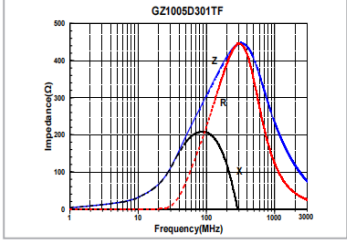
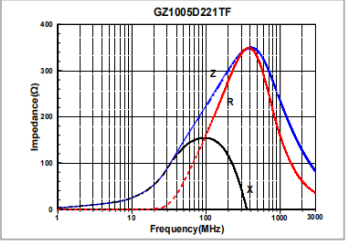
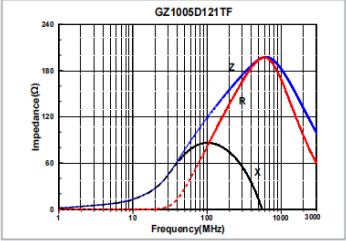
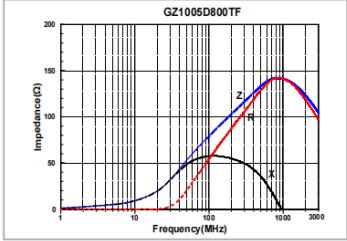
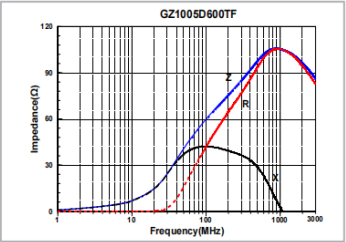
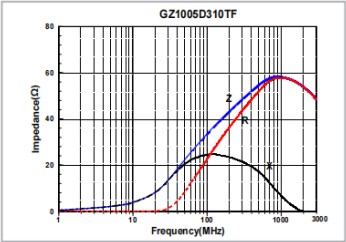
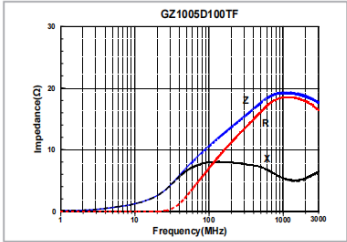
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DETAIL ELECTRICAL
CHARACTERISTICS

GZ0603 TYPE

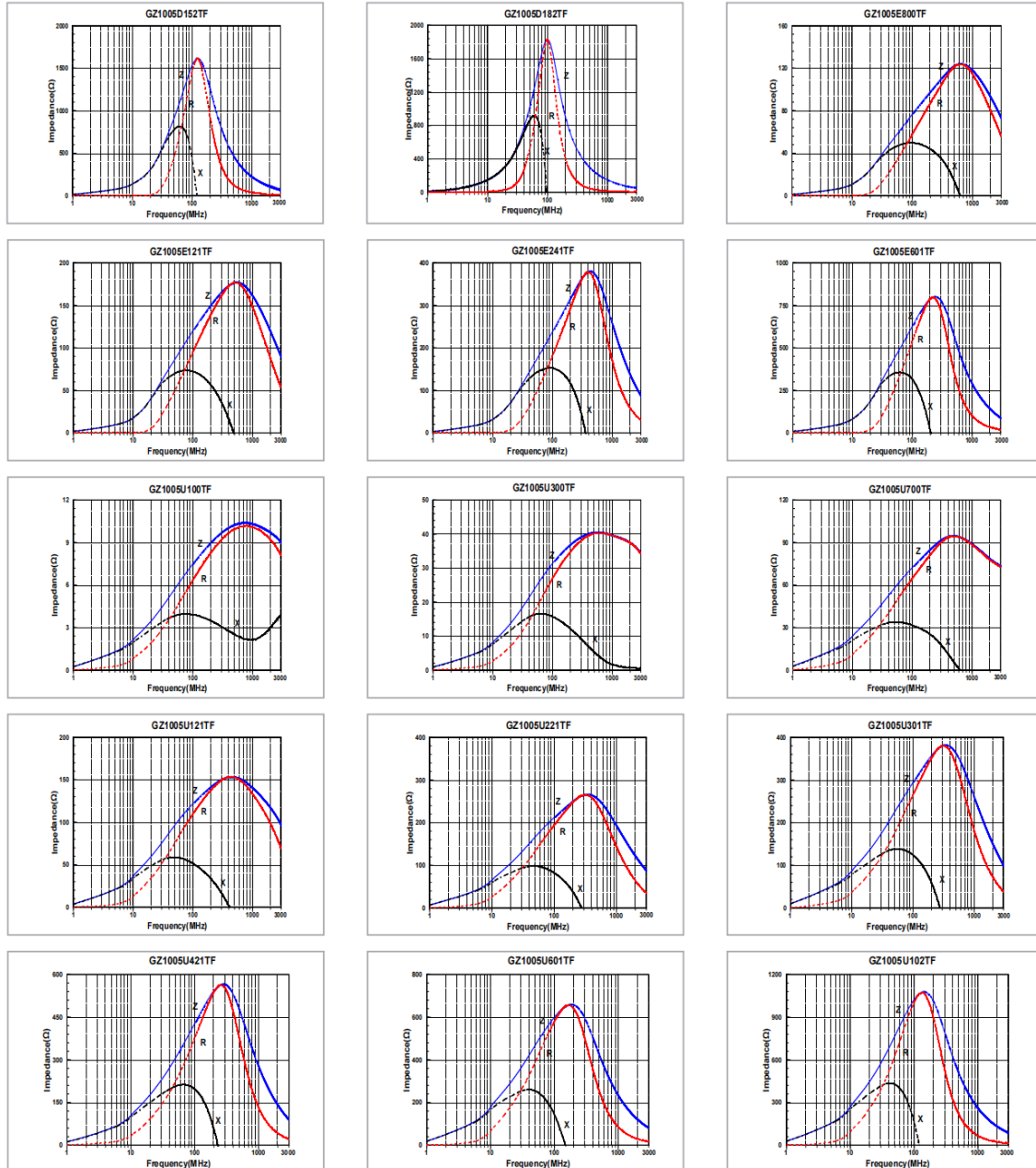


GZ1005 TYPE



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**DETAIL ELECTRICAL
CHARACTERISTICS****GZ1005 TYPE****Sunlord**
expert in e componentsSpecifications subject to change without notice. Please check our website for latest information. Revised 2023/06/01
Sunlord Industrial Park, Dafuyuan Industrial Zone, Guanlan, Shenzhen, China 518110 Tel: 0086-755-29832333 Fax: 0086-755-82269029 E-Mail: sunlord@sunlordinc.comMultilayer Chip Ferrite
Bead

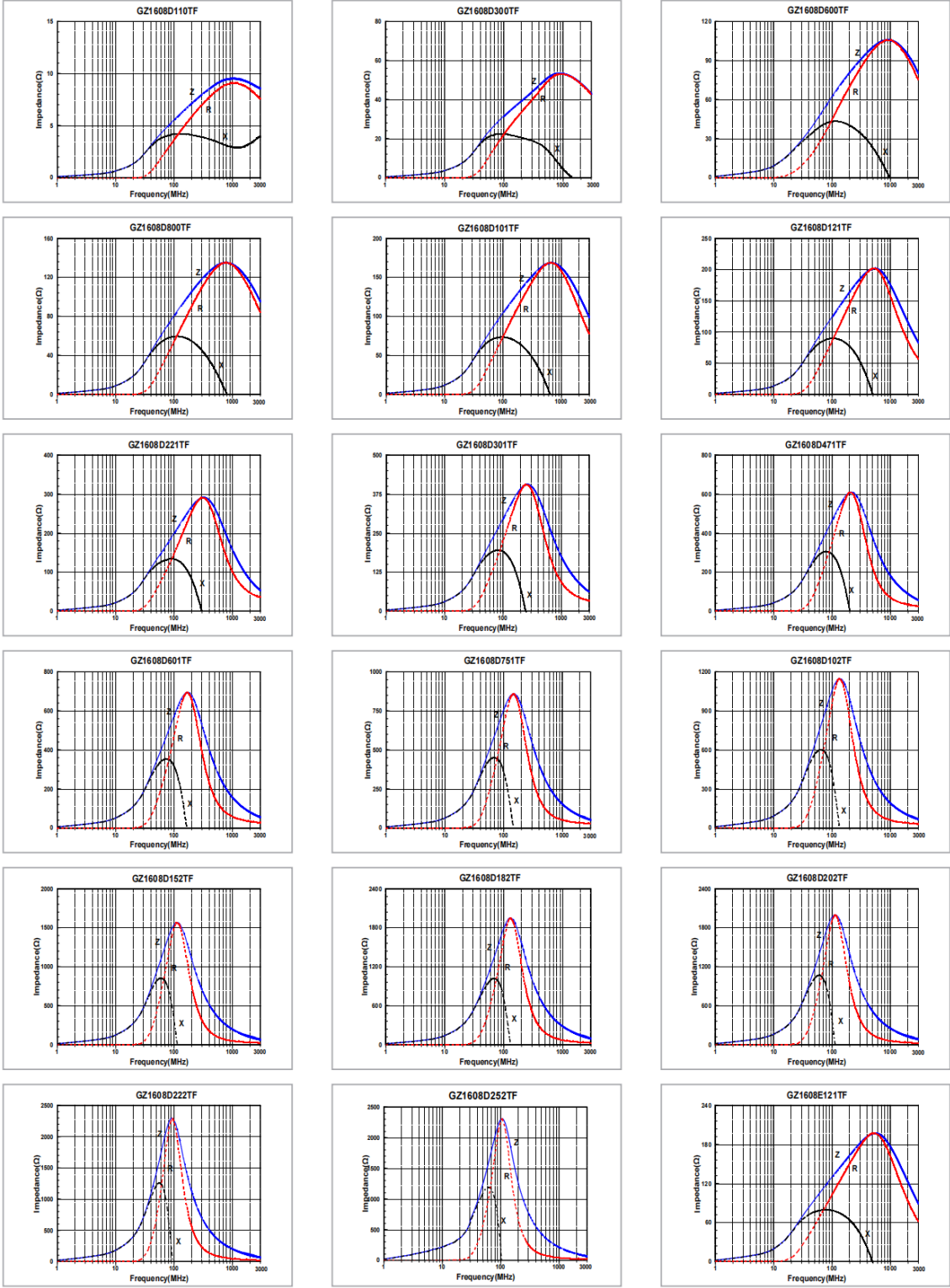
Wire Wound Ferrite Bead

Multilayer Chip Common
Mode FilterWire Wound Chip Common
Mode Choke Coil for Signal Line

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DETAIL ELECTRICAL
CHARACTERISTICS

GZ1608 TYPE



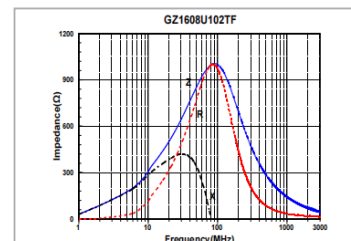
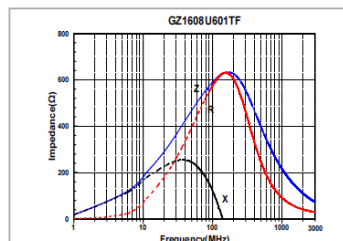
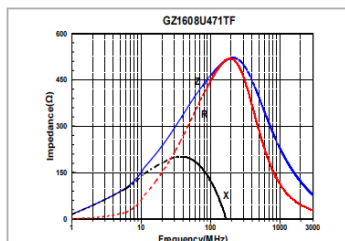
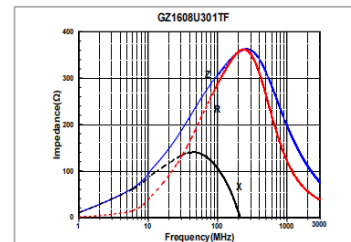
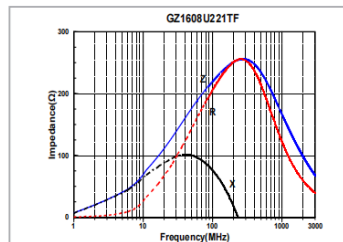
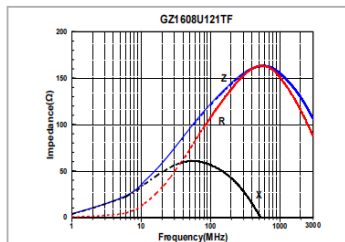
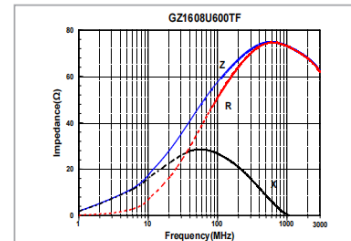
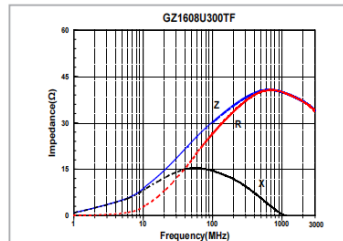
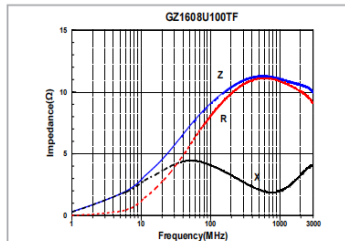
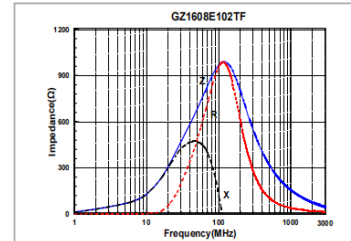
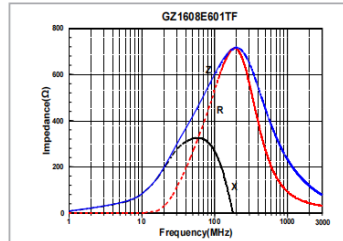
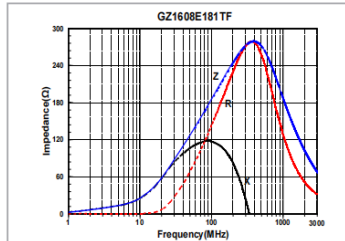
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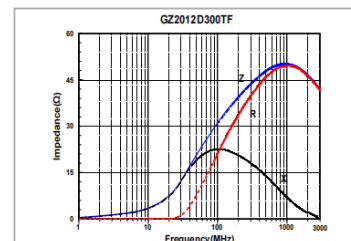
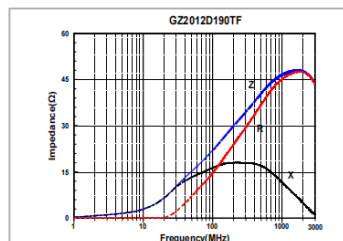
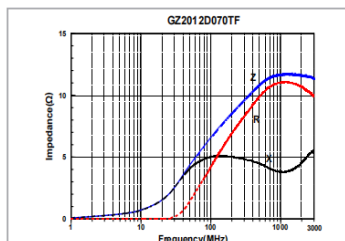
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DETAIL ELECTRICAL CHARACTERISTICS

GZ1608 TYPE



GZ2012 TYPE



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Multilayer Chip Ferrite Bead

Wire Wound Ferrite Bead

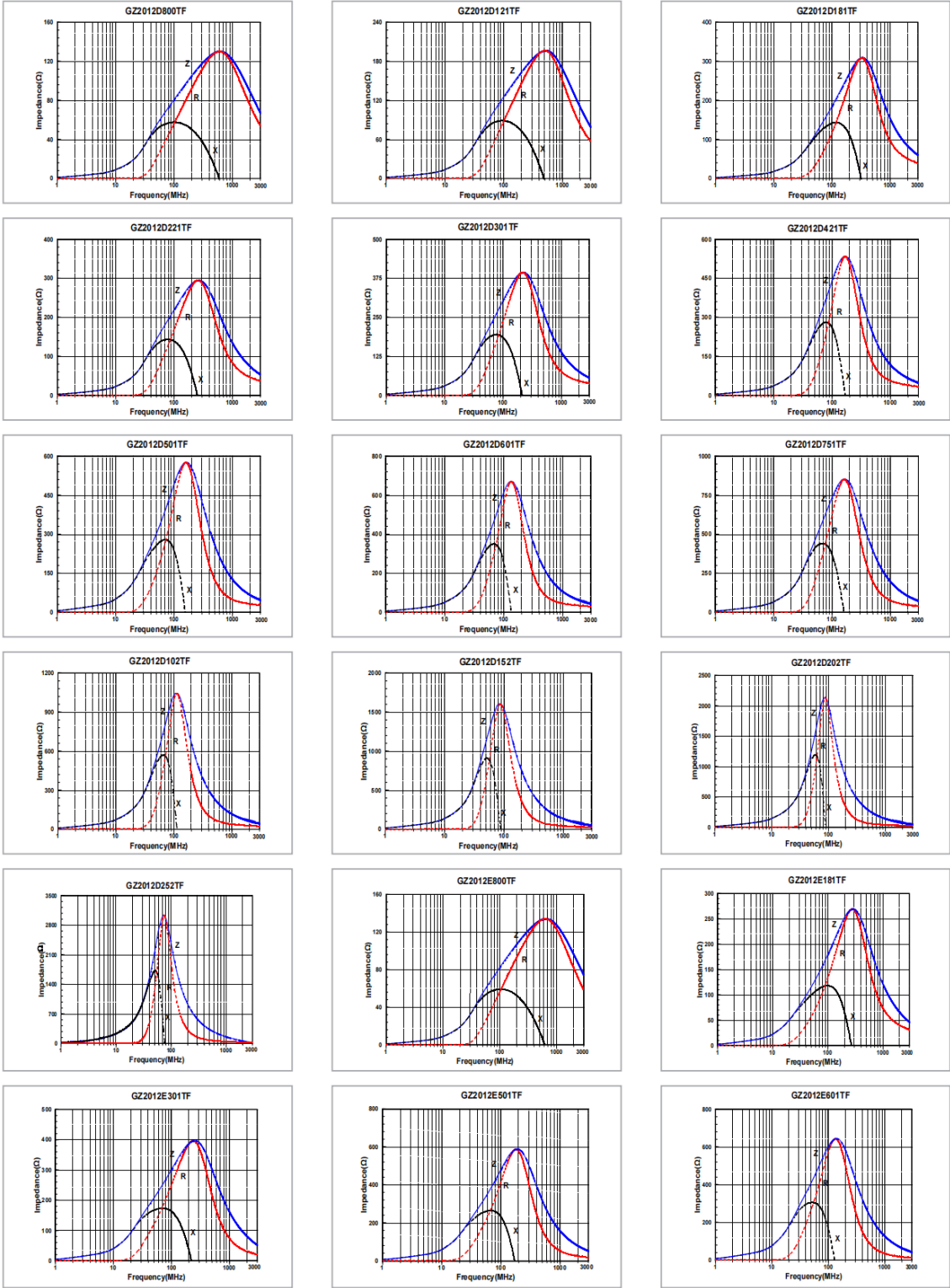
Multilayer Chip Common Mode Filter

Wire Wound Chip Common Mode Choke Coil for Signal Line

Shenzhen Sunlord Electronics Co., Ltd.

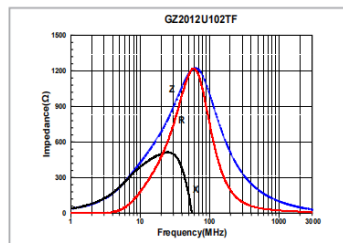
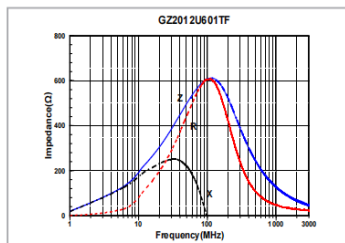
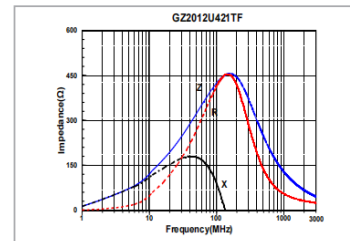
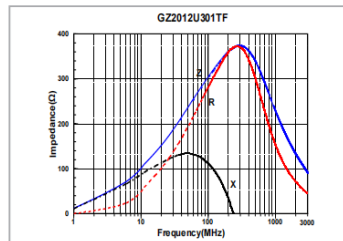
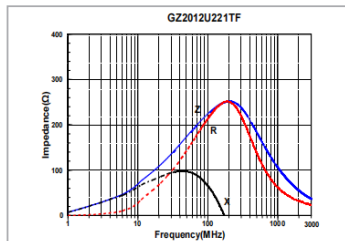
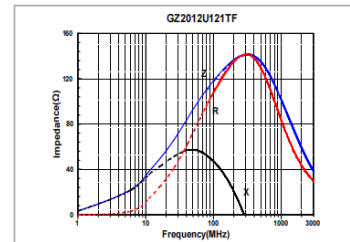
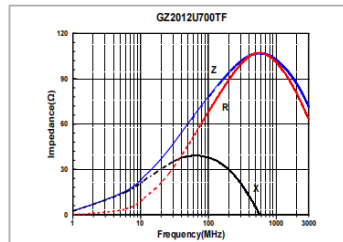
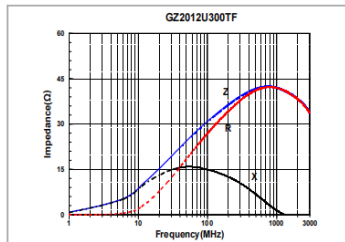
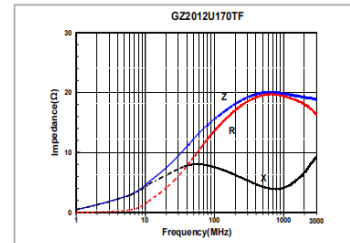
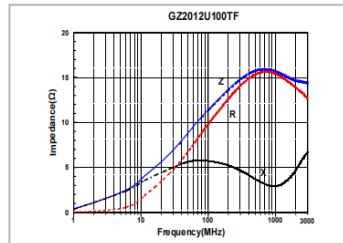
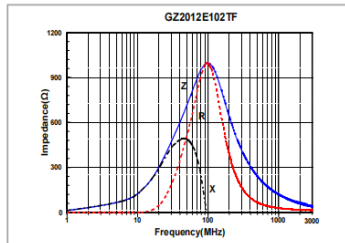
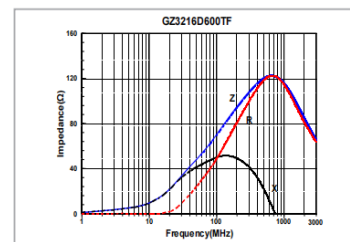
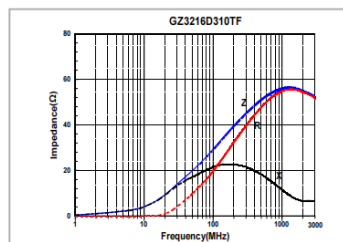
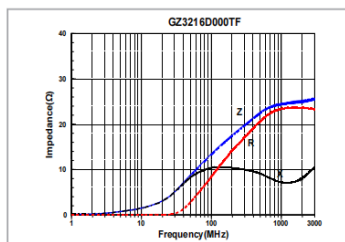
DETAIL ELECTRICAL
CHARACTERISTICS

GZ2012 TYPE



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**DETAIL ELECTRICAL
CHARACTERISTICS****GZ2012 TYPE****GZ3216 TYPE**

Multilayer Chip Ferrite Bead

Wire Wound Ferrite Bead

Multilayer Chip Common Mode Filter

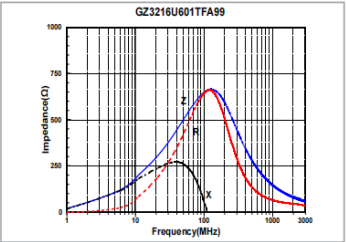
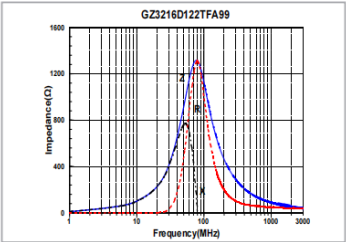
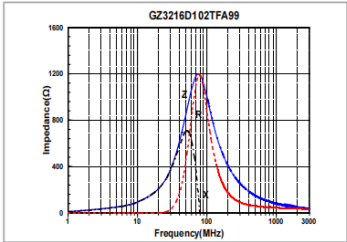
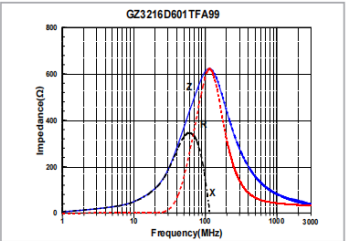
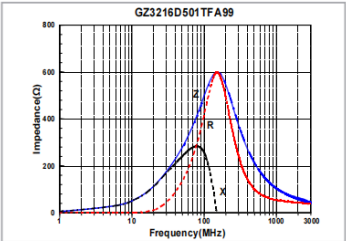
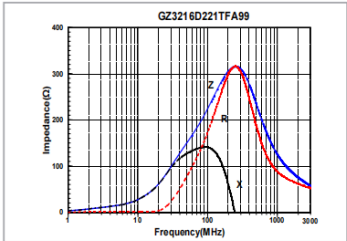
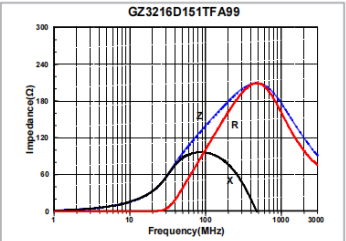
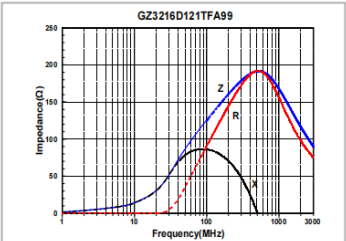
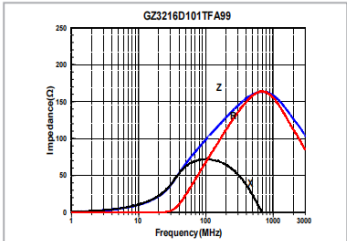
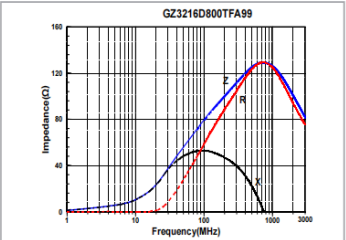
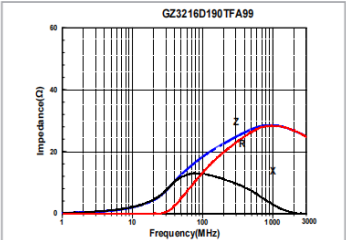
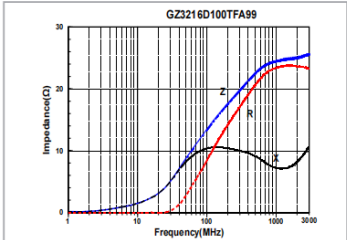
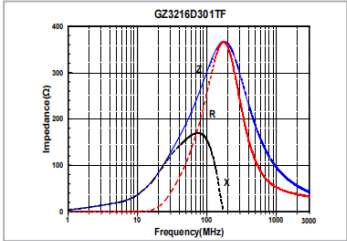
Wire Wound Chip Common Mode Choke Coil for Signal Line

Sunlord
expert in e componentsSpecifications subject to change without notice. Please check our website for latest information. Revised 2023/06/01
Sunlord Industrial Park, Dafuyuan Industrial Zone, Guanlan, Shenzhen, China 518110 Tel: 0086-755-29832333 Fax: 0086-755-82269029 E-Mail: sunlord@sunlordinc.com

Shenzhen Sunlord Electronics Co., Ltd.

DETAIL ELECTRICAL
CHARACTERISTICS

GZ3216 TYPE



All information and data presented in this Catalog are for information only. You are requested to approve our product specification before your ordering.

Section 9

JUSHUO

| Designators | Part / Value | Qty |
|-------------|-------------------|-----|
| J1-J36 | FH34SRJ-14S-0.5SH | 36 |

RoHS evidence document: **FH34SRJ-14S-0.5SH(50)_CL0580-1252-8-50_SpecSheet_0000414509.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

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| | | | | | |
|---|-----------------------------|--|--|--|---|
| APPLICABLE STANDARD | | | | | |
| RATING | OPERATING TEMPERATURE RANGE | -55 °C TO 105 °C | STORAGE TEMPERATURE RANGE | -10 °C TO 50 °C (PACKED CONDITION) | |
| | VOLTAGE | 50 V AC / DC | OPERATING OR STORAGE HUMIDITY RANGE | RELATIVE HUMIDITY 90 % MAX (NOT DEWED) | |
| | CURRENT | 0.5 A | APPLICABLE CABLE | t=0.3±0.03mm, GOLD PLATING | |
| SPECIFICATIONS | | | | | |
| ITEM | | TEST METHOD | REQUIREMENTS | QT | AT |
| CONSTRUCTION | | | | | |
| GENERAL EXAMINATION | | VISUALLY AND BY MEASURING INSTRUMENT. | ACCORDING TO DRAWING. | x | x |
| MARKING | | CONFIRMED VISUALLY. | | x | x |
| ELECTRICAL CHARACTERISTICS | | | | | |
| VOLTAGE PROOF | | 250 V AC FOR 1 min. | NO FLASHOVER OR BREAKDOWN. | x | x |
| INSULATION RESISTANCE | | 100 V DC. | 500 MΩ MIN. | x | x |
| CONTACT RESISTANCE | | AC/DC 20 mV MAX (AC:1 KHz) , 1 mA . | 100 mΩ MAX.
INCLUDING FPC,FPC BULK RESISTANCE (L=8mm) | x | x |
| MECHANICAL CHARACTERISTICS | | | | | |
| VIBRATION | | FREQUENCY 10 TO 55 Hz, HALF AMPLITUDE 0.75 mm, FOR 10 CYCLES IN 3 AXIAL DIRECTIONS. | ① NO ELECTRICAL DISCONTINUITY OF 1 μs. | x | — |
| SHOCK | | 981 m/s ² , DURATION OF PULSE 6 ms AT 3 TIMES IN 3 BOTH AXIAL DIRECTIONS. | ② CONTACT RESISTANCE: 100 mΩ MAX.
③ NO DAMAGE, CRACK AND LOOSENESS OF PARTS. | x | — |
| MECHANICAL OPERATION | | 20 TIMES INSERTIONS AND EXTRACTIONS. | ① CONTACT RESISTANCE: 100 mΩ MAX.
② NO DAMAGE, CRACK AND LOOSENESS OF PARTS. | x | — |
| FPC RETENTION FORCE | | MEASURED BY APPLICABLE FPC. (THICKNESS OF FPC SHALL BE t=0.30mm AT INITIAL CONDITION.) | DIRECTION OF INSERTION : (TOP CONTACT)
0.2N×NUMBER OF CONTACTS MIN. (BOTTOM CONTACT)
0.3N×NUMBER OF CONTACTS MIN. (note 1) | x | — |
| ENVIRONMENTAL CHARACTERISTICS | | | | | |
| CORROSION SALT MIST | | EXPOSED AT 35±2 °C , 5 % SALT WATER SPRAY FOR 96 h. | ① CONTACT RESISTANCE: 100 mΩ MAX.
② NO DAMAGE, CRACK AND LOOSENESS OF PARTS.
③ NO EVIDENCE OF CORROSION WHICH AFFECTS TO OPERATION OF CONNECTOR. | x | — |
| RAPID CHANGE OF TEMPERATURE | | TEMPERATURE -55→+15TO+35→+105→+15TO+35°C 
TIME 30→ 2 TO 3 → 30→ 2 TO 3 min UNDER 5 CYCLES. | ① CONTACT RESISTANCE: 100 mΩ MAX.
② INSULATION RESISTANCE: 50 MΩ MIN.
③ NO DAMAGE, CRACK AND LOOSENESS OF PARTS. | x | — |
| DAMP HEAT (STEADY STATE) | | EXPOSED AT 40±2 °C, RELATIVE HUMIDITY 90 TO 95 %, 96 h. | | x | — |
| DAMP HEAT,CYCLIC | | EXPOSED AT -10 TO +65 °C, RELATIVE HUMIDITY 90 TO 96 %, 10 CYCLES,TOTAL 240 h. | ① CONTACT RESISTANCE: 100 mΩ MAX.
② INSULATION RESISTANCE: 1 MΩ MIN. (AT HIGH HUMIDITY)
③ INSULATION RESISTANCE: 50 MΩ MIN. (AT DRY)
④ NO DAMAGE, CRACK AND LOOSENESS OF PARTS. | x | — |
|  | COUNT | DESCRIPTION OF REVISIONS | DESIGNED | CHECKED | DATE |
| | 3 | DIS-F-00005614 | SE. YOKOYAMA | HS. HIRAHARA | 20200611 |
| REMARK | | | APPROVED | MO. ISHIDA | 20131129 |
| This product is RoHS compliant.
Unless otherwise specified, refer to IEC 60512. | | | CHECKED | HS. SAKAMOTO | 20131129 |
| | | | DESIGNED | YS. EBI | 20131128 |
| | | | DRAWN | NM. SANPEI | 20131128 |
| | | | Note QT:Qualification Test AT:Assurance Test X:Applicable Test | | |
|  | SPECIFICATION SHEET | | PART NO. | FH34SRJ-*S-0. 5SH (50) | |
| | HIROSE ELECTRIC CO., LTD. | | CODE NO. | CL580 |  1/2 |

FORM HD0011-2-1

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| SPECIFICATIONS | | | | | |
|---|---|--|------------------------|----|-----|
| ITEM | TEST METHOD | REQUIREMENTS | QT | AT | |
| DRY HEAT | EXPOSED AT 105±2 °C, 96 h. | ① CONTACT RESISTANCE: 100 mΩ MAX. | x | — | |
| COLD | EXPOSED AT -55±3°C, 96 h. | ② NO DAMAGE, CRACK AND LOOSENESS OF PARTS. | x | — | |
| SULPHUR DIOXIDE
[JIS C 60068-2-42] | EXPOSED AT 40±2 °C ,
RELATIVE HUMIDITY 80±5%
25±5 ppm FOR 96 h. | ① CONTACT RESISTANCE: 100 mΩ MAX.
② NO DAMAGE, CRACK AND LOOSENESS OF PARTS. | x | — | |
| HYDROGEN SULPHIDE
[JIS C 60068-2-43] | EXPOSED AT 40±2 °C ,
RELATIVE HUMIDITY 80±5% ,
10 TO 15 ppm FOR 96 h. | ③ NO EVIDENCE OF CORROSION WHICH AFFECTS TO OPERATION OF CONNECTOR. | x | — | |
| SOLDERABILITY | SOLDERED AT SOLDER TEMPERATURE,
235±5°C FOR IMMERSION DURATION,
2±0.5 sec. | A NEW UNIFORM COATING OF SOLDER SHALL COVER A MINIMUM OF 95 % OF THE SURFACE BEING IMMERSED. | x | — | |
| RESISTANCE TO SOLDERING HEAT | 1) REFLOW SOLDERING :
PEAK TMP. 250 °C MAX .
REFLOW TMP. OVER 230 °C WITHIN 60 sec.
2) SOLDERING IRONS :
TMP. 350 ± 10 °C FOR 5±1 sec . | NO DEFORMATION OF CASE OF EXCESSIVE LOOSENESS OF THE TERMINALS. | x | — | |
| <div>(note1)</div> <div>FASTEN FPC ON PCB OR SOMETHING FIXED IF FORCE IN VERTICAL DIRECTION SHALL BE PREDICTED.</div> <div>DO NOT CLOSE THE ACTUATOR BEFORE INSERTING FPC EVEN AFTER THE CONNECTOR IS MOUNTED ONTO A PCB. CLOSING THE ACTUATOR WITHOUT FPC COULD MAKE THE CONTACT GAP SMALLER, WHICH INCREASES THE FPC INSERTION FORCE.</div> <div>THIS CONNECTOR HAS CONTACTS ON THE BOTH TOP AND BOTTOM.</div> <div> THERE'S A CASE WHICH FPC/FFC RETENTION FORCE DOESN'T FULFILL THE VALUE, BECAUSE FPC SPECIFICATION AFFECTS THE RESULT OF FPC/FFC RETENTION FORCE.</div> | | | | | |
| Note QT:Qualification Test AT:Assurance Test X:Applicable Test | | DRAWING NO. | ELC4-159714-04 | | |
| | SPECIFICATION SHEET | PART NO. | FH34SRJ-*S-0. 5SH (50) | | |
| | HIROSE ELECTRIC CO., LTD. | CODE NO | CL580 | | 2/2 |

FORM HD0011-2-2

Section 10
Hirose

| Designators | Part / Value | Qty |
|-------------|-------------------|-----|
| J37 | FH34SRJ-12S-0.5SH | 1 |

RoHS evidence document: **FH34SRJ-12S-0.5SH(99)_CL0580-1253-0-99_SpecSheet_0000414526.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

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| | | | | | |
|---|-----------------------------|---|-------------------------------------|--|---|
| APPLICABLE STANDARD | | | | | |
| RATING | OPERATING TEMPERATURE RANGE | -55 °C TO 105 °C | STORAGE TEMPERATURE RANGE | -10 °C TO 50 °C (PACKED CONDITION) | |
| | VOLTAGE | 50 V AC / DC | OPERATING OR STORAGE HUMIDITY RANGE | RELATIVE HUMIDITY 90 % MAX (NOT DEWED) | |
| | CURRENT | 0.5 A | APPLICABLE CABLE | t=0.3±0.03mm, GOLD PLATING | |
| SPECIFICATIONS | | | | | |
| ITEM | | TEST METHOD | | REQUIREMENTS | QT AT |
| CONSTRUCTION | | | | | |
| GENERAL EXAMINATION | | VISUALLY AND BY MEASURING INSTRUMENT. | | ACCORDING TO DRAWING. | x x |
| MARKING | | CONFIRMED VISUALLY. | | | x x |
| ELECTRICAL CHARACTERISTICS | | | | | |
| VOLTAGE PROOF | | 250 V AC FOR 1 min. | | NO FLASHOVER OR BREAKDOWN. | x x |
| INSULATION RESISTANCE | | 100 V DC. | | 500 MΩ MIN. | x x |
| CONTACT RESISTANCE | | AC/DC 20 mV MAX (AC:1 KHz) , 1 mA . | | 100 mΩ MAX.
INCLUDING FPC,FPC BULK RESISTANCE (L=8mm) | x x |
| MECHANICAL CHARACTERISTICS | | | | | |
| VIBRATION | | FREQUENCY 10 TO 55 Hz, HALF AMPLITUDE 0.75 mm, FOR 10 CYCLES IN 3 AXIAL DIRECTIONS. | | ① NO ELECTRICAL DISCONTINUITY OF 1 μs. | x — |
| SHOCK | | 981 m/s ² , DURATION OF PULSE 6 ms AT 3 TIMES IN 3 BOTH AXIAL DIRECTIONS. | | ② CONTACT RESISTANCE: 100 mΩ MAX.
③ NO DAMAGE, CRACK AND LOOSENESS OF PARTS. | x — |
| MECHANICAL OPERATION | | 20 TIMES INSERTIONS AND EXTRACTIONS. | | ① CONTACT RESISTANCE: 100 mΩ MAX.
② NO DAMAGE, CRACK AND LOOSENESS OF PARTS. | x — |
| FPC RETENTION FORCE | | MEASURED BY APPLICABLE FPC. (THICKNESS OF FPC SHALL BE t=0.30mm AT INITIAL CONDITION.) | | DIRECTION OF INSERTION :
(TOP CONTACT)
0.2N × NUMBER OF CONTACTS MIN.
(BOTTOM CONTACT)
0.3N × NUMBER OF CONTACTS MIN.
(note 1) | x — |
| ENVIRONMENTAL CHARACTERISTICS | | | | | |
| CORROSION SALT MIST | | EXPOSED AT 35±2 °C , 5 % SALT WATER SPRAY FOR 96 h. | | ① CONTACT RESISTANCE: 100 mΩ MAX.
② NO DAMAGE, CRACK AND LOOSENESS OF PARTS.
③ NO EVIDENCE OF CORROSION WHICH AFFECTS TO OPERATION OF CONNECTOR. | x — |
| RAPID CHANGE OF TEMPERATURE | | TEMPERATURE -55→+15 TO +35→+105→+15 TO +35 °C 
TIME 30→ 2 TO 3 → 30→ 2 TO 3 min UNDER 5 CYCLES. | | ① CONTACT RESISTANCE: 100 mΩ MAX.
② INSULATION RESISTANCE: 50 MΩ MIN.
③ NO DAMAGE, CRACK AND LOOSENESS OF PARTS. | x — |
| DAMP HEAT (STEADY STATE) | | EXPOSED AT 40±2 °C, RELATIVE HUMIDITY 90 TO 95 %, 96 h. | | | x — |
| DAMP HEAT, CYCLIC | | EXPOSED AT -10 TO +65 °C, RELATIVE HUMIDITY 90 TO 96 %, 10 CYCLES, TOTAL 240 h. | | ① CONTACT RESISTANCE: 100 mΩ MAX.
② INSULATION RESISTANCE: 1 MΩ MIN. (AT HIGH HUMIDITY)
③ INSULATION RESISTANCE: 50 MΩ MIN. (AT DRY)
④ NO DAMAGE, CRACK AND LOOSENESS OF PARTS. | x — |
|  | COUNT | DESCRIPTION OF REVISIONS | DESIGNED | CHECKED | DATE |
| | 3 | DIS-F-00005614 | SE. YOKOYAMA | HS. HIRAHARA | 20200611 |
| REMARK | | | | APPROVED | MO. ISHIDA |
| This product is RoHS compliant.
Unless otherwise specified, refer to IEC 60512. | | | | CHECKED | HS. SAKAMOTO |
| | | | | DESIGNED | YS. EBI |
| | | | | DRAWN | NM. SANPEI |
| | | | | | |
| Note QT:Qualification Test AT:Assurance Test X:Applicable Test | | | DRAWING NO. | | ELC4-159714-05 |
|  | SPECIFICATION SHEET | | PART NO. | FH34SRJ-*S-0. 5SH (99) | |
| | HIROSE ELECTRIC CO., LTD. | | CODE NO. | CL580 |  1/2 |

FORM HD0011-2-1

SPECIFICATIONS

| ITEM | TEST METHOD | REQUIREMENTS | QT | AT |
|---|---|--|----|----|
| DRY HEAT | EXPOSED AT 105±2 °C, 96 h.  | ① CONTACT RESISTANCE: 100 mΩ MAX. | x | — |
| COLD | EXPOSED AT -55±3°C, 96 h. | ② NO DAMAGE, CRACK AND LOOSENESS OF PARTS. | x | — |
| SULPHUR DIOXIDE
[JIS C 60068-2-42] | EXPOSED AT 40±2 °C ,
RELATIVE HUMIDITY 80±5%
25±5 ppm FOR 96 h. | ① CONTACT RESISTANCE: 100 mΩ MAX.
② NO DAMAGE, CRACK AND LOOSENESS OF PARTS. | x | — |
| HYDROGEN SULPHIDE
[JIS C 60068-2-43] | EXPOSED AT 40±2 °C ,
RELATIVE HUMIDITY 80±5% ,
10 TO 15 ppm FOR 96 h. | ③ NO EVIDENCE OF CORROSION WHICH AFFECTS TO OPERATION OF CONNECTOR. | x | — |
| SOLDERABILITY | SOLDERED AT SOLDER TEMPERATURE,
235±5°C FOR IMMERSION DURATION,
2±0.5 sec. | A NEW UNIFORM COATING OF SOLDER SHALL COVER A MINIMUM OF 95 % OF THE SURFACE BEING IMMersed. | x | — |
| RESISTANCE TO SOLDERING HEAT | 1) REFLOW SOLDERING :
PEAK TMP. 250 °C MAX .
REFLOW TMP. OVER 230 °C WITHIN 60 sec.
2) SOLDERING IRONS :
TMP. 350 ± 10 °C FOR 5±1 sec . | NO DEFORMATION OF CASE OF EXCESSIVE LOOSENESS OF THE TERMINALS. | x | — |

(note1)

FASTEN FPC ON PCB OR SOMETHING FIXED IF FORCE IN VERTICAL DIRECTION SHALL BE PREDICTED.
DO NOT CLOSE THE ACTUATOR BEFORE INSERTING FPC EVEN AFTER THE CONNECTOR IS MOUNTED ONTO A PCB. CLOSING THE ACTUATOR WITHOUT FPC COULD MAKE THE CONTACT GAP SMALLER, WHICH INCREASES THE FPC INSERTION FORCE.

THIS CONNECTOR HAS CONTACTS ON THE BOTH TOP AND BOTTOM.

 THERE'S A CASE WHICH FPC/FFC RETENTION FORCE DOESN'T FULFILL THE VALUE,
BECAUSE FPC SPECIFICATION AFFECTS THE RESULT OF FPC/FFC RETENTION FORCE.

Note QT:Qualification Test AT:Assurance Test X:Applicable Test

DRAWING NO.

ELC4-159714-05



SPECIFICATION SHEET

PART NO.

FH34SRJ-*S-0.5SH (99)

HIROSE ELECTRIC CO., LTD.

CODE NO

CL580



2/2

FORM HD0011-2-2

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Section 11
Hirose

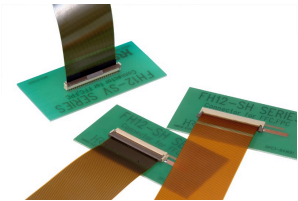
| Designators | Part / Value | Qty |
|-------------|----------------|-----|
| J39 | FH12-30S-0.5SH | 1 |

RoHS evidence document: **FH12-30S-05SH.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

FH12 Serie

FH12-30S-0.5SH(55)CL0586-0525-1-55

RoHS (EU)2015/863



Tatsächliches Produkt kann sich von der Abbildung unterscheiden

[Dokument](#)
[Artikelnummerkonfiguration](#)

| | |
|---------------------|--------------------|
| Artikelnummer | FH12-30S-0.5SH(55) |
| Steckverbindertyp | Buchse |
| Bestückungsabstand | 0.5 mm |
| Steckverbinderlänge | 19.1 mm |
| Steckverbinderhöhe | 2.0 mm |

2D

Fußabdruck (Allegro)

INVENTAR/KAUFEN

KONTAKT

3D-MODELL (IGES)

Fußabdruck (Altium)

ARTIKELNUMMERN NACH SERIEN

3D-MODELL (STEP)

Klicken Sie hier, wenn Sie einen Kauf in Erwägung ziehen. *QA Wie man kauft

Detaillierte Anforderungen

Letztes Update am July. 03, 2024

| | | | |
|---|----------------------|--|------------------|
| Produktinformation | | | |
| Artikelnummer | FH12-30S-0.5SH(55) | CL-Nummer | CL0586-0525-1-55 |
| Grundlegende Informationen | | | |
| Steckverbindertyp | Buchse | Bestückungsabstand | 0.5 mm |
| Steckverbinderlänge | 19.1 mm | Steckverbinderhöhe | 2.0 mm |
| Steckverbinderbreite | 6.4 mm | Kontaktorientierung | Unten |
| ZIF/Non-ZIF/LIF | ZIF | Empfohlenes FFC/FPC | FPC/FFC |
| FPC/FFC Einsteckrichtung | Horizontal | Anwendbare FPC-Kabelstärke (mm) | |
| Tiefenmaß (mm) Aktuator OFFEN | | Tiefenmaß (mm) Aktuator Zustand SCHLIESSEN | |
| Aktuatormethode | Vorne | Anzahl der Kontakte | 30 |
| Anmerkungen zu der Anzahl der Kontakte | | Kontaktabstand | 0.5 mm |
| Anzahl der Reihen (Löt-/Leiterplattenseite) | 1 | Leistungsmerkmale | |
| Übertragungsgeschwindigkeit | | Betriebstemperatur (max.) | 85 °C |
| Betriebstemperatur (min.) | -40 °C | Gleitbereich (XY)(mm) | |
| Effektive Stecklänge (Z) (mm) | | Betriebsspannung (AC) | AC 50.0 V |
| Betriebsspannung (AC) andere | | Betriebsspannung (DC) | DC 50.0 V |
| Betriebsspannung (DC) andere | | Betriebsstrom | 0.5 A |
| Betriebsstrom andere | | RoHS (EU)2015/863 | Konform |
| SVHC | 31th Nicht enthalten | Steckverbindertyp Detailseite | |
| Mechanische Angaben | | | |
| Kontaktbeschichtung | Gold | Beschichtungsstärke | |
| Steckzyklen | 20 | Schutz vor dem Aufsteigen des Lötzinns | |

Elektrische Spezifikationen

| | | | |
|-------------------|--------------|--|--|
| Kontaktwiderstand | 50.0 mΩ Max. | | |
|-------------------|--------------|--|--|

Maßangaben

| | | | |
|----------------|--------|---------------------|--|
| FFC/FPC-Stärke | 0.3 mm | FFC/FPC Anmerkungen | |
|----------------|--------|---------------------|--|

Verpackungsvorschriften

| | | | |
|-------------------------------|----------|----------------|----------|
| Ablebeband | | Verpackungsart | Spule |
| Kaufeinheit | 2000 PCS | Masse | 0.32 g |
| Empfindlich gegen Aufladungen | | Menge / Karton | 2000 PCS |

Andere

| | |
|-----------|--|
| Bemerkung | |
|-----------|--|

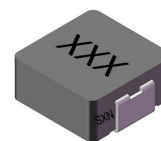
Section 12
SXN

| Designators | Part / Value | Qty |
|-------------|--------------|-----|
| L1,L2 | 2.2uH | 2 |

RoHS evidence document: **2108131530_SXN-Shun-Xiang-Nuo-Elec-SMMS0420-2R2M_C133191.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2 of 28 — RoHS-relevant pages only; the complete document is retained in the project repository.

Molding SMD Power Inductors

SMMS Series Ultra-high current SMT power inductors



◆特征:

- 低直流电阻和超大电流的薄型设计
- 磁屏蔽型抗电磁干扰强适用于高密度安装
- 高可靠性, 通过采用一体成型结构享有卓越的抗震动性
- 由于复合结构, 超低蜂鸣噪声
- 低损耗合金粉末压铸低阻抗, 小寄生电容
- 能效高, 可减少绕线的低直流电阻与磁芯的涡流损耗
- 频率高达 3MHz
- 绝缘最大电压 30VDC
- 符合 RoHS, 无卤和 REACH

◆用途:

- PDA, 笔记本, 台式机, 服务器应用程序
- 大电流 POL 转换器
- 电池供电设备
- 分布式电源系统中的 DC/DC 转换器

◆环境:

- 工作温度: -40℃ 至 +125℃
(包括线圈自身温升)

◆试验设备:

- 电感值: WK3260B 或同等仪器
- 电流: WK3260B+WK3265B
- 直流电阻: Chroma 16502 或同等仪器

◆产品型号:

Features:

- Low RDC and ultra-high current thin design
- Magnetic shielding type, strong anti- electromagnetic Interference, suitable for high- density installation
- High-reliability, High vibration resistance as result of newly developed integral construction
- Ultra Low buzz noise, due to composite construction
- Die-casting by low loss alloy powder low impedance, Small parasitic capacitance
- High efficiency Low DC resistance of winding and low eddy-current loss of the core
- Frequency up to 3MHz
- Absolute maximum voltage 30VDC
- RoHS, Halogen Free and REACH Compliance

Applications:

- PDA , notebook ,desktop ,server applications
- High current POL converters
- Battery powered devices
- DC/DC converters in distributed power systems

Environmental Data:

- Operating Temperature: -40℃ to +125℃
(Including coils self-temperature rise)

Test Equipment:

- L: WK3260B LCR meter or equivalent
- Isat & Irms: WK3260B+WK3265B
- DCR: Chroma 16502 or equivalent

Product Identification:

| <u>SMMS</u> | | <u>0420</u> | <u>100</u> | <u>M</u> | <u>I</u> |
|-------------|----------------------------|---|------------|------------|----------|
| ① | | ② | ③ | ④ | ⑤ |
| ① | | ② | ③ | | |
| 类型 Type | | 外形尺寸(L×W×H) (mm)
External Dimensions (L×W×H)
(mm) | | Inductance | |
| SMMS | 成型贴片功率电感 | 0420 | | 10 uH | |
| | Molding SMT Power Inductor | | | | |
| | | 4.6×4.2×2.0 | | | |

Molding SMD Power Inductors



④

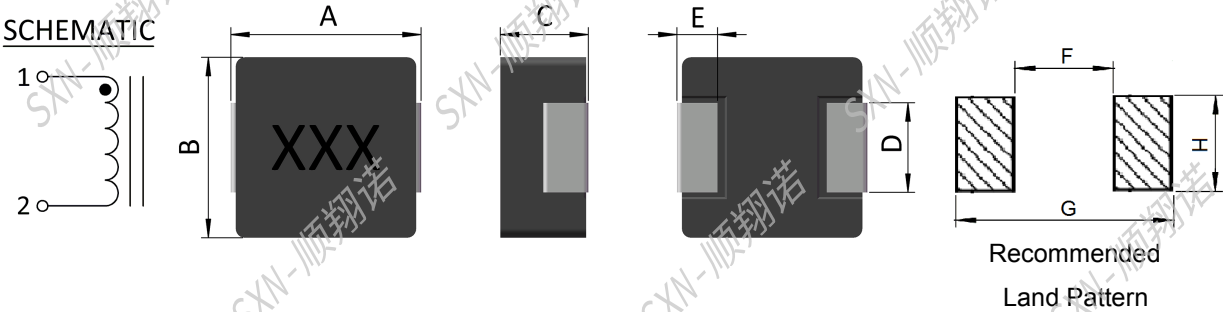
| |
|--------------------------|
| 公差 Inductance Tolerance |
| J:±5%,K: ±10%, L: ±15% |
| M: ±20%,P: ±25%, N: ±30% |

⑤

| | |
|------------|----------------|
| 包装 Packing | |
| B | 散装Bulk Package |
| TF | 编带Tape & Reel |

◆外观尺寸:

Shape and Dimensions(dimensions are in mm):



| Part No | ITEM | | | | | | | |
|----------|------------|------------|---------|----------|----------|------|-------|------|
| | A | B | C | D | E | F | G | H |
| SMMS0420 | 4.60±0.30 | 4.20±0.20 | 2.00Max | 1.50Typ | 0.80 Typ | 2.20 | 5.20 | 2.50 |
| SMMS0520 | 5.50±0.30 | 5.20±0.20 | 2.00Max | 2.30 Typ | 1.20 Typ | 3.00 | 7.00 | 2.50 |
| SMMS0530 | 5.50±0.30 | 5.20±0.20 | 3.00Max | 2.30 Typ | 1.20 Typ | 3.00 | 7.00 | 2.50 |
| SMMS0624 | 7.10±0.30 | 6.60±0.20 | 2.40Max | 3.00 Typ | 1.60 Typ | 3.70 | 8.40 | 3.50 |
| SMMS0630 | 7.10±0.30 | 6.60±0.20 | 3.00Max | 3.00 Typ | 1.60 Typ | 3.70 | 8.40 | 3.50 |
| SMMS0650 | 7.10±0.30 | 6.60±0.20 | 5.00Max | 3.00 Typ | 1.60 Typ | 3.70 | 8.40 | 3.50 |
| SMMS1040 | 11.50 Max | 10.00±0.30 | 4.00Max | 3.00 Typ | 2.00 Typ | 4.10 | 13.60 | 5.40 |
| SMMS1050 | 11.50 Max | 10.00±0.30 | 5.00Max | 3.00 Typ | 2.00 Typ | 5.40 | 13.60 | 4.10 |
| SMMS1350 | 13.80±0.50 | 12.60±0.20 | 5.50Max | 3.70 Typ | 2.50 Typ | 8.00 | 14.60 | 5.00 |
| SMMS1360 | 13.80±0.50 | 12.60±0.20 | 6.50Max | 3.70 Typ | 2.50 Typ | 8.00 | 14.60 | 5.00 |

Section 13

Xinglight

| Designators | Part / Value | Qty |
|-------------|--------------|-----|
| LED1-LED36 | XL-3030RGBC | 36 |

RoHS evidence document: **2402181502_XINGLIGHT-XL-3030RGBC-WS2812B_C5349958.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2 of 18 — RoHS-relevant pages only; the complete document is retained in the project repository.

XINGLIGHT®

Part No. :XL-3030RGBC-WS2812B

XL-3030RGBC-WS2812B**技术数据表** Technical Data Sheet**3030 幻彩 贴片式发光二极管****特点 (Characteristics) :**

- * 外观尺寸 (L/W/H) :3.0*3.0*1.1mm
outline Dimensions (L / w / h): 3.0*3.0*1.1mm
- * 发光颜色及胶体: 七彩幻彩/雾状胶体
Luminous color and colloid: Colorful magic/mist colloid
- * 环保工艺符合 ROHS 要求
Environmental protection products Complied With ROHS Directive
- * 湿气敏感性等级 (MSL) :5a 级
Moisture sensitivity level (MSL) : 5a levels
- * EIA 规范标准包装
EIA standard packaging
- * 适用于 SMT 贴片自动化生产
Suitable for SMT automatic production
- * 适用于红外线回流焊制程
Suitable for infrared reflow soldering process

**应用领域 (Product application) :**

- * LED 全彩发光字灯串
Led full-color luminous word lamp string,
- * LED 全彩模组
led full-color module
- * LED 幻彩软硬灯条, LED 护栏管
Led magic color soft and hard light strip, LED guardrail tube
- * LED 外观, 情景照明
Led appearance, scene lighting
- * LED 异性屏
Led heterosexual screen
- * 各种电子产品, 电器设备跑马灯
All kinds of electronic products, electrical equipment, runninglights





Part No. :XL-3030RGB-WS2812B

WS2812b 功能特点/WS2812b Functional characteristics:

1. 灯珠内部集成高质量外控单线级联恒流 IC 和优质 RGB LED 芯片，体积小，外围简单。

The ball interior integrates high-quality external single-line cascade IC and high-quality RGB LED chips, which are small in size and simple on the periphery.

2. 内置 IC 恒流精度高，内部 RGB 芯片预先分光处理。发光高度一致，白光效果纯正。

BUILT-IN IC constant current high precision, internal RGB chip pre-optical processing. High Degree of uniformity of light, white light effect pure.

3. 整形转发强化技术，单线数据传输，可无限级联。

Plastic forwarding enhancement technology, single-line data transmission, can be cascaded.

4. 数据传输频率 800Kbps/秒，可实现画面刷新速率 30 帧 / 秒时，不小于 1024 点。

The data transmission frequency is 800Kbps per second, and the screen refresh rate can be achieved at 30 frames per second, not less than 1024 points.

5. 输出端口 PWM 控制能够实现 256 级灰度调节，端口扫描频率 1.5KHz / s。

The output port PWM control can achieve 256 levels of grayscale adjustment, and the port scan frequency is 1.5 KHz / S.

6. 采用优化预置 12mA/通道恒流模式，低压驱动级联数量最大化。高恒流精度，片内误差<1.5%，片间误差<3%。Optimized preset 12mA / channel constant current mode is adopted to maximize the number of low-voltage drive cascades. High constant current accuracy, intraslice error < 1.5 %, interslice error < 3 %.

7. 内置低压强化模块，VDD 在 4.5-5.5V 以上 100%正常工作。

With the built-in low-pressure reinforcement module, VDD is 100 % functional above 4.5-5.5 V.

8. 超强数据整形能力：接受完本单元数据自动将后续数据整形输出。

Super data shaping ability: accept this unit data automatically will follow the data shaping output.

Section 14

Yageo

| Designators | Part / Value | Qty |
|------------------------|--------------|-----|
| R1,R2 | 22 0603 | 2 |
| R1_1-R1_36,R15,R20,R25 | 10k 0402 | 39 |
| R8 | 5.6K 0402 | 1 |
| R10,R11,R30 | 390k 0603 1% | 3 |
| R12,R17 | 150k 0603 | 2 |
| R13 | 12k 0603 | 1 |
| R14 | 680k 0603 | 1 |
| R22,R23 | 27 0603 | 2 |
| R24 | 2M2 0603 | 1 |

RoHS evidence document: **2304140030_YAGEO-RC0603JR-0710ML_C141675.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 3 of 10 — RoHS-relevant pages only; the complete document is retained in the project repository.

Innovative Service Around the Globe **YAGEO**

DATA SHEET

GENERAL PURPOSE CHIP RESISTORS

RC_L series

$\pm 0.1\%$, $\pm 0.5\%$, $\pm 1\%$, $\pm 5\%$

Sizes 0075/0100/0201/0402/0603/0805/
1206/1210/1218/2010/2512

RoHS compliant & Halogen free



YAGEO
Phycomp

Product specification — Mar. 7, 2017 V.7



SCOPE

This specification describes RC series chip resistors with lead free terminations made by thick film process.

APPLICATIONS

- All general purpose application

FEATURES

- Halogen Free Epoxy
- RoHS compliant
 - Products with lead free terminations meet RoHS requirements
 - Pb-glass contained in electrodes, resistors element and glass are exempted by RoHS
- Reducing environmentally hazardous wastes
- High component and equipment reliability
- Saving of PCB space
- None forbidden-materials used in products/production

ORDERING INFORMATION - GLOBAL PART NUMBER

Global part numbers are identified by the series, size, tolerance, packing type, temperature coefficient, taping reel and resistance value.

GLOBAL PART NUMBER

RC XXXX X X X XX XXXX L
 (1) (2) (3) (4) (5) (6) (7)

(1) SIZE

0075/0100/0201/0402/0603/0805/1206/1210/1218/2010/2512

(2) TOLERANCE

B = $\pm 0.1\%$

D = $\pm 0.5\%$

F = $\pm 1.0\%$

J = $\pm 5.0\%$ (for jumper ordering, use code of J)

(3) PACKAGING TYPE

R = Paper taping reel

K = Embossed taping reel

S = ESD safe reel (0075/0100 only)

(4) TEMPERATURE COEFFICIENT OF RESISTANCE

- = Based on spec.

(5) TAPING REEL

07= 7 inch dia. Reel

10=10 inch dia. Reel

13=13 inch dia. Reel

7W = 7 inch dia. Reel & 2 x standard power

7N = 7 inch dia. Reel, ESD safe reel (0075/0100 only)

(6) RESISTANCE VALUE

There are 2~4 digits indicated the resistance value.

Letter R/K/M is decimal point

Example:

97R6 = 97.6Ω

9K76 = 9760Ω

1M = $1,000,000\Omega$

(7) DEFAULT CODE

Letter L is the system default code for ordering only.^(Note)

ORDERING EXAMPLE

The ordering code for a RC0402 0.0625W chip resistor value 100K Ω with $\pm 5\%$ tolerance, supplied in 7-inch tape reel of 10,000 units per reel is: RC0402JR-07100KL.

NOTE

1. All our RSMD products meet RoHS compliant and Halogen Free. "LFP" of the internal 2D reel label mentions "Lead Free Process".
2. On customized label, "LFP" or specific symbol can be printed.

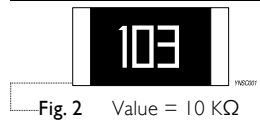
MARKING

RC0075 / RC0100 / RC0201 / RC0402



No Marking

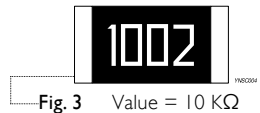
RC0603



E24 series: 3 digits

First two digits for significant figure and 3rd digit for number of zeros

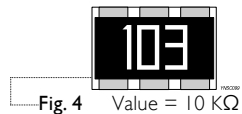
RC0805 / RC1206 / RC1210 / RC2010 / RC2512



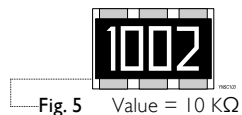
E24/E96 series: 4 digits

First three digits for significant figure and 4th digit for number of zeros

RC1218

E-24 series: 3 digits, $\pm 5\%$

First two digits for significant figure and 3rd digit for number of zeros

Both E-24 and E-96 series: 4 digits, $\pm 1\%$ & $\pm 0.5\%$

First three digits for significant figure and 4th digit for number of zeros

For further marking information, please see special data sheet "Chip resistors marking".

CONSTRUCTION

The resistor is constructed on top of a high-grade ceramic body. Internal metal electrodes are added on each end to make the contacts to the thick film resistive element. The composition of the resistive element is a noble metal imbedded into a glass and covered by a second glass to prevent environmental influences. The resistor is laser trimmed to the rated resistance value. The resistor is covered with a protective epoxy coat, finally the two external terminations (matte tin on Ni-barrier) are added, as shown in Fig.6.

Outlines

For dimensions, please refer to Table I

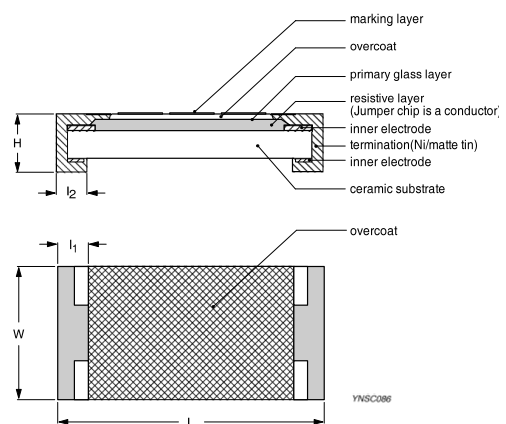


Fig. 6 Chip resistor outlines

Section 15
Yageo

| Designators | Part / Value | Qty |
|-------------|--------------|-----|
| R3,R4,R7,R9 | 5.1K 0402 | 4 |
| R18 | 10M 0603 | 1 |

RoHS evidence document: **50630577_YAGEO-AC0603FR-0710ML_C227547.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 3 of 13 — RoHS-relevant pages only; the complete document is retained in the project repository.

Innovative Service Around the Globe **YAGEO**

DATA SHEET

THICK FILM CHIP RESISTORS
AUTOMOTIVE GRADE

AC series

$\pm 5\%$, $\pm 1\%$, $\pm 0.5\%$, $\pm 0.1\%$ (Low TCR)

Sizes 0201/0402/0603/0805/1206/
1210/1218/2010/2512

RoHS compliant & Halogen free



YAGEO

Product specification – November 17, 2023 V.1.1



YAGEOProduct specification

| |
|----|
| 2 |
| 13 |

Chip Resistor Surface Mount | AC | SERIES | **0201 to 2512**

SCOPE

This specification describes AC0201 to AC2512 chip resistors with lead-free terminations made by thick film process.

APPLICATIONS

- All general purpose applications
- Car electronics, industrial application

FEATURES

- AEC-Q200 qualified
- Moisture sensitivity level: MSL 1
- AC series soldering is compliant with J-STD-020D
- Halogen free epoxy
- RoHS compliant
 - Products with lead-free terminations meet RoHS requirements
 - Pb-glass contained in electrodes, resistor element and glass are exempted by RoHS
- Reduce environmentally hazardous waste
- High component and equipment reliability
- The resistors are 100% performed by automatic optical inspection prior to taping.

ORDERING INFORMATION - GLOBAL PART NUMBER

Part number is identified by the series name, size, tolerance, packaging type, temperature coefficient, taping reel and resistance value.

GLOBAL PART NUMBER

AC XXXX X X X XX XXXX L
 (1) (2) (3) (4) (5) (6) (7)

(1) SIZE

0201 / 0402 / 0603 / 0805 / 1206 / 1210 / 1218 / 2010 / 2512

(2) TOLERANCE

B = $\pm 0.1\%$ for low TCR F = $\pm 1\%$
 D = $\pm 0.5\%$ J = $\pm 5\%$ (for Jumper ordering, use code of J)

(3) PACKAGING TYPE

R = Paper taping reel K = Embossed taping reel

(4) TEMPERATURE COEFFICIENT OF RESISTANCE

— = Base on spec E = ± 50 ppm/ $^{\circ}\text{C}$

(5) TAPING REEL

07 = 7 inch dia. Reel & Standard power 7W = 7 inch dia. Reel & 2 x standard power
 13 = 13 inch dia. Reel 3W = 13 inch dia. Reel & 2 x standard power

(6) RESISTANCE VALUE

1 Ω to 22 M Ω
 There are 2~4 digits indicated the resistance value. Letter R/K/M is decimal point, no need to mention the last zero after R/K/M, e.g. 1K2, not 1K20.
 Detailed coding rules of resistance are shown in the table of "Resistance rule of global part number".

(7) DEFAULT CODE

Letter B is for bulk AOI ordering only. (Note)

| Resistance rule of global part number | |
|---------------------------------------|---|
| Resistance coding rule | Example |
| XXXX
(1 to 9.76 Ω) | 1R = 1 Ω
1R5 = 1.5 Ω
9R76 = 9.76 Ω |
| XXRX
(10 to 97.6 Ω) | 10R = 10 Ω
97R6 = 97.6 Ω |
| XXXR
(100 to 976 Ω) | 100R = 100 Ω
976R = 976 Ω |
| XKXX
(1 to 9.76 K Ω) | 1K = 1,000 Ω
9K76 = 9760 Ω |
| XMXX
(1 to 9.76 M Ω) | 1M = 1,000,000 Ω
9M76 = 9,760,000 Ω |
| XXMX
(10 M Ω) | 10M = 10,000,000 Ω |

ORDERING EXAMPLE

The ordering code for an AC0402 chip resistor, value 100 K Ω with $\pm 1\%$ tolerance, supplied in 7-inch tape reel is: AC0402FR-07100KL.

NOTE

1. All our R-Chip products are RoHS compliant and Halogen free. "LFP" of the internal 2D reel label states "Lead-Free Process".
2. On customized label, "LFP" or specific symbol can be printed.
3. AC series with $\pm 0.5\%$ tolerance is also available. For further information, please contact sales.

YAGEO

Product specification

3

Chip Resistor Surface Mount

AC

SERIES

0201 to 2512

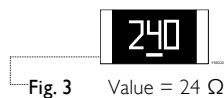
13

MARKING**AC0201 / AC0402**

No marking

AC0603 / AC0805 / AC1206 / AC1210 / AC2010 / AC2512Value = 10 K Ω E-24 series: 3 digits, $\pm 5\%$

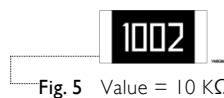
First two digits for significant figure and 3rd digit for number of zeros

AC0603Value = 24 Ω E-24 series: 3 digits, $\pm 1\%$ & $\pm 0.5\%$

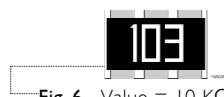
One short bar under marking letter

Value = 12.4 K Ω E-96 series: 3 digits, $\pm 1\%$ & $\pm 0.5\%$

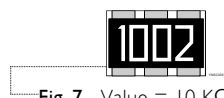
First two digits for E-96 marking rule and 3rd letter for number of zeros

AC0805 / AC1206 / AC1210 / AC2010 / AC2512Value = 10 K Ω Both E-24 and E-96 series: 4 digits, $\pm 1\%$ & $\pm 0.5\%$

First three digits for significant figure and 4th digit for number of zeros

AC1218Value = 10 K Ω E-24 series: 3 digits, $\pm 5\%$

First two digits for significant figure and 3rd digit for number of zeros

Value = 10 K Ω Both E-24 and E-96 series: 4 digits, $\pm 1\%$ & $\pm 0.5\%$

First three digits for significant figure and 4th digit for number of zeros

NOTE

For further marking information, please refer to data sheet "Chip resistors marking". Marking of AC series is the same as RC series.



Nov. 17, 2023 V.1.1

www.yageo.com

Section 16
Panasonic

| Designators | Part / Value | Qty |
|-------------|--------------|-----|
| SW2,SW3 | EVQPUC02K | 2 |

RoHS evidence document: [panasonic_rohs_2ma_2000379.pdf](#)
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863



120003792 Page:1/2

Issue date:2024/10/11

Panasonic Industry Co., Ltd.

PolyFabriq

Non-use Report for Chemical Substances Contained in Products

Application start time : 7/22/2018

Our company (Panasonic Industry Co., Ltd.) hereby report that our products listed in Item I that delivered or to be delivered to your company, subsidiaries or affiliated companies that specified do not contain the substances in Item II below or do not exceed the specified values under RoHS directive as the result of our intentional conduct. The criteria of exemption to be applied to the subject materials and components shall be conformed to the RoHS directive (2011/65/EU, (EU) 2015/863).

I . The Applicable Products: Please refer to the next page.

II . The Applicable Substances (10 substances subject to the RoHS Directive)

| No. | Substance name | Regulated concentration/
Threshold (Homogeneous material unit) |
|-----|---------------------------------------|---|
| 1 | Cadmium and its compounds | 0.01wt%(100ppm) |
| 2 | Lead and its compounds | 0.1wt%(1,000ppm) |
| 3 | Mercury and its compounds | 0.1wt%(1,000ppm) |
| 4 | Hexavalent chromium compounds | 0.1wt%(1,000ppm) |
| 5 | Polybrominated biphenyls (PBBs) | 0.1wt%(1,000ppm) |
| 6 | Polybrominated diphenylethers (PBDEs) | 0.1wt%(1,000ppm) |
| 7 | Di(2-ethylhexyl)phthalate (DEHP) | 0.1wt%(1,000ppm) |
| 8 | Butyl benzyl phthalate (BBP) | 0.1wt%(1,000ppm) |
| 9 | Dibutyl phthalate (DBP) | 0.1wt%(1,000ppm) |
| 10 | Diisobutyl phthalate (DIBP) | 0.1wt%(1,000ppm) |

*1 In this report, EU RoHS Directive means the EU RoHS directive applicable as of the date above.

*2 In the event that your company suffers actual damage because restricted substances in the applicable products exceed the threshold by reason attributable to us, we will assume responsibility only as required by the written agreement(s) of sales and purchase with your company, subject to the limitations therein, and, if such written agreement(s) do not exist, only as required by applicable environmental laws and regulations.
In no event will we be liable for indirect, consequential and/or punitive damages.

Masato Yamasaki, Manager

Quality & Environment Planning Section
Quality Assurance Department
Interface Devices Business Unit
Electromechanical Control Business Division
Panasonic Industry Co., Ltd.

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120003792 Page:2/2
Issue date:2024/10/11

The Applicable Products

| No. | Product Name | Our part No. | Our product No. |
|-----|----------------------------------|--------------|-----------------|
| 1 | Small-sized Side-operational SMD | EVQPUC02K | EVQPUC02K |

- EOF -

Section 17

Kailh

| Designators | Part / Value | Qty |
|----------------|-----------------------|-----|
| SW_K_1-SW_K_36 | Kailh CPG151101S11-16 | 36 |

RoHS evidence document: **2208291600_Kailh-CPG151101S11-16_C5156480.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 9, 10 of 10 — RoHS-relevant pages only; the complete document is retained in the project repository.



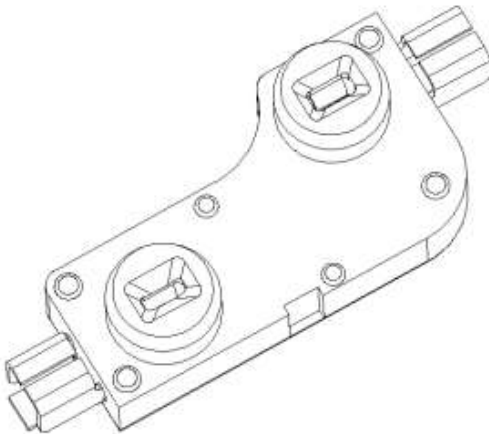


凱華電子
KAIHUA EEELETRONICS

Document Number:

KH-PS2206-43

产品规格书
Product Specification



P/N:

CPG151101S11-16

Title :

1511 Connector

| Rev. | ECN | Release and Revision Description: | Prepared By /Date: | Checked By/Date: | Approved By/Date: |
|------|-------|-----------------------------------|--------------------|------------------|-------------------|
| A | _____ | New releasing 初版发行 | 王枝轩 2022-06-27 | 胡远峰 2022-06-27 | 郑建军 2022-06-27 |
| | | | | | |
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| | | | | | |
|---|---|--------------------------------|----------------------------------|-------------------|---------------------|
| 
Kailh | 凱華電子
KAIHUA EEELETRONICS | Product Specification | | | |
| | | P/N:
CPG151101S11-16 | DOC. No.:
KH-PS2206-43 | Rev.:
A | Page:
9/9 |

11.2 Notes 注意点

(1) Please be cautious not to give excessive static load connector.
注意不要施加超负荷的压力或晃动连接器.

(2) Connector be careful not to stack up P. W. B. after switches were soldered.
连接器焊接以后,印刷基板注意不要叠放.

(3) Preservation under high temperature and high humidity or corrosive gas should be avoided
Especially. When you need to preserve for a long period, do not open the carton.
保管时尤其应注意避开高湿高温和有腐蚀性气体的环境.如需长时间保存,请不要打开包装箱.

(4) The standard storage period is 3 months, with maximum up to 6months, preferably to be used as
soon as possible. After opening the package, you should put the remaining switches in a plastic bag
to prevent from damp and corrosive gas.
保存标准为 3 个月, 限度为 6 个月以内, 请尽早使用. 打开包装后, 有剩余品时, 应将剩余部分以胶袋
包装好以同外界隔离, 请进行合适的防湿, 防腐蚀气体等处理后进行保管.

(5) This Product Specification is considered as the technical agreement on product between the
receiving customer and Kailh. Any information on Product Catalogue which is in conflict with or
different from the corresponding information of this document is considered as invalid.
该规格书为客户与凯华公司产品在技术方面的共识,其他相关数据上与该规格书不一致的内容都是无效的.

(6) If customer issue purchase orders without confirmation by signature of this specification after receipt,
such confirmation will be considered as granted upon receipt of the first purchase order.
如果顾客收到规格书后没有信息反馈而直接向我公司订货, 我们将认为贵客已接受此规格书.

(7) If there is no order or no request for new specification after 1 year upon this specification is issued,
the specification will be regarded as invalid.
本产品规格书从生效日起 1 年后, 如果没有订货或再次申请最新规格书时请做无效处理.

(8) Products meet the ROHS & REACH environmental management substances control standards
产品满足 **ROHS & REACH** 环境管理物质管制标准

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Section 18
TI

| Designators | Part / Value | Qty |
|-------------|--------------|-----|
| U1,U2 | TLV62569DBVR | 2 |

RoHS evidence document: **TexasTLV62569DBVR.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

PACKAGE OPTION ADDENDUM



www.ti.com

15-Dec-2017

PACKAGING INFORMATION

| Orderable Device | Status
(1) | Package Type | Package Drawing | Pins | Package Qty | Eco Plan
(2) | Lead/Ball Finish
(6) | MSL Peak Temp
(3) | Op Temp (°C) | Device Marking
(4/5) | Samples |
|------------------|---------------|--------------|-----------------|------|-------------|----------------------------|-------------------------|----------------------|--------------|-------------------------|---------|
| TLV62569DBVR | ACTIVE | SOT-23 | DBV | 5 | 3000 | Green (RoHS
& no Sb/Br) | CU NIPDAU | Level-1-260C-UNLIM | -40 to 125 | 16AF | Samples |
| TLV62569DBVT | ACTIVE | SOT-23 | DBV | 5 | 250 | Green (RoHS
& no Sb/Br) | CU NIPDAU | Level-1-260C-UNLIM | -40 to 125 | 16AF | Samples |
| TLV62569DRLR | ACTIVE | SOT-5X3 | DRL | 6 | 3000 | Green (RoHS
& no Sb/Br) | CU SN | Level-1-260C-UNLIM | -40 to 125 | 19D | Samples |
| TLV62569DRLT | ACTIVE | SOT-5X3 | DRL | 6 | 250 | Green (RoHS
& no Sb/Br) | CU SN | Level-1-260C-UNLIM | -40 to 125 | 19D | Samples |
| TLV62569PDCCR | ACTIVE | SOT-23-THIN | DDC | 6 | 3000 | Green (RoHS
& no Sb/Br) | CU SN | Level-1-260C-UNLIM | -40 to 125 | 6D9 | Samples |
| TLV62569PDDET | ACTIVE | SOT-23-THIN | DDC | 6 | 250 | Green (RoHS
& no Sb/Br) | CU SN | Level-1-260C-UNLIM | -40 to 125 | 6D9 | Samples |
| TLV62569PDRLR | ACTIVE | SOT-5X3 | DRL | 6 | 3000 | Green (RoHS
& no Sb/Br) | CU SN | Level-1-260C-UNLIM | -40 to 125 | 19E | Samples |
| TLV62569PDRLT | ACTIVE | SOT-5X3 | DRL | 6 | 250 | Green (RoHS
& no Sb/Br) | CU SN | Level-1-260C-UNLIM | -40 to 125 | 19E | Samples |

(1) The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBsolete: TI has discontinued the production of the device.

(2) **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".

RoHS Exempt: TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.

Green: TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of <=1000ppm threshold. Antimony trioxide based flame retardants must also meet the <=1000ppm threshold requirement.

(3) **MSL, Peak Temp. -** The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(4) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

Section 19
Silergy

| Designators | Part / Value | Qty |
|-------------|--------------|-----|
| U3 | SY6280AAC | 1 |

RoHS evidence document: **ROHS-2024-SilergyCorp.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 of 14 — RoHS-relevant pages only; the complete document is retained in the project repository.

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Client Name: Silergy Corp.

Client Address: Oleander Way, 802 West Bay Road, P.O. Box 32052, Grand Cayman KY1-1208, Cayman Islands

Sample Name: Surface Mounted Electric Component

Model No.: DFN4x4

Client Ref. Information: AQFN/BGA/COF/COMBO/CPC/CSP/DFN/DIP/D-SMD/HDIP/GEM/LQFP/LGA/MDFN/MLGA/MQFN/MSOP/ODFN/QFN/PDFN/P-DIP/PLCC/SMB/SMC/SMD/SOD/SOP/SOT/SSOP/TO92/TO220/TO220F/TO252/TO263/TQFP/TSOP/TSSOP/ TSOT See attachment

The above sample(s) and information were provided by the client.

THIS REPORT IS TO SUPERSEDE TEST REPORT NO.SHAEC23021346804, DATE: Jan 03, 2024.

SGS Job No.: SHP23-022732

Sample Receiving Date: Dec 18, 2023

Testing Period: Dec 18, 2023 ~ Jan 03, 2024

Test Requested: Select test(s) as requested by the client.

Test Method(s): Please refer to next page(s).

Test Result(s): Please refer to next page(s).

| Test Requirement | Conclusion |
|--|-------------|
| EU RoHS Directive (EU) 2015/863 amending Annex II to Directive 2011/65/EU- Lead, Mercury, Cadmium, Hexavalent chromium, Polybrominated biphenyls (PBBs), Polybrominated diphenyl ethers (PBDEs), Bis(2-ethylhexyl) phthalate (DEHP), Butyl benzyl phthalate (BBP), Dibutyl phthalate (DBP) and Diisobutyl phthalate (DIBP) | Pass |
| Halogen | See Results |
| Perfluorooctanesulfonate (PFOS) and its derivatives and Perfluorooctanoic Acid (PFOA). | See Results |
| AfPS GS 2019:01 PAK-Polycyclic Aromatic Hydrocarbons (PAHs) | See Results |

Signed for and on behalf of
SGS-CSTC Standards Technical Services (Shanghai) Co., Ltd.

Dora Hu
Approved Signatory

scan to see the report



56E6112B



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Test Result(s):

Test Part Description:

| SN ID | Sample No. | SGS Sample ID | Description |
|-------|------------|-------------------------|-----------------------------|
| SN1 | A1 | SHA23-0213468-0001.C001 | Black body with silvery pin |

Remarks:

- (1) 1 mg/kg = 1 ppm = 0.0001%
- (2) MDL = Method Detection Limit
- (3) ND = Not Detected (< MDL)
- (4) “-” = Not Regulated

EU RoHS Directive (EU) 2015/863 amending Annex II to Directive 2011/65/EU- Lead, Mercury, Cadmium, Hexavalent chromium, Polybrominated biphenyls (PBBs), Polybrominated diphenyl ethers (PBDEs), Bis(2-ethylhexyl) phthalate (DEHP), Butyl benzyl phthalate (BBP), Dibutyl phthalate (DBP) and Diisobutyl phthalate (DIBP)

Test Method: With reference to IEC 62321-4:2013+AMD1:2017, IEC 62321-5:2013, IEC 62321-7-2:2017, IEC 62321-6:2015 and IEC 62321-8:2017, analysis was performed by ICP-OES, AAS, UV-Vis and GC-MS.

| Test Item(s) | Limit | Unit(s) | MDL | A1 |
|------------------------------------|-------|---------|-----|----|
| Cadmium(Cd) | 100 | mg/kg | 2 | ND |
| Lead (Pb) | 1000 | mg/kg | 2 | 5 |
| Mercury (Hg) | 1000 | mg/kg | 2 | ND |
| Hexavalent Chromium (Cr(VI)) | 1000 | mg/kg | 8 | ND |
| Polybromobiphenyl (PBBs) | 1000 | mg/kg | - | ND |
| Monobromobiphenyl (MonoBB) | - | mg/kg | 5 | ND |
| Dibromobiphenyl (DiBB) | - | mg/kg | 5 | ND |
| Tribromobiphenyl (TriBB) | - | mg/kg | 5 | ND |
| Tetrabromobiphenyl (TetraBB) | - | mg/kg | 5 | ND |
| Pentabromobiphenyl (PentaBB) | - | mg/kg | 5 | ND |
| Hexabromobiphenyl (HexaBB) | - | mg/kg | 5 | ND |
| Heptabromobiphenyl (HeptaBB) | - | mg/kg | 5 | ND |
| Octabromobiphenyl (OctaBB) | - | mg/kg | 5 | ND |
| Nonabromobiphenyl (NonaBB) | - | mg/kg | 5 | ND |
| Decabromobiphenyl (DecaBB) | - | mg/kg | 5 | ND |
| Polybromodiphenyl ether(PBDEs) | 1000 | mg/kg | - | ND |
| Monobromodiphenylether (MonoBDE) | - | mg/kg | 5 | ND |
| Dibromodiphenylether (DiBDE) | - | mg/kg | 5 | ND |
| Tribromodiphenylether (TriBDE) | - | mg/kg | 5 | ND |
| Tetrabromodiphenylether (TetraBDE) | - | mg/kg | 5 | ND |
| Pentabromodiphenylether (PentaBDE) | - | mg/kg | 5 | ND |
| Hexabromodiphenylether (HexaBDE) | - | mg/kg | 5 | ND |
| Heptabromodiphenylether (HeptaBDE) | - | mg/kg | 5 | ND |
| Octabromodiphenylether (OctaBDE) | - | mg/kg | 5 | ND |
| Nonabromodiphenylether (NonaBDE) | - | mg/kg | 5 | ND |
| Decabromodiphenylether (DecaBDE) | - | mg/kg | 5 | ND |
| Dibutyl Phthalate(DBP) | 1000 | mg/kg | 50 | ND |
| Benzyl Butyl Phthalate(BBP) | 1000 | mg/kg | 50 | ND |



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| Test Item(s) | Limit | Unit(s) | MDL | A1 |
|------------------------------------|-------|---------|-----|----|
| Bis-(2-ethylhexyl) Phthalate(DEHP) | 1000 | mg/kg | 50 | ND |
| Diisobutyl Phthalate(DIBP) | 1000 | mg/kg | 50 | ND |

Notes:

- (1) The maximum permissible limit is quoted from RoHS Directive (EU) 2015/863.
- (2) IEC 62321 series is equivalent to EN 62321 series.
- (3) The restriction of DEHP, BBP, DBP and DIBP shall apply to medical devices, including in vitro medical devices, and monitoring and control instruments, including industrial monitoring and control instruments, from 22 July 2021.

Halogen

Test Method: With reference to EN 14582:2016, analysis was performed by IC.

| Test Item(s) | Unit(s) | MDL | A1 |
|--------------|---------|-----|----|
| Fluorine(F) | mg/kg | 20 | ND |
| Chlorine(Cl) | mg/kg | 50 | ND |
| Bromine(Br) | mg/kg | 50 | ND |
| Iodine(I) | mg/kg | 50 | ND |

Perfluorooctanesulfonate (PFOS) and its derivatives and Perfluorooctanoic Acid (PFOA).

Test Method: With reference to CEN/TS 15968:2010, analysis was performed by HPLC-MS or LC-MS/MS.

| Test Item(s) | CAS No. | Unit(s) | MDL | A1 |
|---|------------|---------|-----|----|
| Perfluorooctane sulfonates (PFOS) and its derivatives | - | mg/kg | - | ND |
| Perfluorooctanesulfonic acid (PFOS)^ | 1763-23-1 | mg/kg | 10 | ND |
| N-ethylperfluoro-1-octanesulfonamide (EtFOSA) | 4151-50-2 | mg/kg | 10 | ND |
| N-methylperfluoro-1-octanesulfonamide (MeFOSA) | 31506-32-8 | mg/kg | 10 | ND |
| 2-(N-ethylperfluoro-1-octanesulfonamido)-ethanol (EtFOSE) | 1691-99-2 | mg/kg | 10 | ND |
| 2-(N-methylperfluoro-1-octanesulfonamido)- ethanol (MeFOSE) | 24448-09-7 | mg/kg | 10 | ND |
| Perfluorooctane Sulfonamide (PFOSA) | 754-91-6 | mg/kg | 10 | ND |
| Perfluorooctanoic Acid (PFOA) and its salts+ | - | mg/kg | 10 | ND |

Notes:

- (1) + PFOA refer to its salts including PFOA-Na (CAS No.: 335-95-5), PFOA-K (CAS No.: 2395-00-8), PFOA-Ag (CAS No.: 335-93-3), PFOA-F (CAS No.: 335-66-0) and APFO (CAS No.: 3825-26-1);
- (2) ^ PFOS including PFOS-K (CAS No.: 2795-39-3), PFOS-Li (CAS No.: 29457-72-5), PFOS-NH₄ (CAS No.: 29081-56-9), PFOS-NH(OH)₂ (CAS No.: 70225-14-8), PFOS-N(C₂H₅)₄ (CAS No.: 56773-42-3), PFOS-DDA (CAS No.: 251099-16-8) and POSF (CAS No.: 307-35-7).

AfPS GS 2019:01 PAK-Polycyclic Aromatic Hydrocarbons (PAHs)

Test Method: With reference to AfPS GS 2019:01 PAK, analysis was performed by GC-MS.



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| Test Item(s) | CAS No. | Unit(s) | MDL | A1 |
|---|----------|---------|-----|----|
| Benzo(a)pyrene(BaP) | 50-32-8 | mg/kg | 0.1 | ND |
| Benzo(e)pyrene(BeP) | 192-97-2 | mg/kg | 0.1 | ND |
| Benzo(a)anthracene(BaA) | 56-55-3 | mg/kg | 0.1 | ND |
| Benzo(b)Fluoranthene(BbF) | 205-99-2 | mg/kg | 0.1 | ND |
| Benzo(j)fluoranthene(BjF) | 205-82-3 | mg/kg | 0.1 | ND |
| Benzo(k)Fluoranthene(BkF) | 207-08-9 | mg/kg | 0.1 | ND |
| Chrysene(CHR) | 218-01-9 | mg/kg | 0.1 | ND |
| Dibenzo(a,h)Anthracene(DBA) | 53-70-3 | mg/kg | 0.1 | ND |
| Benzo(g,h,i)perylene(BPE) | 191-24-2 | mg/kg | 0.1 | ND |
| Indeno(1,2,3-c,d)pyrene(IPY) | 193-39-5 | mg/kg | 0.1 | ND |
| Phenanthrene(PHE) | 85-01-8 | mg/kg | 0.1 | ND |
| Pyrene(PYR) | 129-00-0 | mg/kg | 0.1 | ND |
| Anthracene(ANT) | 120-12-7 | mg/kg | 0.1 | ND |
| Fluoranthene(FLT) | 206-44-0 | mg/kg | 0.1 | ND |
| Sum of Phenanthrene(PHE),
Pyrene(PYR), Anthracene(ANT),
Fluoranthene(FLT) | - | mg/kg | - | ND |
| Naphthalene(NAP) | 91-20-3 | mg/kg | 0.1 | ND |
| Sum of 15 PAHs | - | mg/kg | - | ND |
| Material Category | - | - | - | - |

Notes:

AfPS (German commission for Product Safety) : PAHs requirements

| Parameter | Category 1 | Category 2 | | Category 3 | |
|----------------------------|--|--|-------------------------------|---|-------------------------------|
| | Materials intended to be placed in the mouth, or materials coming into long-term contact with skin (more than 30s) during the intended use | Materials not covered by category 1, coming into long-term contact (more than 30s) or short-term repetitive contact ^c with skin during the intended or foreseeable use ^d . | | Materials covered neither by category 1 nor by category 2, coming into short-term contact (up to 30s) with skin during the intended or foreseeable use. | |
| | -in toys according to Directive 2009/48/EC or
-for the use by children ^{a,b} up to 3 years of age. | a.
use by children | b.
other consumer products | a.
use by children | b.
other consumer products |
| Benzo(a)pyrene (BaP) mg/kg | < 0.2 | < 0.2 | < 0.5 | < 0.5 | < 1 |
| Benzo(e)pyrene (BeP) mg/kg | < 0.2 | < 0.2 | < 0.5 | < 0.5 | < 1 |



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| | | | | | |
|---|---------|---------|----------|----------|----------|
| Benzo(a)anthracene (BaA) mg/kg | < 0.2 | < 0.2 | < 0.5 | < 0.5 | < 1 |
| Benzo(b)fluoranthene (BbF) mg/kg | < 0.2 | < 0.2 | < 0.5 | < 0.5 | < 1 |
| Benzo(j)fluoranthene (BjF) mg/kg | < 0.2 | < 0.2 | < 0.5 | < 0.5 | < 1 |
| Benzo(k)fluoranthene (BkF)mg/kg | < 0.2 | < 0.2 | < 0.5 | < 0.5 | < 1 |
| Chrysene (CHR) mg/kg | < 0.2 | < 0.2 | < 0.5 | < 0.5 | < 1 |
| Dibenzo(a,h)anthracene (DBA) mg/kg | < 0.2 | < 0.2 | < 0.5 | < 0.5 | < 1 |
| Benzo(g,h,i)perylene (BPE) mg/kg | < 0.2 | < 0.2 | < 0.5 | < 0.5 | < 1 |
| Indeno(1,2,3-cd)pyrene (IPY) mg/kg | < 0.2 | < 0.2 | < 0.5 | < 0.5 | < 1 |
| Phenanthrene (PHE), pyrene (PYR), anthracene (ANT), fluoranthene (FLT), mg/kg | < 1 Sum | < 5 Sum | < 10 Sum | < 20 Sum | < 50 Sum |
| Naphthalene (NAP) mg/kg | < 1 | < 2 | | < 10 | |
| Sum of 15 PAHs | <1 | < 5 | < 10 | < 20 | < 50 |

Notes:

^a A "Child" is legally defined as a person before reaching the age of 14 years.

^b Use by children includes both active and passive contact by children.

^c Definition "short-term repetitive contact" taken from REACH Annex XVII entry 50 amendment (Regulation (EC) No.1272/2013)

^d According to the definition of the German Product Safety Act (ProdSG) (chapter 1 Article 2 No. 28) "foreseeable use" shall mean the use of a product in a manner that the person placing it on the market, has not intended, but which could be reasonably foreseeable.

Remark:

The German committee on Product Safety (AfPS) published a new PAHs document (AfPS GS 2019:01 PAK) on April 10, 2020, which will be binding for the issue of GS mark certificate from July 1, 2020.

This report updates Client Ref. Information.

Unless otherwise stated, the decision rule for conformity reporting is based on Binary Statement for Simple Acceptance Rule (w=0) stated in ILAC-G8:09/2019.



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**Test Report**
ATTACHMENTS

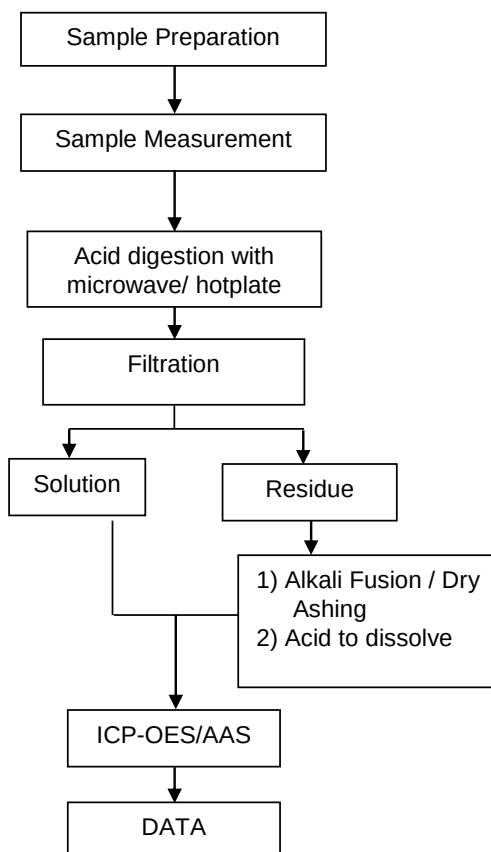
No.: SHAEC23021346804_1

Date: Jan 05, 2024

Page 6 of 14

Elements Testing Flow Chart

These samples were dissolved totally by pre-conditioning method according to below flow chart.



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Test Report

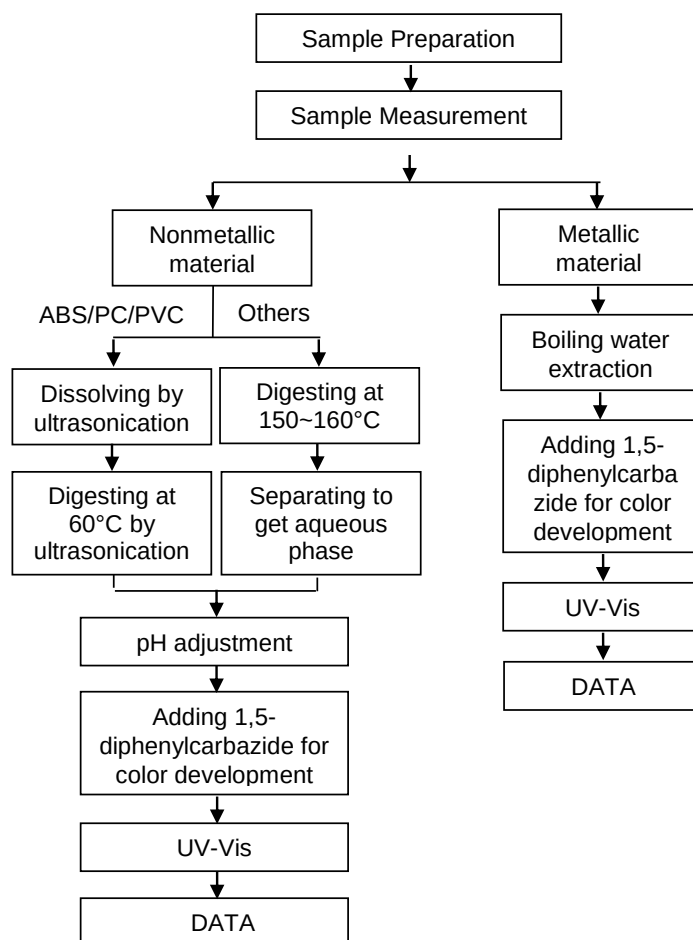
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Hexavalent Chromium (Cr(VI)) Testing Flow Chart



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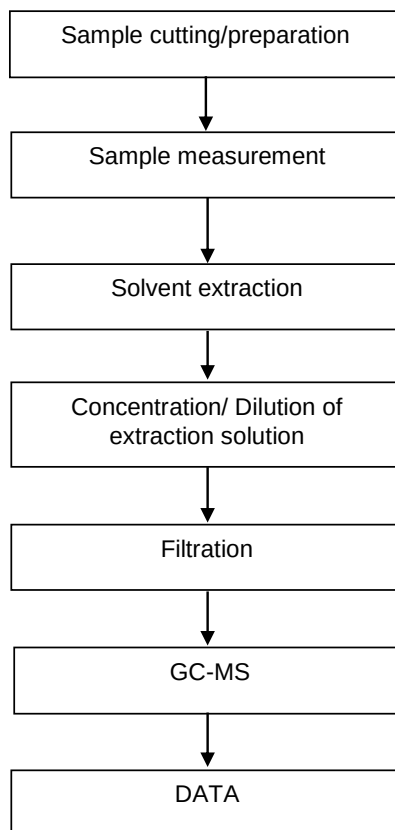
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PBBs/PBDEs Testing Flow Chart

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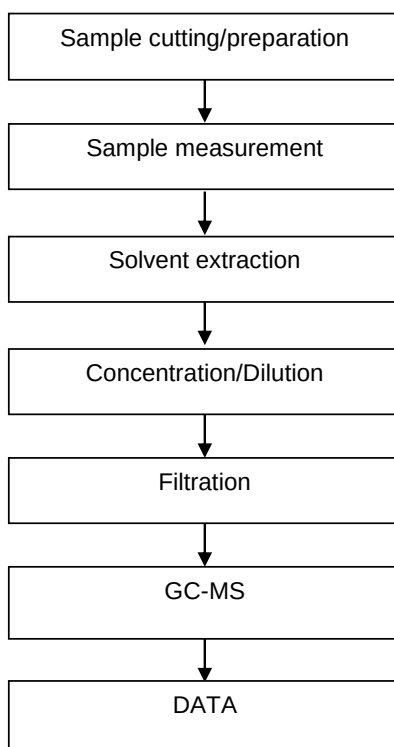
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Phthalates Testing Flow Chart

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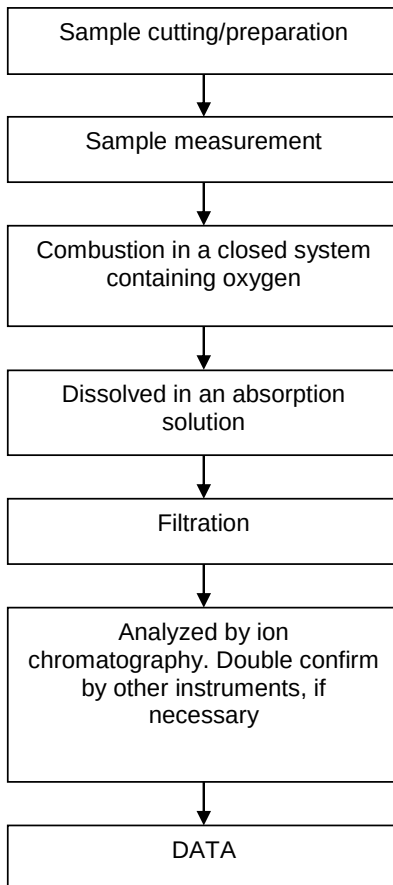
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Halogen Testing Flow Chart

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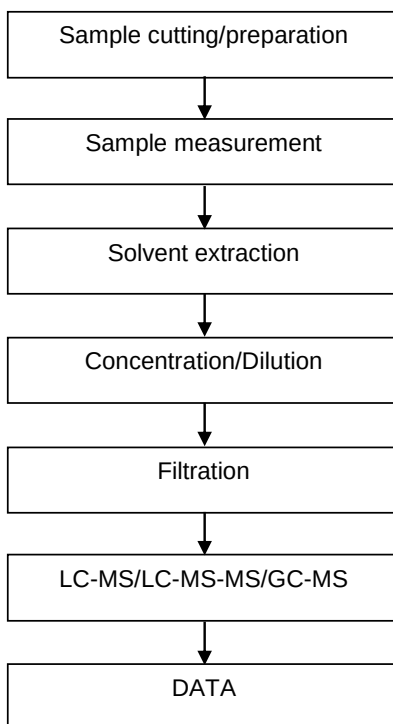
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PFASs/ PFOS/PFOA Testing Flow Chart

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Section 20

Nexperia

| Designators | Part / Value | Qty |
|-------------|------------------|-----|
| U4-U8 | 74HC595BQ | 5 |
| U12-U25 | 74AHC1G125GW,125 | 14 |

RoHS evidence document: **Nexperia - Statement on RoHS 20241008.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863



October 2024

CERTIFICATE OF COMPLIANCE – RoHS Declaration –

Nexperia B.V. declares that its semiconductor products, including all homogeneous materials used in their composition, are designed to be RoHS-compliant by meeting the requirements defined under Directive 2011/65/EU of 2011-07-21, amended by Directive (EU) 2015/863 of 2015-03-31, on the restriction of the use of certain hazardous substances in electrical and electronic equipment (EEE):

| RoHS-Restricted Substance | Permissible Limit on the Homogeneous Material Level |
|--|---|
| Cadmium (Cd) | 100 ppm (0.01 weight %) |
| Mercury (Hg) | 1000 ppm (0.1 weight %) |
| Hexavalent chromium (Cr ⁶⁺) | 1000 ppm (0.1 weight %) |
| Lead (Pb) | 1000 ppm (0.1 weight %) |
| Polybrominated biphenyls (PBBs) | 1000 ppm (0.1 weight %) |
| Polybrominated diphenyl ethers (PBDEs)
including decabromodiphenylether (decaBDE) | 1000 ppm (0.1 weight %) |
| Bis(2-ethylhexyl) phthalate (DEHP) | 1000 ppm (0.1 weight %) |
| Butyl benzyl phthalate (BBP) | 1000 ppm (0.1 weight %) |
| Dibutyl phthalate (DBP) | 1000 ppm (0.1 weight %) |
| Diisobutyl phthalate (DIBP) | 1000 ppm (0.1 weight %) |

All Nexperia products are EU RoHS-compliant and bear the “RoHS-compliant” logo on the box label. Nexperia devices that contain no more than 0.1 % lead by weight per homogeneous material are assigned the RHF indicators G or D. These products can be recognized by the “Lead-free” logo on the box label. Nexperia products that contain RoHS-exempted lead are assigned the RHF indicators E or H and may make use of the following RoHS exemptions:

| RoHS Exemption | RoHS Exemption Description |
|----------------|--|
| 7(a) | Lead in high melting temperature type solders (i.e. lead-based alloys containing 85% by weight or more lead) |
| 7(c)-I | Electrical and electronic components containing lead in a glass or ceramic other than dielectric ceramic in capacitors, e.g. piezoelectronic devices, or in a glass or ceramic matrix compound |

The RHF indicator can be found on the respective product page, such as on the [product page of BAS16-Q](#) under “Environmental information”. Table 1 in the Appendix to this document lists all released packages using RHF indicators E or H and the information on whether the respective package makes use of RoHS exemption 7(a) and/or 7(c)-I, respectively.



To facilitate compliance checks and inform the public, we make product composition declarations available on our public website. The signature below verifies that the above statements, including any product composition data, are valid and accurate to the best of our knowledge and belief. Nexperia has implemented robust systems to ensure that our products comply with global environmental regulations and laws. In the event of any issues arising from information in this document, Nexperia's standard terms and conditions of sale shall apply, unless alternate contracts have been agreed upon in writing by both parties. For further details, please reach out to your Nexperia contact, get in touch with the nearest Sales office, or fill in the technical support form (after registration).

Dr. Timo Stein
Manager ECO-Products
Nexperia B.V.



APPENDIX

Table 1: Packages of Nexperia products that contain lead in concentrations exceeding 0.1 % w/w of any homogeneous material along with the applicability of RoHS exemption 7(a) and/or 7(c)-I, respectively.

| Package | RoHS exemption 7(a) | RoHS exemption 7(c)-I |
|------------|---------------------|-----------------------|
| SOD1001-1 | TRUE | TRUE |
| SOD1002-1 | TRUE | TRUE |
| SOD1003-1 | TRUE | TRUE |
| SOD123FL | TRUE | FALSE |
| SOD123HP | TRUE | FALSE |
| SOD123W | TRUE | FALSE |
| SOD123W-2 | TRUE | FALSE |
| SOD128 | TRUE | FALSE |
| SOD128FL-1 | TRUE | FALSE |
| SOD323HP | TRUE | FALSE |
| SOT1023 | TRUE | FALSE |
| SOT1023A | TRUE | FALSE |
| SOT1205 | TRUE | FALSE |
| SOT1210 | TRUE | FALSE |
| SOT1235 | TRUE | FALSE |
| SOT1289 | TRUE | FALSE |
| SOT1289B | TRUE | FALSE |
| SOT226C | TRUE | FALSE |
| SOT404 | TRUE | FALSE |
| SOT404A | TRUE | FALSE |
| SOT428 | TRUE | FALSE |
| SOT428C | TRUE | FALSE |
| SOT429 | TRUE | FALSE |
| SOT429-2 | TRUE | FALSE |
| SOT669 | TRUE | FALSE |
| SOT78 | TRUE | FALSE |
| SOT78A | TRUE | FALSE |
| SOT8000 | TRUE | FALSE |
| SOT8000A | TRUE | FALSE |
| SOT8005 | TRUE | FALSE |
| SOT8005A | TRUE | FALSE |
| SOT8017 | TRUE | FALSE |
| SOT8018 | TRUE | FALSE |
| SOT8021 | TRUE | FALSE |
| SOT8022 | TRUE | FALSE |
| SOT8070-1 | TRUE | FALSE |
| SOT8071-1 | TRUE | FALSE |

Past Use of Exemptions

Glass diodes belonging to packages SOD27, SOD66, SOD68, and SOD80C were making use of RoHS exemption 7(c)-I in the past. On 2024-03-26, Nexperia issued change notice CN-202310015F, titled "Change to Pb-free Glass Diodes". After the implementation of this change notice on 2024-09-23, Nexperia's glass diodes no longer require RoHS exemption 7(c)-I.

Section 21
BOYAMICRO

| Designators | Part / Value | Qty |
|-------------|-----------------|-----|
| U9 | BY25Q64ESCIG(R) | 1 |

RoHS evidence document: **2006180000_BOYAMICRO-BY25Q64ESCIG-R_C50176394.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863



珠海博雅科技有限公司
ZHUHAI BOYA TECHNOLOGY CO.,LTD

Certificate of Compliance

Company: 珠海博雅科技有限公司

Address: 珠海市唐家湾镇大学路 101 号清华科技园创业大楼 A 座 A1106-1107 单元

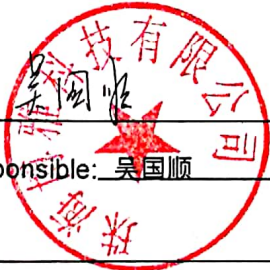
I. We assure that all of Boya's products(Nor flash chips) provided to all of Boya's customers do not contain the Substances of Very High Concern(SVHC)-Reach 209 items

II. We assure that all of Boya's products(Nor flash chips) provided to all of Boya's customers is in conformity with the provisions of regulation, that is RoHS II 2011/65/EU + (EU)2015/863, listed as below Table I.

Table I:

| | Substances | Threshold Level |
|----|---|---|
| 1 | Cadium and Cadimium compounds (Cd) | <100ppm or <0.01% |
| 2 | Lead and Lead compounds (Pb) | <1000ppm or <0.1% |
| 3 | Mercury and Mercury Compounds (Hg) | <1000ppm or <0.1% |
| 4 | Hexavalent chromium compounds (Cr6+) | <1000ppm or <0.1% |
| 5 | Polybrominated biphenyls (PBBs) | <1000ppm or <0.1% |
| 6 | Polybrominated diphenyl ethers (PBDEs) | <1000ppm or <0.1% |
| 7 | Bis(2-ethylhexyl) phthalate (DEHP) | <1000ppm or <0.1% |
| 8 | Butyl Benzyl phthalate (BBP) | <1000ppm or <0.1% |
| 9 | Dibutyl phthalate (DBP) | <1000ppm or <0.1% |
| 10 | Diisobutyl phthalate (DIBP) | <1000ppm or <0.1% |
| 11 | Red Phosphorous (as flame retardant in resin) | <1000ppm or <0.1% |
| 12 | PFOS | 1) textiles or other coated materials:< 1µg/m ²
2) Other: 1000ppm |
| 13 | DIMETHYLFUMARATE (DMF) | <0.1ppm |
| 14 | Ozone Depleting Substances (ODS) | Not used or present |

Company Seal:

Authorized Signature: 

Name of a person responsible: 吴国顺

Title: QA

Phone: 0756-3610791

Date: 2020/06/18

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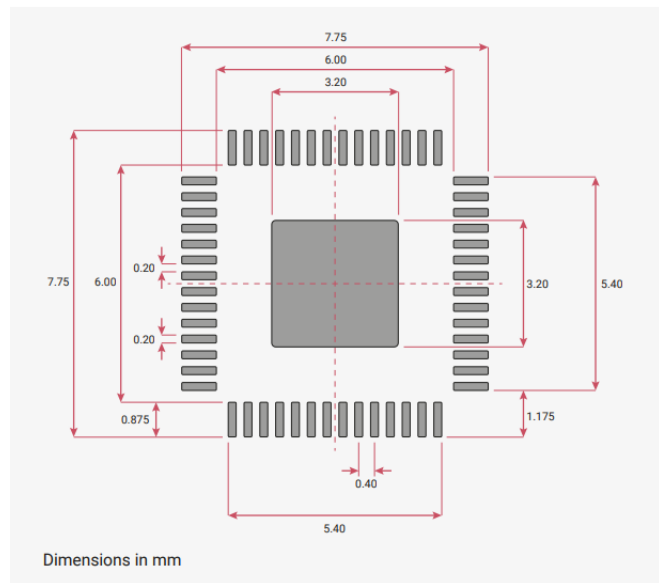
Section 22
Raspberry Pi

| Designators | Part / Value | Qty |
|-------------|--------------|-----|
| U10 | RP2040 | 1 |

RoHS evidence document: [rp2040-rohs.png](#)
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

RP2040 Datasheet

Figure 165.
Recommended PCB
Footprint for the
RP2040 QFN-56
package



5.1.2. Compliance

RP2040 is compliant to Moisture Sensitivity Level 1.

RP2040 is compliant to the requirement of REACH Substances of Very High Concern (SVHC) that ECHA announced on 25 June 2020.

RP2040 is compliant to the requirement and standard of Controlled Environment-related Substance of RoHS directive (EU) 2011/65/EU and directive (EU) 2015/863.

5.2. Pinout

5.2.1. Pin Locations

Section 23

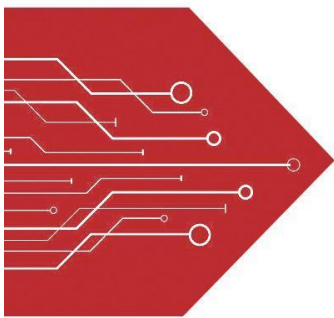
MSKSEMI

| Designators | Part / Value | Qty |
|-------------|----------------|-----|
| U11,U26 | TPD4E05U06DQAR | 2 |

RoHS evidence document: **C2836386-TPD4E05U06DQAR-MS.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 3 of 6 — RoHS-relevant pages only; the complete document is retained in the project repository.

MSKSEMI

SEMICONDUCTOR



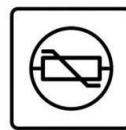
ESD



TVS



TSS



MOV



GDT



PLED

Product data sheet

www.msksemi.com



TPD4E05U06DQAR-MS



Semiconductor Compliance

- Solid-state silicon-avalanche technology
- Low operating and clamping voltage
- Up to four I/O Lines of Protection
- Ultra low capacitance: 0.5pF typical(I/O to I/O)
- Low Leakage
- Low operating voltage:3.3V
- Flow-Through design

IEC COMPATIBILITY (EN61000-4)

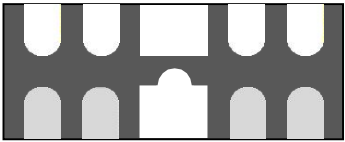
- IEC 61000-4-2 (ESD) ±15kV (air), ±8kV (contact)
- IEC 61000-4-4 (EFT) 40A (5/50ns)
- IEC 61000-4-5 (Lightning) 5A (8/20µs)

Mechanical Characteristics

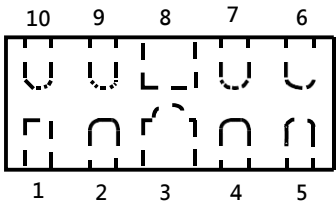
- Molding compound flammability rating: UL 94V-0
- Marking: Marking Code
- Packaging: Tape and Reel
- RoHS/WEEE Compliant

Applications

- Digital Visual Interface(DVI)
- MDDI Ports
- DisplayPort™ Interface
- PCI Express
- High Definition Multi-Media Interface(HDMI)
- eSATA Interfaces



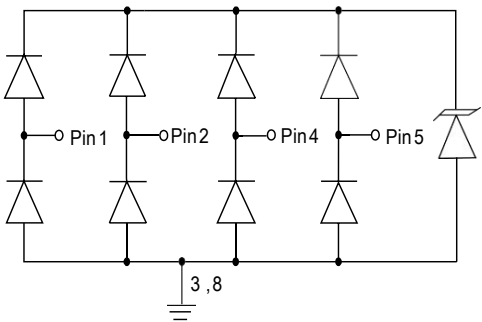
uSON-10



Schematic & PIN Configuration

| Pin | Identificaion |
|----------|--|
| 1,2,4,5 | Input Lines |
| 6,7,9,10 | Output Lines
(No Internal Connection) |
| 3,8 | Ground |

Circuit Diagram

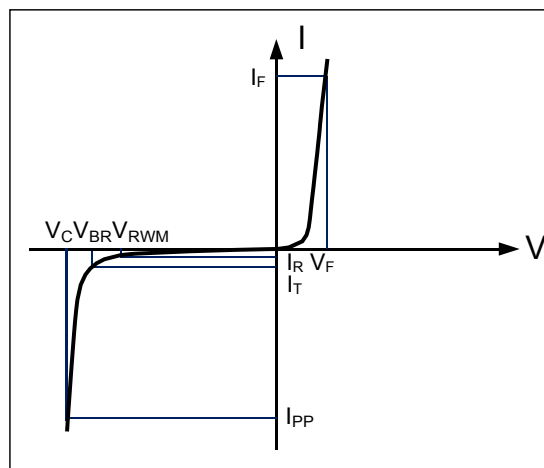


4-Line Protection

| Rating | Symbol | Value | Units |
|--|-----------|--------------|-------|
| Peak Pulse Power ($t_p=8/20\mu s$) | P_{PP} | 150 | Watts |
| Peak Pulse Current ($t_p=8/20\mu s$) | I_{pp} | 5 | A |
| ESD per IEC 61000-4-2(Air) | V_{ESD} | +/-17 | kV |
| ESD per IEC 61000-4-2(contact) | | +/-12 | |
| Operating Temperature | T_J | -55 to + 125 | °C |
| Storage Temperature | T_{STG} | -55 to +150 | °C |

Electrical Parameters (T=25°C)

| Symbol | Parameter |
|-----------|---|
| I_{PP} | Maximum Reverse Peak Pulse Current |
| V_C | Clamping Voltage @ I_{PP} |
| V_{RWM} | Working Peak Reverse Voltage |
| I_R | Maximum Reverse Leakage Current @ V_{RWM} |
| V_{BR} | Breakdown Voltage @ I_T |
| I_T | Test Current |
| I_F | Forward Current |
| V_F | Forward Voltage @ I_F |



Electrical Characteristics

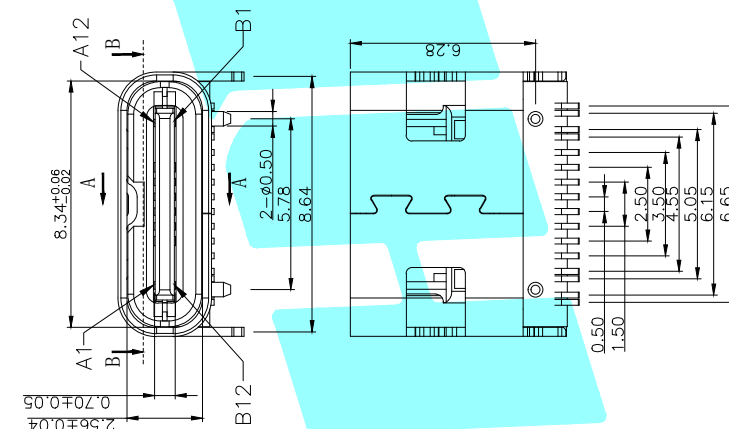
| Parameter | Symbol | Conditions | Minimum | Typical | Maximum | Units |
|---------------------------|-----------|--|---------|---------|---------|---------|
| Reverse Stand-Off Voltage | V_{RWM} | Any I/O pin to ground | | | 3.3 | V |
| Reverse Breakdown Voltage | V_{BR} | $I_T = 1mA$
Any I/O pin to ground | 6.0 | | | V |
| Reverse Leakage Current | I_R | $V_{RWM} = 5V$, $T=25^\circ C$
Any I/O pin to ground | | | 1 | μA |
| Clamping Voltage | V_C | $I_{pp}=5A$, $t_p=8/20\mu s$
Any I/O pin to ground | | | 15 | V |
| Junction Capacitance | C_j | $V_R = 0V$, $f = 1MHz$
I/O pin to GND | | | 0.8 | pF |
| | | $V_R = 0V$, $f = 1MHz$
Between I/O pins | | 0.3 | | pF |

Section 24

Hroparts

| Designators | Part / Value | Qty |
|-------------|----------------|-----|
| USB1,USB2 | TYPE-C-31-M-12 | 2 |

RoHS evidence document: **2410010003_Korean-Hroparts-Elec-TYPE-C-31-M-12_C165948.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863

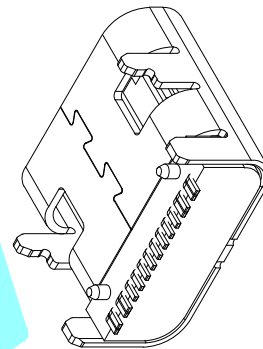
$$\underline{16-0.20\pm0.05}$$


1. MATERIAL SPECIFICATION:
 1. HOUSING: HIGH TEMPERATURE RESISTANT PLASTIC, UL94 V-0.
 2. CONTACTS: COPPER ALLOY
 3. MID PLATE: STAINLESS STEEL
 2. PLATING SPECIFICATION:
 2-1. CONTACTS:
 Ni 50μ" MIN. UNDER PLATED OVER ALL.
 Au PLATED ON THE FUNCTIONAL AREA OF CONTACT.
 (GOLD PLATING THICKNESS FOLLOW THE P/N)
 GOLD FLASH PLATING ON SOLDER AREA
 2-2. FRONT SHELL:

3. MECHANICAL PERFORMANCE,
3-1. INSERTION FORCE: 0.5~2.0kgf.
3-2. REMOVAL FORCE: 0.8kgf~2.0kgf.
3-3. DURABILITY: 10000 CYCLES.
4. ELECTRICAL PERFORMANCE,
4-1. CURRENT RATING: 5A
VOLTAGE RATING: 20V
4-2. INSULATION RESISTANCE: 100MΩ
4-3. DIELECTRIC WITHSTAND VOLTAGE: AC
5. ENVIRONMENTAL PERFORMANCE:
OPERATING TEMPERATURE: -30℃~+80℃
6. IR REFLOW:
THE PEAK TEMPERATURE ON THE BOARD
BE MAINTAINED FOR 10 SECONDS AT 260℃

RECOMMEND P.C.B LAYOUT(COMPONENT SIDE)
TOLERANCE FOR PCB LAYOUT IS ± 0.05

| | | | |
|-----|-------------|-----|-------------|
| A1 | GND | B12 | GND |
| A4 | VBUS | B9 | VBUS |
| A5 | CC1 | B8 | VBUS |
| A6 | DP1 | B7 | DN2 |
| A7 | DN1 | B6 | DP2 |
| A8 | SBU1 | B5 | CC2 |
| A9 | VBUS | B4 | VBUS |
| A12 | GND | B1 | GND |
| PIN | SIGNAL NAME | PIN | SIGNAL NAME |

[illegible]

Section 25

TXC

| Designators | Part / Value | Qty |
|-------------|--------------|-----|
| Y1 | 7M12000044 | 1 |

RoHS evidence document: **TXC-7M12000044.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863
Excerpt: page(s) 1, 2, 3 of 11 — RoHS-relevant pages only; the complete document is retained in the project repository.



4F, NO. 16, Sec. 2 Chung Yang S Rd., Peitou, Taipei, Taiwan.

TEL : 886-2-2894-1202 , 886-2-2895-2201 FAX : 886-2-2894-1206 , 886-2-2895-6207

www.txccorp.com

SPECIFICATION FOR APPROVAL

CUSTOMER : _____

PRODUCT TYPE : SMD SEAM SEALING X'TAL 3.2×2.5

NOMINAL FREQ. : 12.000000MHz

TXC P/N : 7M12000044

REVISION : A1

CUSTOMER P/N : _____

PM / SALES : _____

DATE : _____

CUSTOMER SIGNATURE & Date

- (1) TXC requires one copy returned with signature and title of authorized individual that signifies acceptance of the attached specifications.
- (2) Orders received and accepted by TXC after return of signed copy of specification will be produced per these specifications.
- (3) Any changes to these specifications must be agreed upon by both parties and new revision of the Product Specification Sheet will be issued.
- (4) Any issuance of purchase order prior to consigning back the Approval page of "Specification Sheets" from customers will be regarded as the agreement on the contents of these specifications.

MSL:Level 1
RoHS Compliant



4F, NO. 16, Sec. 2 Chung Yang S Rd., Peitou, Taipei, Taiwan.

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www.txccorp.com

PRODUCT SPECIFICATION SHEET

PRODUCT TYPE : SMD SEAM SEALING X'TAL 3.2×2.5

NOMINAL FREQ. : 12.000000MHz

TXC P/N : 7M12000044

REVISION : A1

| PE/RD | QA | MFG |
|--|---|--|
| 
Robin Huang | 
Samson Xiong | 
Jake Liu |
| 4-Aug-14 | 4-Aug-14 | 4-Aug-14 |

NOTE:

(1)The green product standard set by TXC is based upon the international standards. Related information is publicly described on the TXC's Website, and updated regularly. The document is compliant with the latest green product quality system directives at the time.

(2)Revision "Sx" is for engineering samples only. PE/RD's approval required.

(3)Revision "Ax" is production ready. PE, QA and MFG's approval required

MSL:Level 1
RoHS Compliant

| | | | | | |
|---|--|---------------------|--|--------------|---------|
|  TXC CORPORATION | | TXC P/N: 7M12000044 | | REVISION: A1 | PAGE: 1 |
|---|--|---------------------|--|--------------|---------|

| Rev | Revise page | Revise contents | Date | Ref.No. | Reviser |
|-----|-------------|------------------|-----------|---------|---------------|
| S1 | N/A | Initial released | 10-Jul-14 | N/A | Xiaoyan Jiang |
| A1 | N/A | S Turn A | 4-Aug-14 | N/A | Xiaoyan Jiang |
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Section 26

JLCPCB

| Designators | Part / Value | Qty |
|------------------|--|-----|
| PCB / bare board | Printed circuit board (ENIG & lead-free HASL finish); incl. exposed copper of mounting holes & test points | - |

RoHS evidence document: **rohs-certificate-of-conformity.pdf**
Standard: Directive 2011/65/EU (RoHS) as amended by (EU) 2015/863



DECLARATION OF CONFORMITY (DOC)

NO.: S251120003001-2

The following products have been tested by us and found compliance with the RoHS Directive 2011/65/EU Annex II amending Annex (EU) 2015/863 of CE Directive.

Customer : SHENZHEN JLC TECHNOLOGY GROUP CO.,LTD.

Address : 27TH FLOOR, OLYMPIC BUILDING,SHANGBAO ROAD,FUTIAN DISTRICT,SHENZHEN CITY,GUANGDONG PROVINCE

Manufacturer : SHENZHEN JLC TECHNOLOGY GROUP CO.,LTD.

Sample Name : PCB

Sample Description : Lead-free-HASL

Sample Quantity : 1

Report No : S251120003001-1

Test method : EN 62321-4:2014+A1:2017 & EN 62321-5:2014 & EN 62321-7-2:2017& EN 62321-6:2015 & EN 62321-8:2017



Signature:..



Technical director

Signed for and on behalf of

Soar Testing Certification (Guangdong) Co., Ltd.

Dec 01, 2025

Note: This certification is part of the full test report(s) and should be read in conjunction with it. The certificate is based on a single evaluation of one sample of above-mentioned products. It does not imply an assessment of the whole production and does not permit the use of the test lab logo.

Lab name: Soar Testing Certification (Guangdong) Co., Ltd.

Post Code: 523945

Tel: (86-769) 38937518

Address: Room 1093, No.12, East of Houjie Avenue, Houjie, Dongguan, Guangdong, China

E-mail: service@soar-cert.com

Website: <http://www.soartestlab.com>



DECLARATION OF CONFORMITY (DOC)

NO.: S251120003002-2

The following products have been tested by us and found compliance with the RoHS Directive 2011/65/EU Annex II amending Annex (EU) 2015/863 of CE Directive.

Customer : SHENZHEN JLC TECHNOLOGY GROUP CO.,LTD.

Address : 27TH FLOOR, OLYMPIC BUILDING,SHANGBAO ROAD,FUTIAN DISTRICT,SHENZHEN CITY,GUANGDONG PROVINCE

Manufacturer : SHENZHEN JLC TECHNOLOGY GROUP CO.,LTD.

Sample Name : PCB

Sample Description : ENIG (Electroless Nickel/ Immersion Gold)

Sample Quantity : 1

Report No : S251120003002-1

Test method : EN 62321-4:2014+A1:2017 & EN 62321-5:2014 & EN 62321-7-2:2017& EN 62321-6:2015 & EN 62321-8:2017



Signature:..



Technical director
Signed for and on behalf of
Soar Testing Certification (GuangDong) Co., Ltd.
Dec 01, 2025

Note: This certification is part of the full test report(s) and should be read in conjunction with it. The certificate is based on a single evaluation of one sample of above-mentioned products. It does not imply an assessment of the whole production and does not permit the use of the test lab logo.

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